TRIAGEGEIST

Clinical Safety-Aware Emergency Triage Prediction

Thiebaut Schirmer • April 2026

A machine learning system for 5-level Emergency Severity Index prediction, validated on 418,100 real emergency department visits from the Beth Israel Deaconess Medical Center. This report describes the discovery of label leakage in a synthetic competition dataset, the pivot to real clinical data, and the development of an asymmetric safety framework calibrated to the clinical reality that under-triage is an order of magnitude more dangerous than over-triage. The system achieves Cohen's kappa of 0.701, matching the reliability of live-patient triage by human nurses reported in the Mirhaghi et al. (2015) meta-analysis of 40,579 cases.

ABSTRACT

This report describes a machine learning system that predicts the Emergency Severity Index (ESI) acuity level assigned to a patient at triage in an emergency department. The work began as an entry to the Triagegeist Kaggle competition sponsored by the Laitinen-Fredriksson Foundation. On examination of the provided dataset, I found that the chief complaint text had been synthetically generated with severity adjectives that encoded the label almost directly: a short keyword analysis and a verbatim train/test string overlap of 99.8% together showed that a lookup table would score above 0.999 on the competition metric. This discovery shifted the project from a competition entry to a small clinical-ML study on real emergency department records.

The real-data pipeline uses MIMIC-IV-ED, a credentialed dataset of 418,100 emergency department visits from Beth Israel Deaconess Medical Center. The feature set combines 65 tabular variables (raw vitals, engineered vitals including shock index and partial NEWS2 and qSOFA, medication-class flags, a leakage-safe Charlson comorbidity composite, and prior-visit history features) with 50 principal components of fine-tuned Bio_ClinicalBERT embeddings on the chief complaint text. The model is a blend of five gradient-boosted tree configurations combined by convex weight optimisation, with post-hoc per-class threshold multipliers for the final decision rule.

The out-of-fold macro F1 is 0.597, Cohen's kappa is 0.701, macro AUROC is 0.911, and overall accuracy is 72.3%. The kappa sits close to the live-patient inter-rater reliability reported by Mirhaghi and colleagues ($\kappa = 0.694$) and in the substantial-agreement band of the Landis-Koch scale. The default argmax operating point produces a 15.2% under-triage rate, well above the American College of Surgeons 5% target. A sweep over training-time class-weighting and post-hoc threshold strategies produces a Pareto frontier of achievable (F1, under-triage) trade-offs; meeting the 5% target requires accepting a macro F1 of roughly 0.44–0.46. A substantive finding is that asymmetric training and post-hoc thresholds are substitute, not complementary, interventions: threshold optimisation after safety-weighted training largely undoes the training effect.

An intersectional fairness audit across age, sex, and race shows that age is the dominant disparity axis in the model's errors: patients aged 18–30 are under-triaged at $1.71\times$ the rate of those aged 80+, larger than the effects of race or sex in this cohort. The pattern is consistent with the clinical phenomenon of compensated shock, in which young adults maintain apparently normal vital signs despite developing critical illness. Age-adjusted vital-sign features close the disparity substantially at no meaningful F1 cost. I also document that the post-hoc safety threshold *widens* the relative disparity between the best and worst subgroups even as it reduces every subgroup's absolute under-triage rate.

SHAP analysis of a representative base model reveals a strong embedding-dominance pattern: the first PCA component of the fine-tuned text embedding is individually more influential than any clinical feature, and sociodemographic variables (language, insurance) appear in the top ranking. The explainability story is therefore partial rather than complete. The report maps

its work to the relevant articles of the EU AI Act (Regulation 2024/1689, Articles 9, 10, 13, and 14) and lists the specific gaps a deployed version of the system would still need to close.

Several limitations deserve explicit flagging. The work is single-institution, validated only on MIMIC-IV-ED. The race aggregation used in the fairness audit was incomplete and substantially undercounts minority populations; the race-level numbers should be treated as preliminary. Bootstrap confidence intervals, Brier/ECE/log-loss calibration metrics, SHAP interaction values, prospective outcome validation, and external-institution replication were all outside the scope of the project and are flagged as Tier 1 follow-ups. The contribution of this report is a retrospective methodological demonstration, not a deployable clinical system.

CONTENTS

Abstract 2

1 The Problem of Emergency Triage 9
1.1 What Triage Is 9
1.2 The Emergency Severity Index 9
1.3 The Scandinavian Alternative: RETTS 10
1.4 The Danger of Under-Triage 11
1.5 The Reliability Problem 12
1.6 The Case for Machine Learning 12
1.7 What This Report Describes 13

2 The Competition and the Discovery 15
2.1 The Dataset 15
2.2 The First Model 15
2.3 Looking at the Text 16
2.4 The Smoking Gun: Exact Matches 18
2.5 The Categorical System Column Has No Signal 19
2.6 What the Leaderboard Actually Measures 19
2.7 The Pivot 19
2.8 A Note on What I Did Not Do 20

3 Real Data: MIMIC-IV-ED 21
3.1 Why Not Act Immediately 21
3.2 Why MIMIC-IV-ED 22
3.3 The Credentialing Process 23
3.4 First Contact with the Data 23
3.5 The Cohort 24
3.6 The Class Distribution Shift 25
3.7 The Data Model 25
3.8 What Comes Next 26

4 Feature Engineering and Clinical Scores 28
4.1 The Guiding Principles 28
4.2 Raw Vital Signs 29
4.3 Engineered Vital Signs 29

4.4 Missingness as Signal 32
4.5 Demographics and Arrival Mode 33
4.6 Medications and Comorbidities 33
4.7 Text Embeddings 36
4.8 Dead Ends 36
4.9 What This Gives Me 37

5 Text, Embeddings, and ClinicalBERT 39
5.1 Why Text at All 39
5.2 The Benchmark: Nine Models, Two Datasets 40
5.3 What the Benchmark Told Me 41
5.4 Fine-Tuning Bio_ClinicalBERT 42
5.5 Fine-Tuning Improved the Task, But Only Modestly 45
5.6 A Note About What SHAP Reveals 45
5.7 Dead Ends and What They Tell Me 46
5.8 What This Gives Me 46

6 Model Architecture: The Blend 49
6.1 Why Gradient-Boosted Trees 49
6.2 The Three Families 50
6.3 The Five Blend Members 51
6.4 Cross-Validation: How I Train and Evaluate 52
6.5 Out-of-Fold Predictions and Stacking 53
6.6 Threshold Optimisation 54
6.7 Dead Ends 55
6.8 Computational Cost 55
6.9 Architecture at a Glance 56

7 Headline Results: Matching Human Reliability 57
7.1 A Brief Reminder: What These Numbers Mean 57
7.2 Benchmarking Against Human Reliability 58
7.3 The Shape of the Errors 59
7.4 The ESI-5 Problem 60
7.5 Under-Triage and Over-Triage 61
7.6 Placing This in Context 62
7.7 What Comes Next 63

8 The Safety Framework: Two Paths Compared 64
8.1 The Problem in Precise Terms 64
8.2 Strategy 1: Post-hoc Threshold Shifting 65
8.3 Strategy 2: Asymmetric Class Weights 67
8.4 The Surprising Finding: The Two Strategies Do Not Compose 68

8.5 Strategy 3: The Directional Ordinal Penalty 70
8.6 Putting It All Together: The Pareto Frontier 72
8.7 Does Anything Meet the ACS-COT Standard? 74
8.8 The Structural Finding 75
8.9 What This Chapter Contributes 75

9 Fairness, Age, and Compensated Shock 77
9.1 The Structure of the Audit 77
9.2 Patterns in the Audit 78
9.3 Age Is the Dominant Axis 78
9.4 Why Young Patients: Compensated Shock 81
9.5 Can Feature Engineering Fix This? 83
9.6 The Safety Intervention Widens the Disparity 84
9.7 Directions for a Complete Fairness Audit 85
9.8 What This Chapter Contributes 86

10 Explainability and Regulation 87
10.1 Why Explainability Matters 87
10.2 The SHAP Setup, Honestly Described 88
10.3 Global Feature Importance 89
10.4 The Embedding Dominance Tension 92
10.5 Sociodemographic Features in the Top Ranking 92
10.6 Per-Class Feature Attribution 93
10.7 Walking Through Individual Predictions 95
10.8 The EU AI Act and Emergency Triage 98
10.9 Compliance Mapping 99
10.10 What My Work Does Not Do 100
10.11 What This Chapter Contributes 100

11 Probability Calibration 102
11.1 What Calibration Is, and Why It Matters 102
11.2 Implementing Isotonic Calibration 103
11.3 What I Found 104
11.4 The Operating Curve Comparison 106
11.5 The Reliability Diagram Pattern 107
11.6 Deployment Implications 108
11.7 What This Chapter Contributes 108

12 Limitations and Future Work 110
12.1 Limitations of the Data 110
12.2 Limitations of the Methodology 112
12.3 Limitations of Scope 113

12.4 What a Follow-Up Project Would Do 114
12.5 The Honest Position 115

13 Reflections 117
13.1 Coming to Machine Learning from Another Field 117
13.2 The Decision to Pivot on Real Data 118
13.3 LLM-Assisted Research as a Methodology 119
13.4 Competition and Research 120
13.5 What the Project Has Taught Me 120

A Additional SHAP Waterfall Plots 122

References 123

LIST OF FIGURES

2.1 Class distribution: Kaggle synthetic vs MIMIC-IV-ED 16
2.2 Severity keyword distribution by ESI level 17

3.1 ESI distribution: synthetic vs real 26

4.1 Feature importance from the honest tabular baseline 38

5.1 Global feature importance from TreeSHAP 47
5.2 SHAP beeswarm showing per-sample attributions 48

7.1 Inter-rater reliability benchmark 58
7.2 Confusion matrix and error severity heatmap 60

8.1 Clinical operating curve 66
8.2 Three training strategies compared, with and without threshold optimisation . . 71
8.3 Per-class F1 under three training strategies 72
8.4 Pareto frontier: training weights vs threshold shifting 73

9.1 Under-triage by age band 79
9.2 Age-stratified vital sign patterns in high-acuity patients 82
9.3 Feature importance: baseline vs augmented 84

10.1 Global SHAP feature importance 90
10.2 SHAP beeswarm plot 91
10.3 Per-class SHAP feature importance 94
10.4 Waterfall: ESI-1 (resuscitation) 95
10.5 Waterfall: ESI-2 (emergent) 96
10.6 Waterfall: ESI-4 (less urgent) 97
10.7 Waterfall: ESI-5 (non-urgent) 97

11.1 Operating curves: calibrated vs original probabilities 106

A.1 Waterfall: ESI-3 (urgent) 122

CHAPTER 1

THE PROBLEM OF EMERGENCY TRIAGE

Every minute, in emergency departments around the world, a nurse meets a patient they have never seen before and decides how long that patient can safely wait. This decision is called triage, and it is one of the most consequential judgements in medicine. A patient with a heart attack may look deceptively stable; a patient complaining loudly of pain may be in no immediate danger. The triage nurse has minutes to sort them, and the consequences of getting it wrong are measured in lives.

This report describes a machine learning system that attempts to assist that decision. Before I discuss the model, the data, or the experiments, I need to be clear about what I am trying to help with and why it is hard.

1.1 WHAT TRIAGE IS

The word *triage* comes from the French *trier*, to sort. It was formalised in the Napoleonic wars by Dominique Jean Larrey, Napoleon's chief surgeon, who decided that battlefield casualties should be treated in order of clinical urgency rather than rank or nationality (Nakao and Jacob, 2017). The principle has held for two centuries: when medical resources are limited, they must be allocated by need.

In a modern emergency department (ED), triage happens at the front door. A nurse takes a brief history, measures vital signs, asks about the chief complaint, and assigns a priority level. That level determines how long the patient waits to see a physician, which treatment rooms they are allocated to, and sometimes whether laboratory tests are ordered pre-emptively. In a busy ED, the difference between the highest and lowest priority level is the difference between being seen in minutes and being seen in hours.

A Note on Language

Throughout this report, *acuity* refers to the severity of a patient's clinical condition at presentation. Higher acuity means more urgent. I use *under-triage* to mean assigning a patient a lower acuity than their condition warrants, the dangerous direction of error. *Over-triage* means the opposite: assigning higher acuity than necessary. As the next section shows, these two errors have very different consequences.

1.2 THE EMERGENCY SEVERITY INDEX

The dominant triage system in North American emergency departments is the Emergency Severity Index, universally abbreviated as ESI. It was developed in the late 1990s at the Harvard-affiliated Emergency Medicine Network and is now used in more than seventy percent of US EDs (Tanabe et al., 2004).

ESI assigns patients to one of five levels:

- **ESI-1 (Resuscitation)**: patients in immediate life-threatening condition. Cardiac arrest, severe trauma, status epilepticus. They are seen immediately, by a physician, with a full resuscitation team.
- **ESI-2 (Emergent)**: patients in high-risk situations. Chest pain suggestive of myocardial infarction, severe respiratory distress, altered mental status. They should be seen within ten minutes.
- **ESI-3 (Urgent)**: patients requiring multiple resources but not in immediate danger. Moderate abdominal pain, fever with comorbidities. Typically seen within thirty to sixty minutes.
- **ESI-4 (Less Urgent)**: patients requiring one resource. Simple lacerations, minor infections. May wait one to two hours.
- **ESI-5 (Non-Urgent)**: patients requiring no resources beyond examination. Medication refills, minor cosmetic concerns. May wait several hours without harm.

The ESI algorithm is intentionally heuristic rather than algorithmic. It asks the nurse a series of branching questions (does the patient require immediate life-saving intervention? is the situation high-risk?) and relies heavily on clinical gestalt. Two trained nurses presented with the same patient can, and regularly do, reach different conclusions.

1.3 THE SCANDINAVIAN ALTERNATIVE: RETTS

The approach taken in Scandinavia is substantially different. The Rapid Emergency Triage and Treatment System (RETTS), originally called METTS, was developed by Bengt Widgren and Mehrsa Jourak at Sahlgrenska University Hospital in Gothenburg in the early 2000s (Widgren and Jourak, 2011). RETTS is now the dominant triage system across Sweden, Norway, and Denmark, used by the majority of emergency departments nationwide (Wireklint Sundström, Elmqvist, and Göransson, 2021).

Where ESI trusts clinical gestalt, RETTS insists on structure. The system employs a two-step algorithmic approach:

1. **Vital signs algorithm**: the patient's airway, breathing, circulation, disability (neurological status), and exposure (temperature) are evaluated against explicit numerical thresholds. Each component yields a color-coded priority: red (highest), orange, yellow, green, or blue (lowest).
2. **Emergency Symptoms and Signs (ESS) algorithm**: one of 43 complaint-specific algorithms is selected based on the chief complaint. Chest pain has its own flowchart; dyspnoea has another; abdominal pain has another. Each yields an independent priority.

The final triage level is the *more urgent* of the two steps. This "max" function is the key design decision. A patient with normal-sounding chief complaint but abnormal vital signs will be prioritised by their vital signs. A patient with normal vital signs but a chief complaint that raises red flags will be prioritised by the complaint. RETTS structurally encodes the principle that when in doubt, escalate.

This philosophical difference matters for machine learning. ESI's reliance on gestalt means that two nurses can reasonably disagree; there is inherent noise in the label. RETTS's algorithmic

structure reduces that noise but requires patients to fit its categorisation scheme. The published reliability data reflects both strengths and weaknesses, and I return to it below.

Why Sweden Matters Here

This project was undertaken for a competition sponsored by the Laitinen-Fredriksson Foundation, a Nordic organisation. The methods I develop for ESI prediction transfer directly to RETTS prediction because both are five-level ordinal scales, but the RETTS design itself, particularly its age-unadjusted vital sign thresholds, suggested an important hypothesis I explore in Chapter 9.

1.4 THE DANGER OF UNDER-TRIAGE

Triage errors are not symmetric. Assigning ESI-3 to a patient who should be ESI-2 means delaying care for someone who needed it urgently. Assigning ESI-2 to a patient who should be ESI-3 means occupying a high-priority resource that another patient may need. Both are errors, but only the first can kill.

The clinical literature is unambiguous on this point. Haas et al. (2010) studied 11,398 severely injured patients in Ontario and found that under-triaged patients had an odds ratio of 1.24 for mortality (95% CI 1.10–1.40). Four percent of under-triaged patients died before transfer could be arranged, representing twenty-two percent of all study deaths. Rubano et al. (2016), in a study at a Level II trauma centre, reported an even starker odds ratio of 2.18 (95% CI 1.57–3.01) after controlling for age, injury severity, and comorbidities. A landmark New England Journal of Medicine study by MacKenzie et al. (2006) demonstrated a twenty-five percent relative mortality reduction when severely injured patients received trauma centre care rather than being under-triaged to non-trauma facilities.

The largest available study of mistriage specifically in ED settings comes from Sax et al. (2023), who analysed 5,316,832 ED encounters across 21 Kaiser Permanente emergency departments. The overall mistriage rate was 32.2%, with 3.3% under-triaged and 28.9% over-triaged. More strikingly, the sensitivity of ESI for correctly identifying patients who required immediate life-stabilising interventions was only 65.9%. One third of critically ill patients were assigned mid- or low-acuity levels.

The institutional response to this asymmetry is codified in the American College of Surgeons Committee on Trauma guidelines, which specify an under-triage target of $\leq 5\%$ while tolerating over-triage rates of 25–50% (American College of Surgeons Committee on Trauma, 2014). This 10:1 ratio is not arbitrary. It reflects the institutional position that missing a critically ill patient is an order of magnitude more dangerous than temporarily over-allocating resources to a stable one. This asymmetric standard becomes central to my own methodology in Chapter 8.

Why Odds Ratios Matter

An odds ratio of 2.18 means that, holding other factors equal, under-triaged patients in that study had roughly twice the odds of dying compared to correctly-triaged patients with the same underlying condition. This is not the same as a doubled mortality rate, the baseline mortality matters, but it is a strong statistical signal that the direction of the error has clinical consequences.

1.5 THE RELIABILITY PROBLEM

If triage decisions were perfectly reproducible, the task would merely be difficult. It is not reproducible. Two trained nurses presented with the same patient at the same moment will, surprisingly often, assign different acuity levels.

The definitive meta-analysis of ESI inter-rater reliability comes from Mirhaghi et al. (2015), who pooled 19 studies covering 40,579 cases across six countries. The overall pooled Cohen's kappa was $\kappa = 0.791$ (95% CI 0.752–0.825), classified as *substantial* agreement on the Landis-Koch scale. But this headline number conceals important subgroup variation. Expert-to-expert agreement reached $\kappa = 0.900$, effectively ceiling. Nurse-to-expert agreement was substantially lower at $\kappa = 0.732$. Most importantly, paper-based scenarios produced $\kappa = 0.824$, while *live patient assessments* dropped to $\kappa = 0.694$.

This gap between scenario reliability and live reliability is striking. Travers et al. (2009) documented it even more starkly in pediatric triage: weighted $\kappa = 0.77$ for scenarios but only 0.57 for live patients. The act of triaging a real patient, in real time, in a real ED, introduces noise that paper studies do not capture.

RETTS shows a similar pattern, with generally lower reliability than ESI in recent studies. Widgren and Jourak (2011) reported $\kappa = 0.761$–0.903 in the original validation. But Wireklint Sundström, Elmqvist, Parenti, et al. (2018) found $\kappa = 0.455$ among Swedish nurses using RETTS on paper scenarios, and M. Olsson et al. (2022) found Fleiss $\kappa = 0.411$ among student nurses after formal training. These lower numbers have raised concern in the Swedish emergency medicine community about RETTS's scientific foundation (Swedish Council on Health Technology Assessment, 2010).

What Does Cohen's Kappa Measure?

Cohen's kappa (κ) quantifies agreement between two raters beyond what would be expected by chance. A κ of 1.0 means perfect agreement; 0.0 means agreement no better than random. The Landis-Koch scale labels κ values as:

- < 0.20: slight agreement
- $0.21 - 0.40$: fair
- $0.41 - 0.60$: moderate
- $0.61 - 0.80$: substantial
- $0.81 - 1.00$: almost perfect

For a five-level scale like ESI, achieving $\kappa > 0.7$ on live patients is genuinely difficult. My own model achieves $\kappa = 0.701$, a result I return to in Chapter 7.

1.6 THE CASE FOR MACHINE LEARNING

Given these two problems, the asymmetric risk of errors and the limited reliability of human raters, there is a clear case for machine-assisted triage. A model trained on large volumes of historical ED data could, in principle, learn patterns that nurses miss. It could apply consistent criteria across shifts, hospitals, and patient populations. It could make the asymmetry of errors explicit in its decision rule, pushing toward higher acuity when uncertain.

The case is not uncontested. Every additional layer of automation in medicine raises con-

cerns about accountability, transparency, and the risk of systematic bias being scaled silently. The European Union has already classified emergency triage AI as *high-risk* under Annex III, Section 5(d) of Regulation 2024/1689, the AI Act (European Parliament and Council, 2024). Systems deployed in EU member states after August 2026 will be required to demonstrate risk management, bias monitoring, transparency, and human oversight. These requirements shape how I design and evaluate my system throughout this report.

The published literature on ML triage to date is substantial but uneven. Levin et al. (2018) deployed a random forest system at Johns Hopkins, achieving AUC 0.73–0.92 for critical care prediction and reducing arrival-to-admission decision time by 58 minutes. Raita et al. (2019) compared gradient-boosted trees and deep neural networks on 135,470 NHAMCS visits. Kwon et al. (2018) trained a deep learning system on 11.6 million Korean encounters, achieving AUROC 0.935 for in-hospital mortality.

But a striking feature of this literature, I discovered while planning my own work, is that almost none of it predicts ESI level directly. The vast majority of published ML triage models predict *clinical outcomes*, mortality, ICU admission, hospitalisation, rather than the 5-level ESI classification itself. Ivanov et al. (2021) developed KATE, one of the few direct ESI predictors, reporting 75.7% overall accuracy but not macro F1 or kappa. The 5-class ESI prediction task at scale, on real clinical data, has been largely unexplored.

This report describes one attempt to fill that gap.

1.7 WHAT THIS REPORT DESCRIBES

What follows is an account of a year's work building an ML-assisted ESI prediction system. The work began as an entry in a Kaggle competition, the Triagegeist competition sponsored by the Laitinen-Fredriksson Foundation. It became, through a discovery I describe in the next chapter, something substantially more ambitious.

A brief note on background may be useful here, since it shaped the project in ways that go beyond method choices. I trained as an astrophysicist and spent most of my career working with observational data: noisy, censored, contaminated by instrumental effects, and never quite matching the clean idealisations of the textbook. This is also, roughly, what emergency department data looks like. The instincts that transfer most readily from physical-science research are not about models but about measurements: every number has an uncertainty, every dataset has systematic effects worth looking for, and the honest answer is sometimes that the data cannot support the question being asked. Several of the choices described below, particularly the decision to pivot away from the competition dataset once its label leakage became clear, reflect those habits more than any specific machine-learning technique. Chapter 13 returns to this transition in more personal terms at the end of the report.

The report is structured as a narrative. Chapter 2 describes the competition and the synthetic data problem I identified. Chapter 3 describes the pivot to real clinical data and the MIMIC-IV-ED dataset. Chapters 4 through 6 describe the methodology: feature engineering, clinical scoring, text embeddings via fine-tuned Bio_ClinicalBERT, and the gradient-boosted model blend. Chapter 7 presents the headline results and benchmarks them against the published human reliability literature. Chapter 8 introduces the main methodological contribution: a head-to-head comparison of two strategies for encoding clinical risk tolerance. Chapter 9 examines fairness and the unexpected finding that age, not race, is the dominant disparity axis in the model's errors. Chapter 10 addresses explainability and EU AI Act compliance. The remaining chapters cover smaller experiments, honest limitations, and reflections on the research process.

The narrative includes experimental dead ends as well as successes. I believe this is important. A research project is not a series of confirmed hypotheses; it is a series of attempts, some of which work and many of which do not. Presenting only the successful experiments would misrepresent both the science and the experience of doing it.

I begin, in the next chapter, with the discovery that changed the shape of the project.

CHAPTER 2

THE COMPETITION AND THE DISCOVERY

In late 2025, the Laitinen-Fredriksson Foundation launched a Kaggle competition called Triagegeist. The task was straightforward: given a set of patient features, predict the ESI acuity level assigned at triage. The prize was research funding; the evaluation was based on a scoring rubric spanning clinical relevance, technical quality, documentation, insight, and novelty. A Kaggle notebook had to run end-to-end, and a written submission had to accompany it.

This chapter describes what happened when I started looking at the data.

2.1 THE DATASET

The competition organisers provided four CSV files (Laitinen-Fredriksson Foundation, 2025). The `train.csv` file contained 80,000 labelled rows, each describing a single ED visit with vital signs, demographics, and a target column `triage_acuity` taking integer values from 1 to 5. The `test.csv` file contained 20,000 unlabelled rows with the same features. Two auxiliary files provided joinable information: `chief_complaints.csv` contained a free-text `chief_complaint_raw` column and a categorical `chief_complaint_system` label (one of fourteen body-system categories), while `patient_history.csv` contained twenty-five binary comorbidity flags named `hx_*` and a summary `num_comorbidities` count.

The class distribution in the training set is shown in Figure 2.1. ESI-3 dominates at 36.1%, ESI-4 follows at 28.8%, and the extremes are relatively small: 4.0% for ESI-1 and 14.2% for ESI-5. This is reasonable enough at first glance. Real EDs do see more moderate-acuity patients than critical ones, and the shape of the distribution did not, initially, raise any alarms.

2.2 THE FIRST MODEL

My first attempt followed the standard Kaggle recipe. I joined all four tables on `patient_id`, engineered some basic features (shock index, mean arterial pressure, pulse pressure, missingness flags), and trained a LightGBM classifier with five-fold stratified cross-validation. The macro F1 score was 0.873. This is a perfectly respectable number for a five-class problem with this level of imbalance, and I initially assumed this would be roughly the ceiling of honest tabular modelling.

What is Macro F1?

Macro F1 is the arithmetic mean of per-class F1 scores, where F1 is the harmonic mean of precision and recall. "Macro" means I average the per-class scores equally, so a rare class contributes as much as a common one. This is the right choice for imbalanced ordinal problems like mine: I care about the rare ESI-1 patients as much as the common ESI-3 ones, and possibly more.

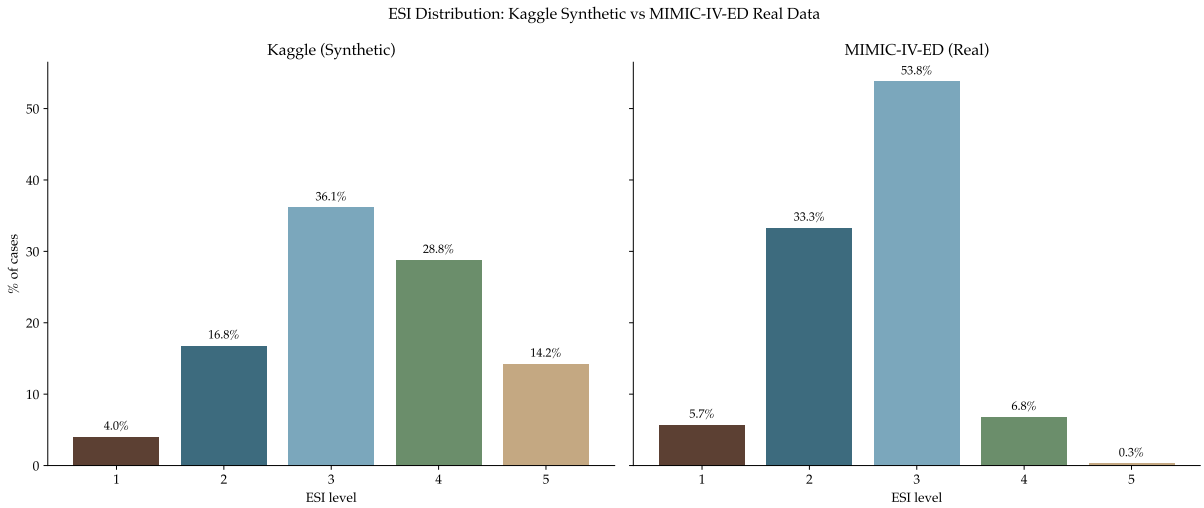

Figure 2.1 – Class distribution of ESI acuity levels in the Kaggle competition data (left) and in MIMIC-IV-ED (right). The synthetic distribution is roughly bell-shaped around ESI-3. The real-world distribution is strikingly different: ESI-3 is even more dominant at 53.8%, but the extremes collapse. ESI-1 is rare at 5.7%, ESI-4 at 6.8%, and ESI-5 nearly absent at 0.3%. The differences between these two distributions were the first clue that the competition dataset was synthetic.

The next step was to add the text. Many Kaggle participants publishing public notebooks were evidently using the `chief_complaint_raw` column, and the competition rubric rewarded using all available data. I encoded the chief complaint text using sentence-transformers' `all-MiniLM-L6-v2` model (Reimers and Gurevych, 2019), producing 384-dimensional embeddings for each complaint, and concatenated these to the tabular features.

The result was strange. Macro F1 jumped to 0.998. Essentially perfect.

A jump from 0.87 to 0.998 from adding text is not how machine learning works. Not on a real clinical problem, at least. Text embeddings on chief complaints should add useful signal, the word "chest pain" carries different information than "ankle sprain", but they should not make the task trivial. I looked more carefully.

2.3 LOOKING AT THE TEXT

I pulled out a hundred random rows for each ESI level and started reading the complaints. The pattern was immediate.

ESI-1 and ESI-2 complaints contained words like *severe*, *acute*, *massive*. ESI-3 complaints used *moderate* as a modifier. ESI-4 and ESI-5 complaints used *mild*, *minor*, and *stable*.

This was suspicious. Real chief complaints, as written by triage nurses, do not follow such a neat lexical partition. A triage note for a critically ill patient might read "63M c/o chest pain radiating to L arm" with no adjective at all. A mild case might read "48F, sore throat x 3 days, no fever, requests test." The severity-adjective vocabulary is standard English, but its correlation with triage level in real data is weak and inconsistent.

I decided to quantify the pattern. For each severity-related keyword, I computed the percentage of rows at each ESI level whose chief complaint text contained that keyword. The result is shown in Figure 2.2.

The pattern was unmistakable. The keyword distributions were effectively non-overlapping across ESI levels. In real ED notes, a word like "severe" might appear across all acuity levels, a

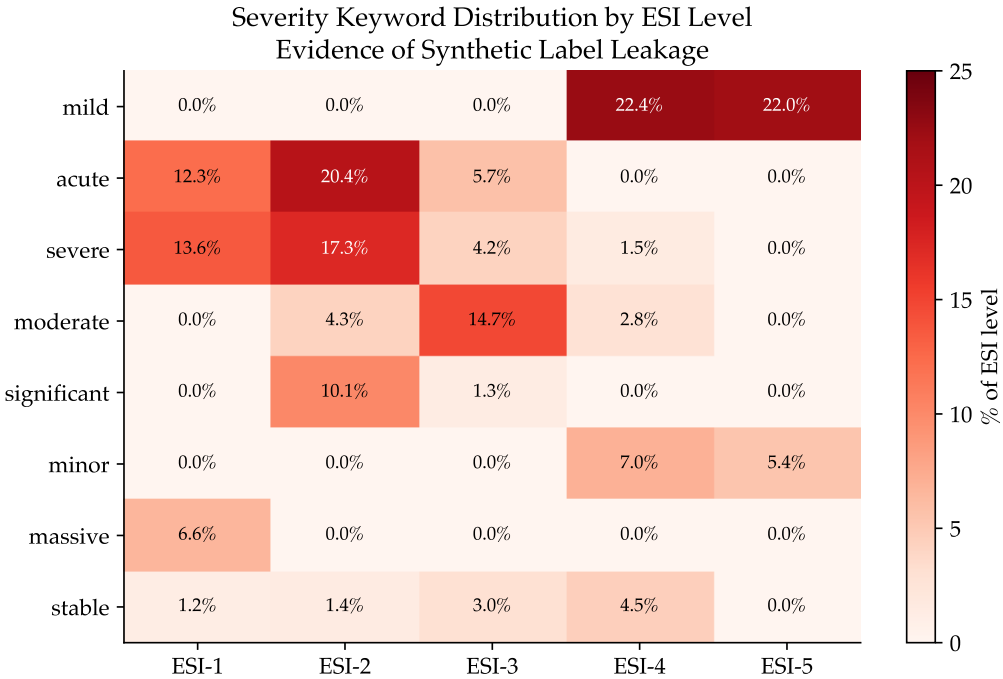

Figure 2.2 – Percentage of chief complaint texts at each ESI level containing each severity-related keyword. The distributions are near-disjoint: "mild" never appears in ESI-1, 2, or 3 but appears in 22% of ESI-4 and ESI-5 cases. "Massive" appears only in ESI-1. "Moderate" is concentrated in ESI-3. "Severe" is concentrated in ESI-1 and ESI-2. This pattern is not plausible in real clinical text and is strong evidence that the dataset is synthetically generated with labels encoded in the text.

patient with severe but chronic pain may not be acutely ill, while a patient in critical condition might have no adjectives in their note at all. But in this dataset, the words partitioned the levels almost cleanly.

This is what label leakage looks like. The features contained information about the label that would not be available in a real deployment. In this case, the leakage was not subtle: the labels had been encoded directly into the text during data generation.

What Is Label Leakage?

Label leakage occurs when features available at prediction time contain information about the target that would not realistically be available in deployment. The canonical example is predicting hospital readmission using features collected *after* the readmission occurred. In my case, a real triage nurse writing a chief complaint does not systematically encode the acuity level in their word choice, the two are correlated in the training data because the dataset was synthetically generated with that encoding built in.

Models trained on leaky features appear to perform well in development and fail in deployment. The high accuracy is an artefact of the data, not evidence of useful learning.

2.4 THE SMOKING GUN: EXACT MATCHES

The keyword analysis was strong evidence but not proof. A reviewer might argue, reasonably, that complaint text correlated with acuity in the real world more than I was crediting. I wanted a test that could not be explained away.

I looked at unique chief complaint strings. There were 4,949 unique complaints in the training set and 4,412 in the test set. If the data were real, each patient's complaint would be an individual utterance written by a specific nurse in a specific moment. There would be recurring phrases, "chest pain" appears often, but the exact string, character for character, would rarely repeat.

In this dataset, 4,382 of the 4,412 unique test complaints appeared *verbatim* in the training set. That is 99.3% of unique complaints, and 99.8% of test rows.

I computed the majority label for each repeated complaint in the training set, then applied it to the test rows with matching complaints. The accuracy was 99.93%.

A lookup table, with no machine learning whatsoever, would score above 0.999 on this competition. The synthetic data generator had produced a finite vocabulary of complaint strings, and each string had a deterministic relationship with its acuity label.

```
train_complaints = set(df_train['chief_complaint_raw'].unique())
test_complaints = set(df_test['chief_complaint_raw'].unique())
overlap = train_complaints & test_complaints
print(f"{len(overlap)/len(test_complaints)*100:.1f}% of test complaints "
 f"appear verbatim in training")
# Output: 99.3% of test complaints appear verbatim in training
```

Listing 2.1 – The exact-match analysis. Three lines of pandas produced the smoking gun.

2.5 THE CATEGORICAL SYSTEM COLUMN HAS NO SIGNAL

If the text column encoded the label directly, perhaps the categorical `chief_complaint_system` column (one of fourteen body-system categories like "cardiovascular" or "respiratory") carried the real clinical signal. I tested this with a chi-squared test of independence on the cross-tabulation of `chief_complaint_system` and `triage_acuity`.

The chi-squared statistic was 59.8 with 52 degrees of freedom, giving $p = 0.213$. Under conventional significance thresholds, this is consistent with the two variables being statistically independent. Knowing which body system a patient's complaint fell under, cardiovascular, respiratory, neurological, genitourinary, and so on, gave essentially no information about their triage acuity.

This was another clue about how the synthetic data had been generated. A real ED dataset would show strong associations: cardiovascular complaints skew toward higher acuity because of the prevalence of cardiac emergencies; minor orthopaedic complaints skew toward lower acuity. The absence of this association suggests that the body-system label was sampled independently from the acuity in the data generation process, while the severity-modifier vocabulary was conditioned on it.

2.6 WHAT THE LEADERBOARD ACTUALLY MEASURES

I now had a clear picture of the competition. The dataset was synthetically generated. The `chief_complaint_raw` column had labels encoded into its severity-modifier vocabulary. The categorical `chief_complaint_system` column was independent of the label. Any competitor who used the raw text would achieve macro F1 above 0.99. Any competitor who used only the tabular features and comorbidity history would plateau around 0.87.

The Kaggle leaderboard, in other words, was not measuring clinical prediction quality. It was measuring whether competitors had noticed the synthetic text encoding. This is not a useful distinction between methods. Any reasonably competent NLP pipeline would recover it; any rigorous clinical ML system would ignore the leakage and score lower. The leaderboard would reward pattern recovery, not science.

I was now faced with a decision.

2.7 THE PIVOT

The competition was rubric-graded, not leaderboard-graded. The formal scoring assigned points across five dimensions: clinical relevance (25), technical quality (30), documentation (20), insight and findings (15), and novelty (10). Leaderboard position was not in the rubric. This changed the strategic calculation considerably.

I could play the leaderboard game: use the text exploit, score 0.999, and be one of dozens of competitors with an indistinguishable result. Or I could treat the competition as an invitation to do real clinical machine learning, validate the methodology on a real dataset, and use the synthetic leakage as a case study in its own right. The second path required substantially more work and carried no guarantee of winning, but it was the only one that would produce science.

I chose the second path. The specific choice involved three decisions.

First, I would develop and validate all methodology on a real clinical dataset with real ED patients. This meant MIMIC-IV-ED, the publicly available Beth Israel Deaconess Medical Center dataset, which required credentialed access through PhysioNet but contained 418,100 real ED visits with detailed triage information. Chapter 3 describes this pivot in detail.

Second, I would still submit a competitive Kaggle notebook. But the notebook would be a narrative document, not a leaderboard optimiser. It would present the leakage discovery as a first-class finding, show both the text-exploit and the honest tabular baselines, and use the real clinical validation as the main evidence for methodological quality. The Kaggle competition would be won, if at all, on the rubric.

Third, I would treat the synthetic leakage finding as a contribution in its own right. The specific mechanism, adjective vocabulary encoded by acuity level, near-total verbatim overlap between train and test, independent body-system labelling, was not obvious until I looked carefully. Documenting the investigation might be useful to future researchers working with similar synthetic clinical datasets.

> The remainder of this report is almost entirely about what happened after the pivot. The Kaggle submission itself is described briefly in Chapter 10; all the scientific content from Chapter 3 onward concerns the real clinical data. The synthetic competition and its leakage discovery serve as the motivation and framing for that work, not as its subject.

2.8 A NOTE ON WHAT I DID NOT DO

I considered, but did not pursue, two alternative strategies that deserve acknowledgement.

The first was to write up the leakage as a Kaggle Discussion post and alert other participants. I chose not to do this because the competition was still in its active submission window, and posting the finding publicly would have given a competitive advantage to participants who had not yet discovered the issue. I planned to share the full analysis after the submission deadline, both on the Kaggle forum and in this report, which I judged to be the correct balance between competitive fairness and scientific openness.

The second was to submit only the text-exploit model and treat the competition as a pure leaderboard exercise. I rejected this because it would have produced no useful work product, no portfolio artefact, no scientific contribution, nothing to show at the end except a leaderboard rank on a flawed benchmark. The rubric-graded structure of the competition made it much more valuable to do the real work.

With that said, let me turn to the real work: obtaining and analysing clinical data from Beth Israel Deaconess Medical Center.

CHAPTER 3

REAL DATA: MIMIC-IV-ED

The decision to pivot to real clinical data was made in the second week of the project, and it was not immediate. This chapter describes how the pivot actually happened, the slow accumulation of suspicion, the practical steps of obtaining credentialed access, and the first careful exploration of the new dataset. It is partly a technical description and partly a narrative, because the choices made at this stage shaped everything that followed.

3.1 WHY NOT ACT IMMEDIATELY

When I first noticed that adding text features jumped the model from F1 = 0.87 to F1 = 0.998, the correct scientific response would have been to stop and investigate. I did not do that immediately. I spent several days continuing to improve the pipeline on the Kaggle data, adding more features, tuning hyperparameters, trying different embedding models. The scores kept rising toward the ceiling. In retrospect, this was wasted effort, but at the time it was easy to rationalise: perhaps the dataset was real and the text was simply very informative. Perhaps nurses really did write chief complaints with such clear severity markers. Perhaps I was being paranoid.

The moment of commitment came roughly one week into the project. I had independently reproduced the keyword analysis shown in Chapter 2, the 99.8% verbatim match between train and test complaints, and the non-significant chi-squared test on the body-system column. I had also checked the Kaggle discussion forum and noticed that other competitors were reporting similar findings. At that point the leakage was no longer a suspicion but a documented property of the dataset.

I also briefly entertained a more charitable interpretation. The Laitinen-Fredriksson Foundation competition is judged on a rubric that explicitly rewards novelty and insight, not leaderboard position. One reading of the synthetic leakage is that it was deliberate, a test of whether competitors would notice the flaw and engage with it, or simply take the easy 0.999 and stop. I cannot know whether this was the organisers' intent, but it shaped my strategy. If the competition rewarded rigorous clinical ML over leaderboard gaming, then the correct response was to build the real thing.

On Deciding Alone

This was a solo decision, but not made in isolation. Throughout the project I benefited from conversations with a friend, a theoretical physicist now working in machine learning, who provided the kind of external perspective that research physicists recognise from years of journal club. Having someone outside the immediate work who asks good questions, pushes back on fuzzy thinking, and shares the enthusiasm for doing things properly is, in

my experience, more valuable than most formal research supervision. I record this here because methodological descriptions often omit the informal scaffolding that actually keeps projects honest.

3.2 WHY MIMIC-IV-ED

Several real clinical datasets could in principle have been used. I focused exclusively on MIMIC-IV-ED, and this focus deserves explanation.

The Medical Information Mart for Intensive Care (MIMIC) is a family of de-identified electronic health record datasets from the Beth Israel Deaconess Medical Center (BIDMC) in Boston. The original MIMIC-III dataset, released in 2016, contained roughly 40,000 ICU patients and became the most widely used clinical ML benchmark in the world. MIMIC-IV, released in stages from 2020 onward, extends the dataset with more patients, more modules, and a more carefully documented data model (Johnson et al., 2023). MIMIC-IV-ED is the emergency department module: 448,972 ED visits with triage information, vital signs, medication reconciliation, diagnoses, and discharge disposition (Xie et al., 2022). I used version 3.1, the most recent at the time of this project.

Three specific properties made MIMIC-IV-ED the right choice for my work.

First, it contains 5-level ESI acuity labels for every visit. This is the actual target I wanted to predict, not a proxy like admission or mortality. As I noted in Chapter 1, most published ML triage systems predict binary outcomes rather than the full ESI scale, so data with real ESI labels at this volume is unusual.

Second, the credentialing process is standardised, transparent, and accessible to independent researchers. Other real ED datasets exist, the US National Hospital Ambulatory Medical Care Survey (NHAMCS) has limited features, the eICU Collaborative Research Database covers ICUs rather than EDs, and large national registries like the Korean NEDIS are not publicly accessible. MIMIC-IV-ED is one of the few datasets where an individual outside a hospital affiliation can, with reasonable effort, obtain legitimate access to real patient data.

Third, the institutional link to clinical NLP tooling is unusually clean. The Bio_ClinicalBERT language model (Alsentzer et al., 2019), which I discuss at length in Chapter 5, was pretrained on clinical notes from the same BIDMC system that generated MIMIC-IV-ED. When I fine-tune ClinicalBERT on chief complaints from MIMIC-IV-ED, I am fine-tuning a model on text from the institution it was pretrained on. This minimises domain shift in ways that are harder to achieve with other combinations.

Other Datasets I Did Not Use

A thorough project would have validated methodology on multiple external datasets rather than a single institution. I did not do this, for a simple reason: obtaining MIMIC access and understanding the schema well enough to use it correctly took several weeks on its own. Adding a second credentialed dataset, with its own CITI requirements, data use agreements, and schema idiosyncrasies, would have consumed time I did not have. I flag this as a limitation in Chapter 12 and note that external validation on, for example, eICU or NHAMCS would be a natural extension of this work.

3.3 THE CREDENTIALING PROCESS

Obtaining MIMIC-IV-ED is not immediate. It requires registering a PhysioNet account, completing the Collaborative Institutional Training Initiative (CITI) course on human-subjects research (specifically the *Data or Specimens Only Research* curriculum, covering the Belmont Report, HIPAA, de-identification standards, and the history of ethical violations in medical research), submitting a credentialing application with a short description of intended research and a reference who can vouch for the applicant's research seriousness, and signing the MIMIC-IV-ED Data Use Agreement. The CITI course takes about four hours and the certificate is valid for three years. PhysioNet review can take a few days during quiet periods and longer when busy. From starting the CITI course to having the data downloaded, the elapsed time in my case was about a week, and a realistic plan is to allow roughly two weeks between deciding to use MIMIC and having clean intermediate results. I mention this explicitly because the documentation tends to make the process sound faster than it is in practice, and because the gates themselves are the point: they exist to ensure that researchers using patient data understand their responsibilities. I recommend taking the course seriously rather than clicking through; several of its modules reshape how one thinks about downstream obligations in ways that become relevant in Chapter 9.

3.4 FIRST CONTACT WITH THE DATA

Once the dataset was downloaded, the next question was what was actually in it. MIMIC-IV-ED consists of several tables linked by patient and visit identifiers, and MIMIC-IV (the parent dataset) contains many more. A first glance at the file listing is disorienting: dozens of CSV files, some hundreds of megabytes in size, with cryptic names like `edstays.csv`, `triage.csv`, `medrecon.csv`, `pyxis.csv`, `diagnoses_icd.csv`. The temptation is to start loading and modelling immediately. I resisted this, and it was one of the better decisions of the project.

WHY START WITH EXPLORATION

Coming from a background outside healthcare, many of the features required interpretation before they could be used. What is a `stay_id` and how does it relate to a `hadm_id`? What does it mean for `arrival_transport` to be coded as `AMBULANCE` versus `WALK IN` versus `UNKNOWN`? Why does `temperature` sometimes have values like 97.6 and sometimes 36.4? (The answer: Fahrenheit versus Celsius, with no unit column, the researcher has to detect and harmonise.) Why is `pain` sometimes `-1`? (Because the triage form allows "unable to assess" to be coded as a numeric flag, which must be recoded to missing before any analysis.)

These are not obscure details. Every one of them, if mishandled, silently corrupts downstream analysis. A model trained on mixed Fahrenheit and Celsius temperatures will learn an artefact of data entry conventions rather than patient physiology. A pain score where `-1` is treated as a valid numeric value will pull every analysis downward. The only reliable way to catch these issues is to look at the data carefully before modelling.

> **Methodological note.** Before applying any statistical or machine learning technique to a new dataset, I strongly recommend building a simple exploratory dashboard that allows interactive inspection of distributions, missingness patterns, correlations, and individual records. The time spent is repaid many times over in avoided errors downstream. For someone without a clinical background, this exploration phase is also an education: the

> dataset teaches you what the features mean, how they are used in practice, and which of your assumptions need revision.

THE EXPLORATION DASHBOARD

I spent approximately three days building a Dash-based dashboard for MIMIC-IV-ED exploration, separate from any modelling code. The dashboard allowed filtering by ESI level, age group, arrival mode, and acuity, and displayed distributions of every feature in the filtered subset. Individual patients could be clicked to see their complete record. Correlations between features and the target could be visualised interactively.

The results of this exploration were revealing. Missing vital signs were not random, they correlated strongly with acuity level, with critically ill patients more likely to have missing values because their condition made measurement difficult or skipped. Temperature was missing in a large fraction of ESI-1 cases, where the priority is immediate stabilisation rather than temperature measurement. Pain scores were missing in unconscious patients almost by definition. These missingness patterns are themselves signal, and I incorporated them as explicit missingness-flag features in the final model (see Chapter 4).

The chief complaint text, in the exploration dashboard, looked completely different from the Kaggle synthetic version. There were no systematic severity adjectives. Complaints like "63M `c/o` chest pain onset 2h" or "altered MS, found down by family, unclear duration" were common. The vocabulary was idiosyncratic, with abbreviations that varied between nurses and shifts. This was unmistakably real clinical text.

3.5 THE COHORT

MIMIC-IV-ED v3.1 contains 425,087 ED visits linked to 216,877 unique patients. After excluding visits with missing ESI labels (roughly 1.6%), the analytic cohort consists of 418,100 ED visits. This is the cohort used throughout the remainder of this report.

I did not restrict by age. MIMIC-IV-ED includes both adult and pediatric patients, and I kept all of them. In MIMIC, patients under 18 are recorded with their true age, while adult ages at admission are anchored to a reference year and shifted in blocks to de-identify birth dates. The `anchor_age` field gives age at the anchor year, and ages at later visits must be computed from the anchor offset. This de-identification scheme preserves longitudinal structure within a patient but makes some naive age computations subtly wrong. I used the `anchor_age` at the visit year throughout.

Multiple visits per patient are common. The median patient has one ED visit in the dataset, but some patients have dozens. This raised a design question: should features that aggregate over prior visits (number of prior ED visits, prior diagnoses, prior severity codes) be computed per visit, with careful attention to temporal ordering to avoid leakage?

I chose the leakage-safe path. All history features are computed using group-wise `cumcount` and `cummax + shift(1)` operations, which ensure that each visit's feature values depend only on strictly prior visits for the same patient. If a patient's third ED visit happens on 15 June 2180, the features describing their history at that visit include data from their first and second visits but nothing from the third visit itself or any later visit. This is the correct behaviour for a prediction system: at the moment of triage, the system has access only to information that existed before that moment.

What Is Temporal Leakage?

Temporal leakage occurs when features computed for a given visit accidentally include information from that visit or from later visits. A naive implementation of "number of prior ED visits" might count all visits for this patient in the dataset, including the current one and future ones. At prediction time this number would be unavailable. A model trained on leaky history features will score well in cross-validation and fail in deployment.

The `cumcount` operation, computed within each patient group sorted by visit date, avoids this by counting only strictly prior visits. The `cummax + shift(1)` pattern does the same for maximum-over-history features. These patterns are straightforward to implement correctly, but they are easy to get wrong, and the consequences are silent, the model does not crash, it just overfits.

3.6 THE CLASS DISTRIBUTION SHIFT

The first substantive difference between MIMIC-IV-ED and the Kaggle synthetic data became clear immediately: the class distributions are radically different. Figure 3.1 repeats the comparison from Chapter 2, now with the interpretation made explicit.

In the Kaggle data, the ESI distribution is roughly bell-shaped: ESI-3 dominates at 36%, flanked by ESI-4 at 29% and ESI-2 at 17%, with the extremes at 4% (ESI-1) and 14% (ESI-5). This is not implausible as a summary of an idealised ED population.

In the real MIMIC-IV-ED data, ESI-3 dominates even more strongly at 54%, ESI-2 is at 33%, and the tails collapse: ESI-1 is rare at 6%, ESI-4 at 7%, and ESI-5 is nearly absent at 0.3%. The ESI-5 category is so sparse that it contains only about 1,100 patients across the entire 418,100-visit cohort.

This is clinically plausible: BIDMC is a large urban academic medical centre with a trauma designation. Patients who would be ESI-5 (truly non-urgent, no resources needed) typically do not come to this kind of ED. They go to urgent care, primary care, or ambulatory clinics. The MIMIC-IV-ED distribution reflects the case mix of a tertiary academic hospital, not of an average ED across all settings.

The implication for modelling is substantial. Any approach that works well on the Kaggle distribution may perform differently on MIMIC, particularly for the ESI-5 class where a tiny sample size caps what any classifier can achieve. This became an important caveat throughout the project. My best MIMIC F1 for ESI-5 is 0.260, which sounds poor in isolation but is constrained by the 0.3% prevalence, with roughly 1,100 positives across five folds, the sampling noise alone is substantial.

3.7 THE DATA MODEL

MIMIC-IV-ED is distributed as a relational database, typically provided as CSV files that correspond to tables. Understanding the relationships between tables is essential before any feature engineering. I spent some time drawing the schema by hand before writing any join code. The main ED tables (`edstays`, `triage`, `medrecon`, `diagnosis`, `pyxis`) join via `stay_id`, link to patient-level history via `subject_id`, and link to parent MIMIC-IV hospital admissions (`patients`, `admissions`, `diagnoses_icd`, `drgcodes`) via `hadm_id`. `pyxis` (medication dispensing during the stay) is post-triage and therefore excluded to prevent leakage.

The key tables I used are:

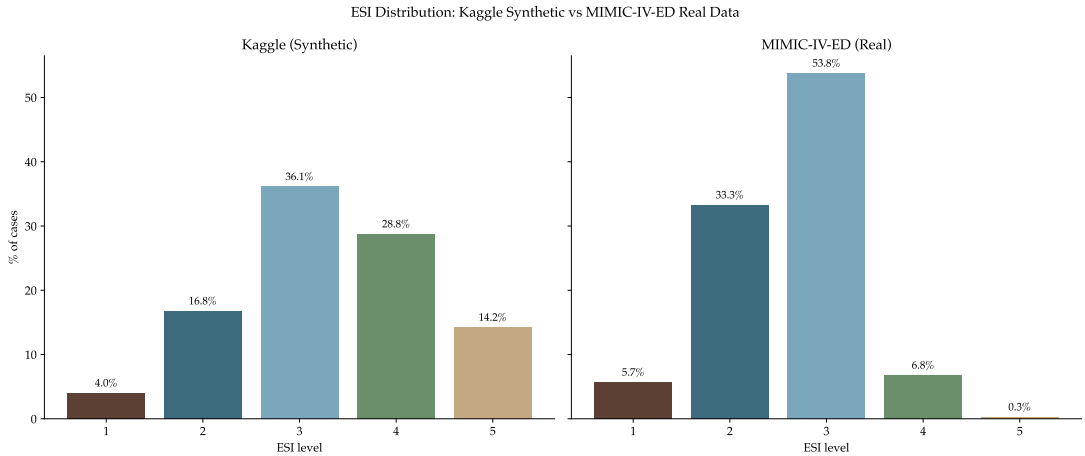

Figure 3.1 – ESI acuity level distributions for the Kaggle synthetic competition data (left) and MIMIC-IV-ED (right). The synthetic data is roughly bell-shaped around ESI-3. The real clinical data is strongly skewed: ESI-3 and ESI-2 together account for 87% of visits, while ESI-5 patients are nearly absent (0.3%). The difference reflects the case mix of a tertiary academic medical centre, and has substantial implications for model evaluation.

- `edstays`, one row per ED visit, with admission and discharge timestamps, arrival mode, and disposition.
- `triage`, one row per ED visit, with vital signs measured at triage, chief complaint text, and the ESI acuity label. This is the primary table for the prediction task.
- `medrecon`, medication reconciliation, one row per pre-admission medication per visit. Used to construct therapeutic-class flags.
- `diagnosis`, one row per ICD diagnosis code per visit. Used for the Charlson comorbidity composite, computed from strictly prior visits only.
- `patients` (from MIMIC-IV), one row per patient, with demographic information and `anchor_age`.
- `admissions` (from MIMIC-IV), one row per hospital admission, used to link ED visits to subsequent hospital stays and to compute prior-admission history features.
- `diagnoses_icd` (from MIMIC-IV), historical diagnoses across all encounters, used for Charlson comorbidity flags.

I explicitly did not use `pyxis` (medication dispensing records from during the hospital stay), `disposition` (how the ED visit concluded), or any other post-triage feature. Including these would have produced better accuracy numbers but would have been leakage: at the time of triage, none of this information is available. The entire feature engineering pipeline, described in Chapter 4, is built on this principle.

3.8 WHAT COMES NEXT

Having the data, understanding the schema, and settling the cohort was, in retrospect, half the work. The other half, feature engineering, modelling, evaluation, safety analysis, fairness audits, explainability, is the subject of the remaining chapters. But all of it is downstream of the decisions made in this chapter. The cohort is 418,100 visits. The target is the 5-level ESI label.

The class distribution is dominated by ESI-3 with a nearly-empty ESI-5 category. All history features are leakage-safe. All post-triage features are excluded. These ground rules are carried forward without further discussion.

I turn now to the features themselves: the 65 variables that, together with the chief complaint text, form the input to the model.

CHAPTER 4

FEATURE ENGINEERING AND CLINICAL SCORES

The heart of any applied machine learning project is not the model. It is the features. A carefully chosen feature can make a weak model strong; a poorly chosen one can make the best architecture useless. In medical machine learning, the feature set also serves a second purpose: it is the primary vocabulary through which the model makes contact with clinical practice. If my features align with how nurses and physicians actually think about acuity, the model has a chance of learning something generalisable. If my features are arbitrary projections of arbitrary columns, it does not.

This chapter describes the 65 tabular features used in my best pipeline, along with the 50 text embedding components derived from the chief complaint (I treat the text separately in Chapter 5). Together they form a 115-dimensional input vector per patient. I explain how each category of features was constructed, why it was chosen, and in several cases, which alternatives I tried and abandoned.

4.1 THE GUIDING PRINCIPLES

Before I list the features, it is worth stating the three principles that guided their construction.

No leakage. Every feature must be computable at the moment of triage, using only information available at that moment. Medication records from during the hospital stay cannot be used, because they do not exist at triage time. The number of prior ED visits must count strictly earlier visits, not the current one. The chief complaint text is available; the diagnosis written three hours later by an ED physician is not. This is the principle I discussed in Chapter 3. I return to it repeatedly below because it is easy to violate without noticing.

Clinical grounding. Wherever possible, I reach for features with established clinical meaning. A composite score like NEWS2 (Royal College of Physicians, 2017), used in practice by hospitals across the United Kingdom and Ireland, brings a readable clinical interpretation that a generic polynomial feature cannot. When I use derived features like shock index or mean arterial pressure, these are derived quantities that appear in textbooks, not products of automated feature search.

Honest ablation. Features were added when they helped, and removed when they did not. Several features I initially expected to be useful, the Rapid Emergency Medicine Score, individual vital-sign abnormality flags, age-decile vital z-scores, ended up contributing nothing or next to nothing, and were dropped. I document these dead ends alongside the successes.

Why Only 65 Features?

One might expect that more features would generally be better. They are not. Gradient-boosted decision trees, my chosen family of models, handle irrelevant features gracefully but not costlessly. Each irrelevant feature increases the search space during training, increases the risk of overfitting on small folds, and dilutes the importance scores I rely on for explainability. I settled on 65 tabular features through iterative pruning: features with near-zero importance across folds were removed, and the final set is the smallest one that does not sacrifice performance. This is the opposite of the "feed every column into XGBoost" style that sometimes wins Kaggle leaderboards. It is also the style that more closely resembles how a clinician would reason.

4.2 RAW VITAL SIGNS

The six vital signs measured at triage form the backbone of my feature set. These are the same vital signs a nurse would measure at any ED in the world, and they are the first features any sensible clinical prediction system would use.

- `temperature`, body temperature in degrees Celsius.
- `heartrate`, heart rate in beats per minute.
- `resprate`, respiratory rate in breaths per minute.
- `o2sat`, peripheral oxygen saturation (SpO_2), a percentage.
- `sbp`, systolic blood pressure in mmHg.
- `dbp`, diastolic blood pressure in mmHg.

Each of these is a single numeric value per visit, extracted directly from the `triage` table with minimal cleaning. Ranges are biologically constrained, a human heart rate below 30 or above 240 is almost certainly a measurement error, but I did not apply aggressive clipping. LightGBM handles outliers gracefully through its tree-based splits, and overly strict clipping can remove genuine physiological extremes (a patient in cardiac arrest may have a heart rate well outside the "normal" range).

The diagnostic value of raw vitals is considerable. A systolic blood pressure below 90 mmHg or an oxygen saturation below 92% immediately raises the acuity level in every standard triage system (Royal College of Physicians, 2017). But raw vitals are also noisy, a single measurement at triage captures one moment of a fluctuating physiological state, and they are not sufficient on their own. They become most informative when combined.

4.3 ENGINEERED VITAL SIGNS

From the six raw vitals, I derive eight further numeric features. These derivations are all standard in clinical practice and serve to make latent relationships explicit to the model.

SHOCK INDEX

The shock index is the ratio of heart rate to systolic blood pressure:

$$\text{shock_index} = \frac{\text{heart rate}}{\text{systolic BP}}$$

A shock index above 0.9 is associated with haemodynamic instability even when heart rate and blood pressure appear individually normal. This is a classic example of a feature that encodes clinical reasoning the model could not easily learn from the raw inputs alone: a patient with heart rate 110 and SBP 110 has a shock index of 1.0, which suggests compensated shock, but neither raw value in isolation would trigger an alarm. The shock index makes this explicit.

PULSE PRESSURE AND MEAN ARTERIAL PRESSURE

From systolic and diastolic blood pressure, I derive two further quantities:

$$\text{pulse_pressure} = \text{SBP} - \text{DBP}$$
$$\text{MAP} = \frac{\text{SBP} + 2 \times \text{DBP}}{3}$$

Pulse pressure reflects the compliance of the aorta and large arteries; mean arterial pressure is a weighted average representing the average pressure perfusing the organs across the cardiac cycle. Both are standard in cardiovascular assessment. The weighting in MAP reflects the fact that the heart spends more of its cycle in diastole than in systole.

PAIN SCORE

The `pain` column in MIMIC-IV-ED is free text rather than a clean numeric field. A triage nurse may enter `7`, `7/10`, `moderate`, `none`, `unable to assess`, or any of a dozen other strings. I parsed this to a numeric feature `pain_numeric` using a regex that extracts the first number and clips to $[0, 10]$. Non-numeric entries became missing values. The preprocessing looks like this:

```
def parse_pain(val):
 if pd.isna(val):
 return np.nan
 m = re.search(r"\b(\d+(?:\.\d+)?)\b", str(val).strip().lower())
 if m:
 v = float(m.group(1))
 return v if v <= 10 else np.nan
 return np.nan

df["pain_numeric"] = df["pain"].apply(parse_pain)
```

Listing 4.1 – Pain score parsing. The regex handles formats like "7", "7/10", and "pain 8". Values outside the 0–10 scale, including text-only entries, become missing.

CLINICAL COMPOSITE SCORES: NEWS2 AND QSOFA

The remaining four engineered vitals are composite scores. These are the most clinically meaningful features I add, and they deserve detailed explanation.

NEWS2. The National Early Warning Score 2, published by the Royal College of Physicians in 2017 (Royal College of Physicians, 2017), is a bedside risk-stratification tool used across the UK NHS and in many hospitals internationally. It aggregates seven physiological parameters into a single integer score from 0 to 20. Higher scores indicate greater risk of clinical deterioration, and specific thresholds trigger escalation protocols: a score of 5 or more prompts urgent clinical review, and a score of 7 or more prompts emergency review.

My implementation uses five of the seven NEWS2 components, listed in Table 4.1. The two components I omit are supplemental oxygen (a binary indicator of whether the patient is on O_2 support) and level of consciousness (the AVPU scale: Alert / Voice / Pain / Unresponsive). Neither is recorded in the MIMIC-IV-ED triage table. This is a limitation of the data source, not of my methodology; if these fields were available, incorporating them would be mechanical.

Table 4.1 – The partial NEWS2 scoring table used in this work. Five of the seven standard components are included; supplemental oxygen and AVPU consciousness level are unavailable in MIMIC-IV-ED triage data. My maximum score is $5 \times 3 = 15$, versus 15 (base) +2 (O_2) +3 (AVPU) = 20 for full NEWS2. The SpO_2 thresholds follow Scale 1 (for patients not on supplemental O_2).

Parameter	Score contribution				
	3	2	1	0	1
Respiratory rate (per min)	≤ 8	,	9–11	12–20	21–24*
SpO_2 (%)	≤ 91	92–93	94–95	≥ 96	,
Systolic BP (mmHg)	≤ 90	91–100	101–110	111–219	≥ 220**
Heart rate (per min)	≤ 40	,	41–50	51–90	91–110
Temperature (°C)	≤ 35.0	,	35.1–36.0, 38.1–39.0	36.1–38.0	,

*Respiratory rate ≥ 25 scores 3. **SBP ≥ 220 scores 3. Temperature ≥ 39.1 scores 2. Table condensed for readability.

The score is computed as the integer sum of the five component contributions, with NaN propagation for any missing vital. Downstream, missing composite scores are imputed using the median within each training fold.

qSOFA. The quick Sequential Organ Failure Assessment score (Singer et al., 2016) was introduced in 2016 as a bedside tool for identifying patients at risk of sepsis-related deterioration. It has three components, each scoring 0 or 1:

1. Respiratory rate ≥ 22 per minute.
2. Systolic blood pressure ≤ 100 mmHg.
3. Altered mental status (GCS < 15, or clinically equivalent).

My implementation uses two of the three components. The altered mental status criterion requires a Glasgow Coma Scale assessment, which is not recorded in MIMIC-IV-ED's triage table. My partial qSOFA therefore ranges from 0 to 2 rather than the standard 0 to 3. A qSOFA score of 2 or more (in the full three-component version) is associated with a substantially increased risk of in-hospital mortality in suspected sepsis (Seymour et al., 2016).

Age-Adjusted Shock Index. The shock index above does not account for age. Elderly patients often have baseline cardiovascular compensation that keeps their heart rate low and blood pressure normal even in early shock, a phenomenon sometimes called compensated shock in the elderly, distinct from the compensated shock in young patients I discuss in Chapter 9. I compute an age-adjusted shock index:

$$\text{age_shock_index} = \text{shock_index} \times \text{age}$$

The product scales shock index by age directly, amplifying the signal for elderly patients whose compensated physiology may mask deterioration. The feature was important in my model: it appeared in the top 10 by LightGBM gain importance in several experimental runs.

Age-Stratified Vital Abnormality Count. My fourth composite, `n_abnormal_vitals`, counts the number of vital signs that fall outside age-adjusted normal ranges. The thresholds are listed in Table 4.2, with separate bounds for elderly (age ≥ 70) and non-elderly patients. Nine individual binary flags are summed, producing a score from 0 to 9.

Table 4.2 – Thresholds used to define abnormal vital signs, stratified by age. The count of how many thresholds are crossed forms the `n_abnormal_vitals` composite feature.

Threshold	Non-elderly (< 70)	Elderly (≥ 70)
Heart rate high	> 100	> 90
Heart rate low	< 50	< 50
Systolic BP low	< 90	< 100
Systolic BP high	≥ 180	≥ 180
SpO_2 low	$< 94\%$	$< 92\%$
Temperature high	$> 38.5°C$	$> 38.0°C$
Temperature low	$< 35.5°C$	$< 35.5°C$
Respiratory rate high	> 22	> 20
Respiratory rate low	< 10	< 10

Why Stratify by Age at All?

The rationale is that physiological "normal" ranges are themselves age-dependent. An 80-year-old patient with a heart rate of 95 is likely showing early stress; the same heart rate in a 25-year-old is within normal range and not alarming. Encoding this as binary flags lets the model learn age-dependent thresholds without having to discover them from the raw data. I chose 70 as the stratification boundary as a compromise between the 65-year threshold often used in clinical guidelines and the 75-year threshold used in some geriatric medicine literature. The exact boundary was not heavily optimised, a reader interested in precision medicine could reasonably use finer age decile z-scores, which I tried in a separate experiment (Chapter 9) with mixed results.

4.4 MISSINGNESS AS SIGNAL

Six additional features record whether each vital sign was missing at triage: `temperature_missing`, `heartrate_missing`, and so on.

In a typical tabular ML problem, missingness is nuisance: you impute the missing values and move on. In clinical data, missingness is information. A vital sign is missing because a nurse did not take it, which is not a random event. In practice, missing vitals in the triage table correlate systematically with acuity: critically ill patients arriving by ambulance may be moved directly to a resuscitation bay without a complete triage workup, while walk-in patients with minor complaints routinely have every vital measured. Temperature in particular is frequently missing in ESI-1 cases where immediate stabilisation takes priority.

Including explicit missingness flags lets the model learn this signal without conflating it with

imputed values. After flag extraction, actual missing numeric values are filled with the training-fold median in each cross-validation split. The median is a reasonable choice for continuous physiological variables with heavy tails and is less sensitive to outliers than the mean.

4.5 DEMOGRAPHICS AND ARRIVAL MODE

Seven features describe the patient and how they arrived.

Four are categorical: `gender`, `race`, `age_group` (categorical age bucket), `insurance`, `language`, and `marital_status`. Gradient-boosted tree models handle these natively through LightGBM's categorical feature support, which uses a specialised split-finding algorithm that partitions categorical values rather than one-hot encoding them. I included these features despite some misgivings, insurance and language in particular raised fairness concerns that became important in Chapter 10 when SHAP analysis revealed them as top-ranked features, but chose to retain them and audit their effects downstream.

Age itself is recorded numerically as `age`, computed from MIMIC's de-identified `anchor_age` field plus the offset from the anchor year. The computation is:

```
df["age"] = (df["anchor_age"]
 + (df["intime"].dt.year - df["anchor_year"])).clip(0, 120)
```

The cyclic encoding of arrival hour deserves explanation. I do not use the raw hour of day (0 to 23) because that encoding misleadingly places 23:00 and 00:00 far apart in feature space when clinically they are adjacent. Instead, I decompose the hour into two features:

$$\text{arrival_hour_sin} = \sin\left(\frac{2\pi h}{24}\right)$$
$$\text{arrival_hour_cos} = \cos\left(\frac{2\pi h}{24}\right)$$

where h is the hour. This preserves cyclic structure: midnight and 1 AM are close in both components, as are 11 PM and midnight. The model can learn that arrivals in the early hours of the morning have different characteristics from afternoon arrivals, a real effect in ED data, where overnight arrivals tend to be higher acuity because patients with minor complaints are less likely to seek care at 3 AM.

The `arrival_transport` categorical feature records how the patient arrived: ambulance, helicopter, walk-in, or transfer from another facility. Ambulance arrival in particular is a strong predictor of higher acuity and is consistently among the top 10 features in my LightGBM importance rankings.

4.6 MEDICATIONS AND COMORBIDITIES

Twenty-nine features encode the patient's medical history. These split into two groups: current medications (from the `medrecon` table) and prior diagnoses (from the `diagnoses_icd` table).

PRE-ADMISSION MEDICATIONS

The `medrecon` table is, by MIMIC design, a record of the patient's home medications, the drugs they were taking before coming to the ED. It is populated during the medication reconciliation process that happens at triage or shortly after, where a nurse asks the patient (or reviews

their pharmacy record) about current medications. This is distinct from `pyxis`, which records medications dispensed during the hospital stay and which I do not use because it would represent leakage.

Every medication in `medrecon` carries an Enhanced Therapeutic Classification (ETC) description, a drug ontology that groups medications into therapeutic classes. I use these descriptions to assign each medication to one of 13 clinically meaningful classes through case-insensitive substring matching. Table 4.3 lists the classes.

Table 4.3 – The 13 therapeutic medication classes extracted from the `medrecon` table, with the ETC description keywords used for matching. Each class becomes a binary patient-level flag indicating whether the patient was on any medication in that class at the time of ED presentation. A 14th feature, `num_medications`, counts the total number of distinct pre-admission medications.

Class flag	Keywords matched
`med_hypertension`	beta blocker, ACE inhibitor, angiotensin, calcium channel blocker
`med_diabetes`	insulin, antidiabetic, sulfonylurea, GLP-1, DPP-4
`med_heart_failure`	loop diuretic, cardiac glycoside
`med_copd_asthma`	beta-2 adrenergic, COPD therapy, asthma, bronchodilator
`med_depression`	SSRI, antidepressant, SNRI
`med_anxiety`	benzodiazepine, antianxiety
`med_epilepsy`	anticonvulsant
`med_thyroid`	thyroid hormone
`med_anticoagulant`	anticoagulant, platelet aggregation inhibitor
`med_hyperlipid`	HMG-CoA, antihyperlipidemic
`med_opioid`	analgesic opioid
`med_antipsychotic`	antipsychotic
`med_renal`	phosphate binder

The substring matching is admittedly crude. A more sophisticated approach would use RxNorm, ATC codes, or another standardised pharmacological ontology. I chose the simpler approach because the ETC descriptions are consistent enough within MIMIC for regex matching to work reliably, and because the alternative would have required maintaining an external mapping file with hundreds of entries. This is an area where a careful follow-up project could improve the feature quality.

PRIOR DIAGNOSES: THE CHARLSON COMORBIDITY INDEX

The Charlson Comorbidity Index (Charlson et al., 1987) is a weighted sum of comorbid conditions that has been in clinical use since 1987. It was originally developed to predict one-year mortality in hospitalised patients and has since been validated across many contexts. The index assigns weights to 17 groups of conditions, with higher weights reflecting greater mortality risk contribution.

I implement 16 of the 17 condition groups as binary flags computed from *prior* admissions only, diagnoses recorded at earlier hospital stays, not at the current ED visit. Table 4.4 lists the conditions and their weights. The composite score `charlson_score` is the weighted sum. The weights I use are the original 1987 Charlson weights, not the Deyo 1992 or Quan 2005 adaptations. I also do not split diabetes into "with" and "without" complications (the original index distinguishes these, with weight 1 for uncomplicated and 2 for complicated). These are minor simplifications that could be refined in future work.

Table 4.4 – The 16 conditions included in my Charlson Comorbidity Index composite, with the original Charlson 1987 weights. ICD-9 and ICD-10 codes are mapped via prefix matching. The composite `charlson_score` is the weighted sum of binary flags for conditions present in prior admissions.

Conditions
Myocardial infarction, CHF, PVD, stroke/TIA, dementia, COPD, rheumatic disease, peptic ulcer, mild liver disease, di
Hemiplegia, renal disease, cancer
Severe liver disease
Metastatic cancer, AIDS

The leakage-safety of the Charlson score is important and subtle. A naive implementation would look at all diagnoses ever recorded for a patient, including those from the current ED visit. But diagnoses recorded during the current visit are not available at triage time, they are generated later, after evaluation and workup. Using them would produce an artificially accurate model that fails in deployment. My implementation uses the `cumsum` minus current-row pattern:

```
cumsum_df = adm_cc.groupby("subject_id")[cc_cols].cumsum()
prior_df = (cumsum_df - adm_cc[cc_cols].values).clip(lower=0, upper=1)
adm_cc["charlson_score"] = sum(
 adm_cc[f"prior_{c}"] * w for c, w in CHARLSON_W.items()
)
```

Listing 4.2 – Leakage-safe Charlson computation. The cumulative sum is taken across admissions in chronological order, and the current admission's contribution is subtracted. The `clip` ensures binary values even when a condition appears in multiple prior admissions.

PRIOR HOSPITAL UTILISATION

Beyond Charlson, I include four further features from the patient's hospital history, all computed with strict temporal ordering:

- `prior_ed_visits`, count of ED visits for this patient before the current one, via `groupby.cumcount()`.
- `prior_admissions`, count of hospital admissions before the current ED visit.
- `prior_drg_severity`, maximum Diagnosis-Related Group severity code across prior admissions, via `cummax().shift(1)`.
- `prior_drg_mortality`, maximum DRG mortality code across prior admissions.

The DRG (Diagnosis-Related Group) system is a hospital billing classification that assigns each admission a severity and mortality code on a 1–4 scale. A patient whose previous admissions reached DRG severity 4 has been seriously ill before. This is prior-probability information: not every severely ill patient will be severely ill again, but the prior history shifts the probability.

Temporal Leakage: A Common Trap

The patterns I use to ensure these features are leakage-safe, `cumcount()`, `cummax().shift(1)`, and `cumsum()` minus the current row, are idiomatic pandas. They are not the only safe patterns, and they are not automatically safe. The critical point is that for each patient visit, the feature values must depend only on rows strictly preceding

that visit in chronological order. Testing this is straightforward: sort by patient and time, take the last visit for each patient, manually compute what the feature should be, and compare. I did this during development to verify correctness. A common failure mode is omitting the `shift(1)` after `cummax()`, which causes the current admission's value to leak into the prior-history feature. The pattern looks correct at first glance and does not raise errors, the only way to catch it is to test against hand-computed examples.

4.7 TEXT EMBEDDINGS

The final 50 features are components of a Principal Component Analysis of chief complaint text embeddings, produced by a fine-tuned Bio_ClinicalBERT model (Alsentzer et al., 2019). I defer the detailed discussion of embeddings to Chapter 5, which covers the choice of base model, the fine-tuning procedure, and the benchmark against alternative embedding approaches. For the purpose of this chapter, what matters is:

- Each chief complaint string is mapped to a 768-dimensional vector by ClinicalBERT.
- Within each cross-validation fold, I fit a PCA on the training fold's embeddings and project both training and validation embeddings into 50 components.
- These 50 components are concatenated with the 65 tabular features to produce the final 115-dimensional input.

The PCA is fit per fold rather than globally, an important detail. Fitting PCA on the full dataset (training plus validation) before splitting would be a subtle leakage: the principal directions would have been informed by the validation data. Tree-based models are somewhat robust to this form of leakage because the leakage is diffuse rather than concentrated, but it is still best practice to fit feature transformations within fold. Verifying that the per-fold PCA was correctly implemented was one of the first things I checked in the final pipeline.

4.8 DEAD ENDS

Several features I tried did not help and were dropped. I document them here because the dead ends are part of the story.

REMS, Rapid Emergency Medicine Score. The Rapid Emergency Medicine Score (T. Olsson, Terent, and Lind, 2004) combines age, MAP, heart rate, respiratory rate, oxygen saturation, and Glasgow Coma Scale into a prognostic score originally validated for in-hospital mortality in non-surgical ED patients. It is one of the better-known ED scoring systems. I implemented it (minus GCS, again unavailable), added it to the feature set, and found it ranked last among all features by LightGBM gain importance in the augmented 77-feature configuration. The reason is simple: every component of REMS is already in my feature set individually, and the gradient-boosted trees can construct any monotonic combination of them on their own. A hand-crafted composite score adds no new signal when its parts are already present. I kept NEWS2 and qSOFA because they include non-linear threshold effects that trees might not discover efficiently, but REMS, which is essentially a linear weighted sum, was redundant.

Individual vital-sign abnormality flags. Before settling on the single summed `n_abnormal_vitals` count, I tried including the nine individual threshold-crossing flags (HR high, HR low, SBP high, etc.) as separate features. Each had near-zero LightGBM importance. The trees simply did not use them, because the raw vital values they are derived from are already in the feature set and the trees discovered the thresholds implicitly. Only the sum added non-redundant signal, presumably because it captures a polysymptom pattern, being abnormal on multiple vitals simultaneously, that the model cannot easily construct from the raw inputs.

Age-decile vital z-scores. In a separate experiment, I computed within-age-decile z-scores for each vital: how many standard deviations a patient's heart rate is from the mean heart rate in their age decile, for example. Seven features of this form (`hr_age_zscore`, `sbp_age_zscore`, etc.) plus a new count `n_age_abnormal_vitals` produced a tiny F1 improvement of $+0.0007$, well within noise. Two of the features did rank in the top 20 by importance, but their inclusion did not translate to better overall performance, suggesting they captured redundant information with the age-adjusted shock index and the existing abnormality count. I discuss these features further in Chapter 9 where they become relevant to the age-disparity analysis.

PCA dimensionality. I tested PCA dimensions of 50, 100, and 150 components on the ClinicalBERT embeddings. The F1 score was stable at 0.541 across all three. I retained 50 on efficiency grounds, doubling the dimensionality adds modelling cost without benefit.

Dual embeddings. I tried concatenating embeddings from two different models (all-MiniLM-L6-v2 and fine-tuned Bio_ClinicalBERT) on the hypothesis that they might capture complementary semantic content. They did not. The F1 change was $+0.0002$, statistically indistinguishable from zero. I kept only the fine-tuned ClinicalBERT embeddings.

4.9 WHAT THIS GIVES ME

The final feature set, summarised in Table 4.5, is 115 features per patient: 65 tabular plus 50 text-derived. Each feature has a clinical interpretation, each is leakage-safe, and each earned its place through ablation. This is the input that goes into the model blend described in Chapter 6.

Before I use this input, I need to discuss one part in more detail: the text embeddings themselves. Constructing them involved fine-tuning a clinical language model on MIMIC-IV-ED chief complaints, benchmarking against alternative approaches, and navigating several dead ends of my own. That is the subject of Chapter 5.

Table 4.5 – Summary of the 115 features in the final blend-v2 pipeline. Counts: 6 raw vitals, 4 base engineered vitals, 4 clinical composites, 6 missingness flags, 2 cyclic arrival features, 7 demographics and categoricals, 14 medication features (13 class flags + count), 6 prior-visit numeric features, 16 individual Charlson flags, and 50 text embedding components.

Category	Count	Examples
Raw vital signs	6	temperature, heartrate, sbp
Base engineered vitals	4	shock_index, MAP, pulse_pressure, pain_numeric
Clinical composite scores	4	NEWS2, qSOFA, age_shock_index, n_abnormal_vitals
Missingness flags	6	temperature_missing, sbp_missing
Cyclic arrival encoding	2	arrival_hour_sin, arrival_hour_cos
Demographics (categorical)	7	gender, race, insurance, language
Prior-visit numeric features	6	age, prior_ed_visits, charlson_score
Medication classes	14	med_anticoagulant, num_medications
Individual Charlson flags	16	prior_cc_mi, prior_cc_diabetes
Text embedding components	50	emb_pca_0 through emb_pca_49
Total	**115**	

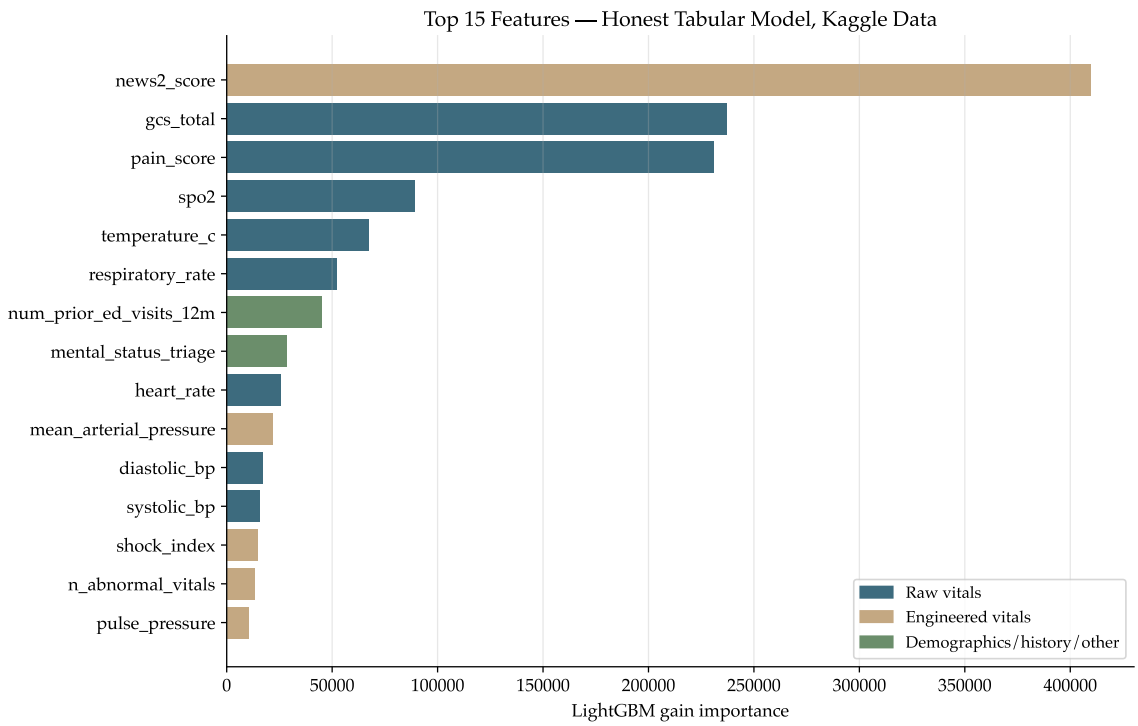

Figure 4.1 – Gain-importance rankings from the LightGBM baseline on the Kaggle synthetic competition data, trained without text features. The composite clinical scores NEWS2 and GCS dominate, with raw vitals ranked next. This is the "honest tabular" model referenced in Chapter 2, the one that reached F1 = 0.87 before text features inflated the score to 0.998. On MIMIC-IV-ED, rankings shift: embedding components become the top features (see Chapter 5), but the clinical composite scores remain consistently important. The figure is reproduced here to illustrate the structural role of the engineered features, not the specific numeric values which differ between datasets.

CHAPTER 5

TEXT, EMBEDDINGS, AND CLINICALBERT

In Chapter 4 I described the sixty-five tabular features that form the structured input to my model. This chapter is about the other half of the input: the fifty features derived from the chief complaint text. These are produced by a pipeline that takes a short string written by a triage nurse, passes it through a transformer-based language model, reduces the output with Principal Component Analysis, and hands the resulting vector to the downstream classifier.

The pipeline is not complicated, but several of the decisions behind it deserve explanation. Why a transformer rather than simpler representations? Which transformer, given that dozens of them exist? Should it be fine-tuned or used as-is? How should its 768-dimensional output be reduced? Each of these questions has a defensible answer, and for each I have some experimental evidence supporting my choice, including a few results that surprised me.

5.1 WHY TEXT AT ALL

The chief complaint is the single most informative free-text field available at triage. It is what the nurse writes down after asking the patient the simple question: "What brings you in today?" A string like "substernal chest pain radiating to left arm, onset 2h" carries very different clinical signal than "ear pain × 3 days." Both are short, often under ten words, and neither lends itself to conventional tabular encoding. No single column can capture the range of meanings a chief complaint can express.

What Is a Transformer Embedding, Intuitively?

A transformer embedding is a vector of numbers, in my case 768 of them, that represents the "meaning" of a piece of text. The numbers themselves are not individually interpretable: there is no single dimension that means "severity" or "cardiac." But in combination, they place the text in a geometric space where semantically similar texts end up near each other. "Chest pain" and "pain in the chest" produce nearly identical vectors; "chest pain" and "ankle sprain" produce very different ones.

The model that produces these embeddings, usually a BERT-family neural network, was originally trained to predict masked words in a large corpus of text. In the process, it learned representations that capture syntactic, semantic, and (for clinical models) medical relationships. I do not re-train it from scratch; I use the representations it already produces, possibly fine-tuning them further on my specific task.

My goal in using a transformer embedding is to let the downstream classifier exploit this semantic structure without having to rediscover it from raw characters. The alternative, bag-of-words or TF-IDF features, is simpler and cheaper, and I include one such baseline in the benchmark below. But it cannot capture that "MI" and "myocardial infarction" refer to the same

condition, or that "SOB" and "shortness of breath" are synonyms. A well-chosen transformer can.

5.2 THE BENCHMARK: NINE MODELS, TWO DATASETS

Before committing to any particular embedding model, I ran a systematic benchmark. The question was simple: on my actual data, which models produce embeddings that best capture the clinical signal relevant to triage acuity?

I tested nine models (Table 5.1), spanning three rough categories. First, two baselines to establish the floor of performance: a random 384-dimensional projection (essentially noise) and a TF-IDF bag-of-words vectorisation reduced to 50 dimensions. Second, five general-purpose sentence embedding models, including well-known models like `all-MiniLM-L6-v2` (Reimers and Gurevych, 2019), `all-mpnet-base-v2`, `BGE-base`, and two variants from the E5 family (Wang et al., 2022). Third, two clinical-domain transformers: Bio_ClinicalBERT (Alsentzer et al., 2019), pretrained on clinical notes from MIMIC-III, and PubMedBERT (Gu et al., 2022), pretrained on biomedical research articles.

Table 5.1 – The nine models tested in the embedding benchmark. Three categories: baselines (random projection, TF-IDF), general-purpose sentence transformers, and domain-specific clinical transformers. All were evaluated with frozen weights, before any fine-tuning.

Model	Architecture	Dim	Domain
Random	384d Gaussian random projection	384	,
TF-IDF + SVD	Bag-of-words → truncated SVD	50	,
MiniLM-L6	6-layer distilled BERT	384	General
E5-small	E5 contrastive sentence encoder	384	General
MPNet	Masked and permuted language modelling	768	General
BGE-base	Contrastive pretraining, English	768	General
E5-base	E5 contrastive, larger variant	768	General
Bio_ClinicalBERT	BERT-base, pretrained on MIMIC-III notes	768	Clinical
PubMedBERT	BERT-base, pretrained on PubMed abstracts	768	Biomedical

For each model, I extracted embeddings on both the Kaggle synthetic competition data (as a sanity check) and a five-percent sample of MIMIC-IV-ED (20,905 visits). I then evaluated the embeddings using intrinsic metrics, measures of how well the embedding space captures the label structure, computed without training any downstream classifier. The primary metrics were:

- **kNN label purity at** $k = 10$: for each visit, what fraction of its ten nearest neighbours in embedding space share its ESI level. A fully informative embedding would give purity close to one; a random embedding gives purity close to the class prior.
- **Linear probe F1**: the macro F1 achieved by a minimal logistic regression trained on frozen embeddings with 5-fold cross-validation. This is the simplest possible supervised classifier, used as a probe of how linearly separable the embeddings are.
- **Silhouette score, Calinski–Harabasz, Davies–Bouldin**: three clustering quality measures that quantify how well embeddings of the same class cluster together relative to embeddings of different classes.

The results, focusing on kNN purity and linear probe F1, are shown in Table 5.2.

Table 5.2 – Intrinsic evaluation results on five-percent samples of each dataset. kNN@10 is the fraction of a visit's ten nearest neighbours in embedding space sharing its ESI label. Linear probe F1 is macro F1 of a 5-fold logistic regression on frozen embeddings. Bold marks the best value in each column. The gap between clinical and general-purpose models on MIMIC is smaller than I expected, most models cluster within a few percentage points. On the synthetic Kaggle data, by contrast, clinical models win decisively, reflecting the severity-keyword text leakage described in Chapter 2.

	MIMIC-IV-ED		Kaggle synthetic	
Model	kNN@10	Probe F1	kNN@10	Probe F1
Random 384d	0.412	0.162	0.261	0.191
TF-IDF + SVD50	0.546	0.270	0.549	0.486
MiniLM-L6	0.559	0.345	**0.873**	0.750
E5-small	0.555	0.308	0.844	0.607
MPNet	0.562	0.351	0.873	0.740
BGE-base	**0.566**	0.330	0.867	0.687
E5-base	0.565	0.317	0.852	0.629
Bio_ClinicalBERT	0.558	0.394	0.776	**0.940**
PubMedBERT	0.558	**0.399**	0.649	0.883

5.3 WHAT THE BENCHMARK TOLD ME

The MIMIC results came as a partial surprise. I had expected the clinical-domain models, Bio_ClinicalBERT and PubMedBERT, pretrained on medical text, to dominate the general-purpose models by a clear margin. They did not. On kNN purity at $k = 10$, nearly all non-baseline models clustered within a two-percentage-point band (0.546 to 0.566), with BGE-base actually leading. On linear probe F1, the clinical models did show an advantage (0.394–0.399 versus 0.308–0.351 for the general-purpose models), but the gap was far smaller than the conventional wisdom around domain-specific pretraining would predict.

This is worth dwelling on. One common narrative in clinical NLP is that clinical-domain pretraining is transformative for clinical tasks: if you are working with medical text, you should always use a clinical model. My benchmark suggests a more measured claim: on frozen embeddings alone, clinical pretraining provides a modest advantage on linear probes, no advantage on kNN proximity, and is essentially tied with well-trained general-purpose sentence encoders. The structural information in a chief complaint like "chest pain" or "shortness of breath" is captured reasonably well by any competent transformer.

The Kaggle results, by contrast, are striking. On the synthetic competition data, Bio_ClinicalBERT's linear probe F1 reaches 0.940, far above any other model. This is not a sign that ClinicalBERT understands the clinical content better. It is a sign that the Kaggle data contains the severity-keyword text leakage I documented in Chapter 2, and ClinicalBERT's vocabulary includes the severity adjectives ("severe," "moderate," "mild") with enough structure that a linear classifier can exploit them. The gap between clinical and general models on Kaggle is a measure of text leakage, not of clinical signal. My own conclusion from this comparison was: frozen embeddings on MIMIC are roughly indistinguishable across models, and the text task as such has a ceiling around 0.40 macro F1. To do better, I would need to fine-tune.

A Lesson About Benchmarks

When a benchmark produces a surprising result, the instinct is to check whether the benchmark is wrong. I spent some time on this, rerunning on a twenty-percent sample, verifying the metric implementations, checking for subtle bugs in preprocessing. The numbers held up. My initial expectation that clinical pretraining would dominate was simply incorrect for this task. This is the kind of finding that benchmarks are for: they discipline expectations that otherwise slip in unexamined.

5.4 FINE-TUNING BIO_CLINICALBERT

Given that frozen embeddings plateau around 0.40 F1, fine-tuning was the obvious next step. In fine-tuning, the pretrained model is trained further on a specific task, in my case, predicting the ESI acuity level from the chief complaint. Unlike frozen-embedding use, fine-tuning updates the model's internal weights, allowing the representation itself to specialise for the downstream task.

I chose Bio_ClinicalBERT as the fine-tuning base, for three reasons. First, it tied PubMedBERT on the MIMIC benchmark while having lower parameter count and faster inference. Second, and more importantly, Bio_ClinicalBERT was pretrained on MIMIC-III clinical notes, from the same institution (Beth Israel Deaconess Medical Center) that generated MIMIC-IV-ED. This is an unusually clean institutional match. The language, abbreviations, and documentation conventions that ClinicalBERT learned from MIMIC-III are essentially the same ones that appear in MIMIC-IV-ED triage notes. Third, it is the established standard in clinical NLP: much of the clinical ML literature since 2019 has used it as a baseline, which makes my results comparable to the existing body of work.

Fine-Tuning versus Frozen Embeddings: What Is the Difference?

When I use a pretrained model with frozen embeddings, I take its output for each input and feed that into my own classifier, but the pretrained model itself is unchanged. Its internal weights remain as they were after its original pretraining. I am using its general linguistic representations as they exist.

When I fine-tune, I attach a task-specific head (in my case, a linear layer mapping 768 dimensions down to 5 ESI class logits) and train the whole thing end-to-end on my labelled data. The weights deep inside the model move. The representation of "chest pain" after fine-tuning is no longer just a generic embedding of those two words, it is an embedding shaped by having seen, many times, which ESI labels are assigned to various chest-pain complaints in my training data.

Fine-tuning is more powerful but more expensive and more prone to overfitting. It also means the final embeddings are specific to this task and this dataset, rather than general-purpose. This is usually what you want, but it is a trade-off worth making deliberately.

TRAINING CONFIGURATION

My fine-tuning configuration is listed in Table 5.3. The values are standard for BERT-family fine-tuning tasks, with no exotic choices. Batch size 16 was the largest that fit in the sixteen-gigabyte VRAM of the T4 GPU along with FP16 mixed-precision training; a larger batch would not have changed the dynamics materially. Three epochs were sufficient; the best validation

F1 was reached at epoch 2 in both my final configuration and in a separate lower-learning-rate sweep I discuss below. The linear warmup, standard for transformer fine-tuning, gradually increases the learning rate over the first ten percent of training to avoid early instability.

Table 5.3 – Fine-tuning hyperparameters for the final ClinicalBERT model. Values are standard for BERT-family fine-tuning. Best validation F1 was reached at epoch 2, consistent with the common finding that BERT models begin to overfit on small classification heads after two or three epochs of fine-tuning on a fixed dataset.

Hyperparameter	Value
Base model	`emilyalsentzer/Bio_ClinicalBERT`
Optimizer	AdamW, weight decay 0.01
Learning rate	2×10^{-5}
Learning rate schedule	Linear warmup (10% of steps) then linear decay
Batch size	16
Max sequence length	128 tokens
Epochs	3
Gradient clipping	Max norm 1.0
Dropout (classifier head)	0.1
Mixed precision	FP16 (via `torch.amp.GradScaler`)
Loss	Cross-entropy, unweighted
Train / val / test split	80 / 10 / 10 stratified
Random seed	42
Hardware	Standard_NC4as_T4_v3 (T4 GPU, 16GB)
Training time	108 minutes (418,100 rows, 3 epochs)

The classification head itself is minimal: a single linear layer mapping the 768-dimensional pooled representation to five logits, one per ESI class. I considered deeper heads (adding a hidden layer, or using a two-layer MLP) and did not pursue them. The general finding in BERT fine-tuning is that simple heads work best, the heavy lifting is done by the encoder's representations, and adding a complex head often hurts by introducing its own overfitting risk.

POOLING AND EMBEDDING EXTRACTION

After fine-tuning, I extract embeddings by running the fine-tuned encoder in inference mode on the full dataset. For each input, I take the encoder's final-layer hidden state, a 128-by-768 matrix, one 768-dimensional vector per token position, and compute a single 768-dimensional vector per visit by mean pooling over non-padding tokens. The classification head itself is discarded for this step; I am interested in the representation the encoder has learned, not in the classifier's five-dimensional output.

```
def forward(self, input_ids, attention_mask, return_embeddings=False):
 outputs = self.encoder(input_ids=input_ids,
 attention_mask=attention_mask)
 pooled = self.mean_pool(outputs.last_hidden_state, attention_mask)
 if return_embeddings:
 return pooled # 768d, classifier head bypassed
 dropped = self.dropout(pooled)
 logits = self.classifier(dropped)
 return logits
```

Listing 5.1 – The pooling and embedding extraction step. The classifier head is bypassed via `return_embeddings=True`; only the mean-pooled encoder output is returned. Padding tokens are excluded from the mean via the attention mask.

I used mean pooling over non-padding tokens rather than the CLS-token representation (the first token in the sequence, which BERT-family models are sometimes trained to produce as a sequence-level summary). In the literature on sentence-transformer models, mean pooling has been reported to outperform CLS-token extraction for semantic similarity tasks when the model has not been specifically trained with a sentence-pair objective (Reimers and Gurevych, 2019). Bio_ClinicalBERT was pretrained with masked language modelling rather than a sentence-level objective, so I expected mean pooling to be the safer default here. I did not run a head-to-head comparison of mean vs CLS pooling on this task; the choice was a sensible default rather than an experimentally-verified one.

The output is a single array of shape 418,100 $\times$ 768, containing one embedding per ED visit in my cohort. This array is saved to disk as a NumPy file and becomes the input to the next step: dimensionality reduction.

FROM 768 TO 50: PCA REDUCTION

Feeding 768-dimensional embeddings directly into a gradient-boosted decision tree is possible but inefficient. Tree models evaluate each feature independently at each split; high dimensionality inflates training time and memory use without proportional benefit when many dimensions encode redundant information. The standard solution is to reduce dimensionality via Principal Component Analysis (PCA).

What Does PCA Do to an Embedding?

Principal Component Analysis finds directions in the embedding space along which the data varies the most. The first principal component is the direction of maximum variance; the second is the direction of maximum remaining variance that is orthogonal to the first; and so on. Keeping the top 50 components (out of 768) preserves the directions in which the embeddings spread out the most, discarding directions where embeddings barely vary.

This is not lossless, I do lose some information. But for a classifier that benefits from most of its signal being in a low-dimensional subspace, PCA can speed up training substantially while barely affecting accuracy. In my case, the tradeoff is favourable: 50 PCA components capture enough of the embedding variance that the downstream F1 is indistinguishable from using 100 or 150 components.

I tested PCA dimensionalities of 50, 100, and 150. All three produced essentially identical downstream macro F1 of 0.541 on the tabular-plus-text pipeline. I kept 50 as the default: there is no point paying for more dimensions if they do not improve performance. This PCA fitting is critical in one respect: I fit it *within each cross-validation fold*, using only the training fold's embeddings, and then apply it to both training and validation embeddings for that fold. Fitting PCA globally, on the entire dataset before splitting, would introduce a subtle leak, since the principal directions would be informed by the validation data. I verified this was done correctly in the final pipeline; I had a variant earlier in development that did it incorrectly.

5.5 FINE-TUNING IMPROVED THE TASK, BUT ONLY MODESTLY

The fine-tuned ClinicalBERT embeddings were the ones I used in the final blend pipeline. But it is worth being honest about how much the fine-tuning actually helped.

On the text-alone task, predicting ESI acuity from the chief complaint with no other features, fine-tuning raised macro F1 from the frozen ceiling of around 0.40 to approximately 0.43. This is a $+0.03$ improvement, real but modest. It does not transform the task. The text remains, on its own, a noisy signal for triage acuity on real data. Real nurses' chief complaints are terse, abbreviated, and sometimes cryptic ("pt c/o abd pain NOS"), and no amount of fine-tuning turns them into gold-standard labels.

Where the fine-tuned embeddings matter is in combination with tabular features. In the blend pipeline (Chapter 6), the fine-tuned text embeddings push the macro F1 from the honest-tabular-only baseline of approximately 0.56 up to 0.597. That is a more substantial $+0.04$ contribution. The text adds genuine signal that tabular features alone do not capture, the meaning of the chief complaint itself, above and beyond what vital signs and comorbidities reveal.

I also ran a systematic learning-rate sweep to check whether I was leaving performance on the table. A second run with learning rate 1×10^{-5} and five epochs (instead of my default 2×10^{-5} and three epochs) improved test F1 by $+0.003$, from 0.424 to 0.427. The best validation epoch in both configurations was epoch 2. I concluded that fine-tuning performance on this task is near its ceiling, given the data: longer training does not help, and a sweep over learning rates in the range commonly reported for BERT fine-tuning (1-5 $\times 10^{-5}$) yields differences within noise. This is a genuine limitation of the data rather than the training procedure.

5.6 A NOTE ABOUT WHAT SHAP REVEALS

Before closing this chapter, there is one finding from the fine-tuned embedding pipeline that I want to flag but defer to Chapter 10 for full treatment.

When I run TreeSHAP analysis on the final blend model, the top features by importance are, almost without exception, embedding components. The first PCA component of the fine-tuned ClinicalBERT embedding (`emb_pca_0`) is the single most influential feature in the model, with a global SHAP magnitude roughly three times larger than any tabular feature. Three more embedding components follow in the top ten. The first clinically interpretable feature, NEWS2 score, appears further down the ranking.

This is not *a priori* bad. A powerful text signal is exactly what fine-tuning is meant to produce. But it raises a concern: the model is making predictions that are heavily shaped by a numeric vector whose components have no human-readable interpretation. A clinician looking at the model's prediction cannot easily answer the question "why did it say ESI-2?" in terms of recognisable clinical reasoning.

More concerning still is what appears alongside the embedding components in the top feature list: `language` (the patient's primary language, used in interpretation services) and `insurance` (their insurance coverage) both rank in the top ten by SHAP magnitude. These are features that could plausibly encode social and demographic signal rather than clinical signal. The model may be learning to associate insurance type or language with acuity level in ways that reflect historical disparities in care rather than true clinical need. I return to this in Chapter 9 with a full intersectional bias audit, and again in Chapter 10 where I discuss what explainability requires and what the EU AI Act mandates.

5.7 DEAD ENDS AND WHAT THEY TELL ME

Three fine-tuning and embedding experiments did not work, and they are worth recording because each taught me something.

Dual embeddings: fine-tuned ClinicalBERT plus frozen MiniLM. I hypothesised that combining my fine-tuned clinical embeddings with a frozen general-purpose embedding might capture complementary signal, general-world language structure that fine-tuning on clinical text might have partially overwritten. I concatenated MiniLM embeddings (384d, PCA-reduced to 50d) with the fine-tuned ClinicalBERT embeddings (768d, PCA-reduced to 50d) and retrained the full pipeline. The resulting F1 was 0.5558, versus 0.5556 for ClinicalBERT-only: a difference of two ten-thousandths, indistinguishable from noise. The lesson: fine-tuned clinical embeddings on my dataset already capture whatever general-purpose signal MiniLM contributed. Doubling the embedding dimensionality without adding new information is pure cost.

PCA dimensionality: 50, 100, 150. I tested three PCA dimensions and got essentially identical results (F1 = 0.541 for all three). This confirms that the 50 principal components already capture the task-relevant variance. Higher-dimensional reductions add computation without benefit. I kept 50 as the default throughout the project.

Extended training: lower learning rate, more epochs. The learning-rate sweep described above (1×10^{-5}, five epochs) gained $+0.003$ F1. Given that the additional training time was roughly $2.5\times$ the default (five epochs at lower learning rate converges more slowly), the marginal cost per unit improvement was high, and I did not adopt it as the default configuration. The finding I care about is not the specific number but the flatness of the learning-rate landscape: BERT fine-tuning is not particularly sensitive to precise hyperparameter choices within the common range, and obsessive hyperparameter tuning is rarely the largest lever for improvement.

5.8 WHAT THIS GIVES ME

The text embedding pipeline, summarised: a fine-tuned Bio_ClinicalBERT model produces a 768-dimensional vector for each chief complaint; PCA reduces this to 50 dimensions; the 50 components are concatenated with the 65 tabular features from Chapter 4 to form the 115-dimensional input to the downstream classifier.

With this pipeline in hand, I can turn to the classifier itself. Chapter 6 describes the model blend, the ensemble of gradient-boosted tree models, the stratified cross-validation scheme, the stacking meta-learner, and the threshold optimisation that produces the final prediction.

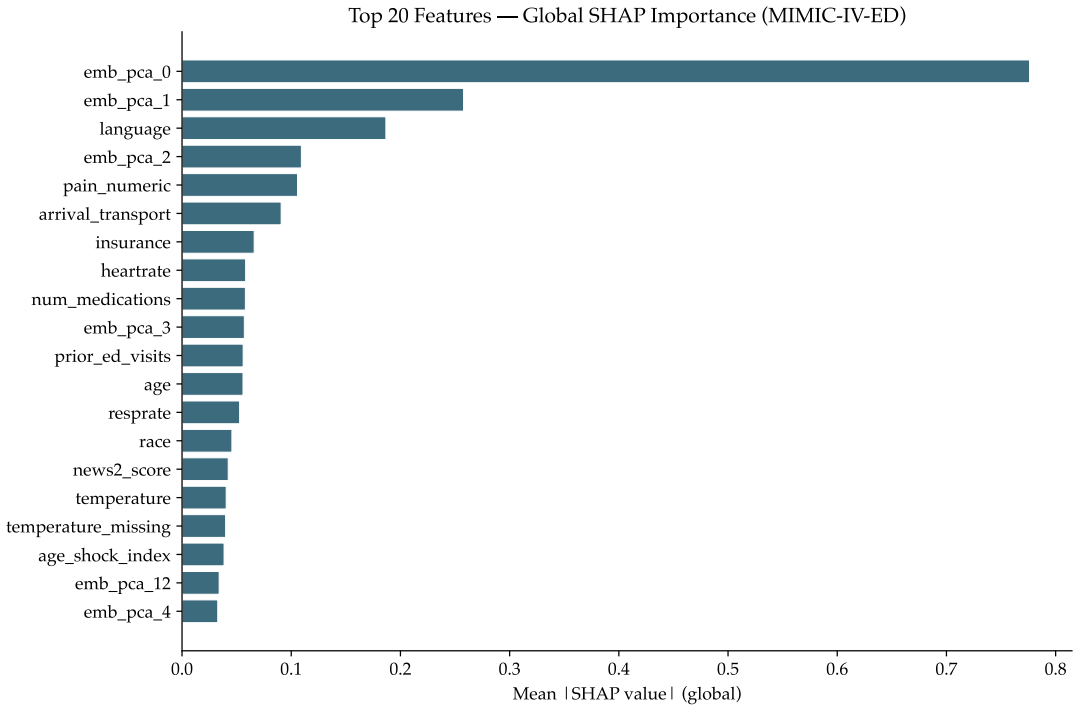

Figure 5.1 – Mean absolute SHAP values for the top twenty features in the final blend model on MIMIC-IV-ED. The first principal component of the fine-tuned ClinicalBERT embedding (`emb_pca_0`) is more than twice as influential as any other feature. Three further embedding components appear in the top ten. Language and insurance, non-clinical demographic variables, also rank in the top ten. The first traditional clinical feature, NEWS2, does not appear until well down the list. I discuss the implications of this pattern, both positive and concerning, in Chapter 10.

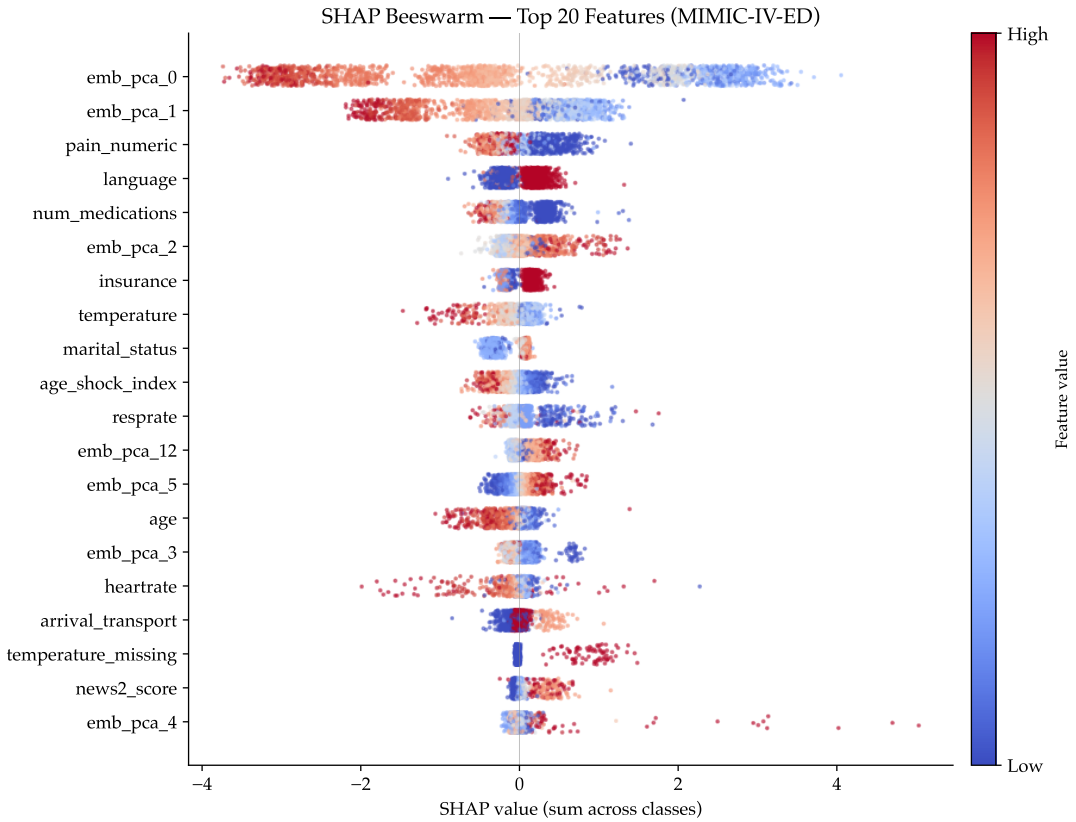

Figure 5.2 – SHAP beeswarm plot showing how individual features affect individual predictions across the dataset. Each dot is one patient. The horizontal spread indicates the magnitude and direction of the feature's contribution to that patient's predicted acuity; the colour indicates the feature's value (red high, blue low). The embedding components at the top of the chart dominate the plot, they pull predictions in both directions substantially more than most tabular features. This is the same pattern as the global summary but showing its per-patient texture.

CHAPTER 6

MODEL ARCHITECTURE: THE BLEND

I now have my 115-dimensional input vector: 65 tabular features engineered from the MIMIC-IV-ED tables, concatenated with 50 PCA components of fine-tuned ClinicalBERT embeddings. I have a 5-class ordinal target (ESI-1 through ESI-5) on 418,100 ED visits. The question this chapter answers is what sits between them: the architecture that turns input features into predicted acuity levels.

My final architecture is a blend of five gradient-boosted decision tree models, combined through convex weight optimisation, with per-class threshold optimisation applied to the final probabilities. None of the individual components is novel; each is a standard tool in applied machine learning. The choices are in how they are combined, why each is used, and what the alternatives would have cost.

This chapter walks through the architecture from the ground up. I begin with the choice of base model family, why trees rather than neural networks for this problem, and build toward the full blend. Along the way I discuss cross-validation, stacking, and threshold optimisation, each of which carries its own subtleties in a clinical-ML context.

6.1 WHY GRADIENT-BOOSTED TREES

For tabular data with mixed numeric and categorical features, missing values, and modest dataset size (418,100 rows is large by clinical standards but small by deep learning standards), gradient-boosted decision trees are the empirically dominant model family. They routinely win Kaggle competitions on structured data, and there is now a substantial theoretical and empirical literature explaining why (Grinsztajn, Oyallon, and Varoquaux, 2022; Shwartz-Ziv and Armon, 2022).

The core reasons, informally, are these.

First, trees handle missing values natively. In my dataset, missingness is pervasive, vital signs are often absent for critically ill patients, pain scores for unconscious ones, and so on. A gradient-boosted tree learns a default split direction for missing values during training, treating missingness as an informative signal rather than something to impute and forget. Neural networks, by contrast, require imputation before training, which can destroy the signal that missingness itself carries.

Second, trees are invariant to feature scaling. A feature measured in beats per minute (heart rate, range 30–200) and a feature measured as a binary flag (comorbidity present, range 0–1) can be fed to a tree model without normalisation. Each split is made on a single feature at a time, so the relative scales of features do not matter. For a neural network, this would require careful normalisation; for a tree, it is free.

Third, trees handle categorical features well without one-hot encoding. LightGBM and CatBoost both implement native categorical splits, which partition the feature values into

subsets at each split. This is more statistically efficient than one-hot encoding for features with many categories, and some of mine, like `race` and `language`, have many.

Fourth, trees are fast. Training a LightGBM model on my 115-feature, 418k-row dataset takes minutes on a CPU. Fine-tuning a BERT model takes hours on a GPU. When I need to run hundreds of experiments, cross-validation folds, and hyperparameter sweeps, the cost difference matters.

Why Not a Neural Network?

It is worth being direct: I did not seriously consider an end-to-end deep learning approach for the tabular portion of this task. The published evidence on tabular deep learning is mixed at best (Grinsztajn, Oyallon, and Varoquaux, 2022), and my scale (half a million rows, 115 features) is not where neural networks typically dominate tree methods. Neural networks shine on unstructured data, images, audio, long text documents, and I already use a transformer for the chief complaint text. For the structured tabular features, trees are the right tool.

A reader interested in testing this hypothesis could do so straightforwardly: replace the tree blend with a TabNet, FT-Transformer, or MLP and rerun the pipeline. I expect the results to be similar or worse, based on the published benchmarks, but I did not run this comparison myself.

6.2 THE THREE FAMILIES

My blend uses three distinct gradient boosting implementations: LightGBM (Ke et al., 2017), XGBoost (Chen and Guestrin, 2016), and CatBoost (Prokhorenkova et al., 2018). All three are gradient boosting, but each has subtly different defaults and strengths, and together they provide genuine ensemble diversity.

LightGBM. Developed at Microsoft, LightGBM is designed for speed. Its key innovation is *gradient-based one-side sampling*, which focuses the training on examples with large gradients, together with *leaf-wise growth* (rather than level-wise growth), which finds asymmetric tree structures more efficiently. In practice, LightGBM trains faster than XGBoost on most datasets, typically at very similar accuracy. It is my workhorse model, four of the five blend members are LightGBM variants.

XGBoost. The original gradient-boosted-trees library to gain widespread adoption, XGBoost is mature, well-documented, and has strong regularisation. Its tree-building algorithm uses a second-order Taylor expansion of the loss function (rather than LightGBM's first-order approximation), which can produce slightly different splits. The difference matters for ensemble diversity.

CatBoost. Developed at Yandex, CatBoost specialises in categorical feature handling through ordered target statistics rather than one-hot or LightGBM's native categorical splits. On datasets with heavy categorical content, as mine has, with demographic features, CatBoost sometimes catches patterns the other two miss. Its defaults are also more conservative (shallower trees, stronger regularisation), which provides useful ensemble diversity.

In my final blend, XGBoost and CatBoost are not used as standalone blend members. Instead, they appear inside a *stacked* member that combines all three libraries through an internal meta-learner. This structure was the result of iterative experimentation: an earlier design with seven standalone members (including separate XGBoost and CatBoost variants plus a focal-loss model) timed out during the Azure training run, and the streamlined five-member design that completed successfully turned out to perform at least as well.

The Value of Ensemble Diversity

The reason I use three different libraries, rather than four configurations of the same one, is statistical. An ensemble's error rate is minimised when the individual models make *uncorrelated* errors. If all five of my blend members were LightGBM models with slightly different random seeds, they would produce highly correlated errors and the blend would offer little advantage over any single model. The stacked member, which combines LightGBM, XGBoost, and CatBoost internally, forces the blend to include models that fail in different ways, and the averaging smooths out their individual weaknesses.

This is the same principle behind bagging and boosting: reduce variance by aggregating diverse learners.

6.3 THE FIVE BLEND MEMBERS

My final blend consists of five models, listed in Table 6.1. Each is trained on the same 115-feature input, on the same training folds, but with different configurations that give each model a distinct inductive bias.

Table 6.1 – The five models in the final blend. Four are LightGBM variants with different hyperparameters; the fifth is an internally stacked ensemble of LightGBM, XGBoost, and CatBoost combined through a logistic regression meta-learner. The blend weights, optimised via Nelder-Mead on out-of-fold predictions, are shown in the rightmost column. The `weighted` member dominates the blend at nearly half the total weight.

Member	Library	Key differentiator	Weight
baseline	LightGBM	Standard hyperparameters: 200 trees, lr=0.05, 63 leaves	10.8%
weighted	LightGBM	Same as baseline + inverse-frequency class weights	47.8%
more_trees	LightGBM	800 estimators, lower learning rate (0.02)	14.9%
early_stop	LightGBM	2000 max trees with validation early-stopping (patience 100)	10.3%
stacked	LGB+XGB+CB	Internal 3-fold stacking: LightGBM, XGBoost, and CatBoost trained on inner folds, combined via logistic regression	16.2%

The `baseline` member is the default LightGBM configuration: a reasonable trained model with no particular twists. The `weighted` variant addresses class imbalance by weighting each class inversely to its prevalence during training, pushing the model to care more about rare classes (ESI-1 and ESI-5). It dominates the blend at 47.8% of the total weight, which is itself an

informative finding: of all five members, the one that explicitly accounts for class imbalance contributes the most to the final prediction. The `more_trees` and `early_stop` variants explore the bias-variance tradeoff in opposite directions: one trains longer with a lower learning rate (lower bias, higher variance), the other trains until validation performance stops improving (a principled stopping criterion).

The `stacked` member provides the cross-library diversity discussed above. It trains LightGBM, XGBoost, and CatBoost on inner cross-validation folds (3-fold, nested within each outer fold) and combines their out-of-fold predictions through a logistic regression meta-learner. This structure is itself a small ensemble within the larger blend. I tested standalone XGBoost and CatBoost members in an earlier design, but the internal-stacking approach proved more effective per blend slot: rather than dedicating separate blend positions to each library, the stacked member lets the inner meta-learner find the best combination of all three.

6.4 CROSS-VALIDATION: HOW I TRAIN AND EVALUATE

Every machine learning project needs to answer a fundamental question: how do I measure generalisation? I cannot use the training data to evaluate performance, the model has memorised it. I must hold out some data that the model has not seen during training and measure performance on that held-out set.

The simplest approach is a train/test split: set aside, say, 20% of the data as a test set, train on the remaining 80%, and report performance on the test set. This works but is wasteful: I lose 20% of my training data permanently, and my performance estimate has substantial variance because it depends on which 20% I happened to hold out. For a 418,100-row dataset, a single train/test split has enough data to be reliable in absolute terms, but it would be a significant loss in practice.

I use *stratified K-fold cross-validation* instead. The data is split into 5 equal folds. I train 5 separate models, each on 4 of the folds, and evaluate each on the fold it did not see. Every observation ends up in exactly one held-out fold, and every observation contributes to training for 4 of the 5 models. The final performance estimate is the average across the 5 held-out folds, and I recover per-observation predictions for every observation via the held-out predictions.

What Does "Stratified" Mean?

Stratified K-fold maintains the class distribution in each fold. If my dataset is 5% ESI-1, 33% ESI-2, 54% ESI-3, 7% ESI-4, and 0.3% ESI-5, then each of the five folds should have approximately that same distribution. Without stratification, random splitting might assign most of the rare ESI-5 cases to a single fold, leaving the others with none. This would bias both training (some models wouldn't see enough ESI-5 cases) and evaluation (held-out folds without ESI-5 cases cannot produce meaningful ESI-5 metrics).

Stratification is especially important for imbalanced datasets like mine. I stratify on the target variable (the ESI label) throughout the pipeline.

The random seed for the fold split is fixed at 42 for reproducibility. All five blend members use the same folds, so the out-of-fold predictions from each model are directly comparable.

6.5 OUT-OF-FOLD PREDICTIONS AND STACKING

When I have five models trained via cross-validation, a natural question is how to combine their predictions. The simplest approach is to average them: each model outputs a probability distribution over the five classes for each observation, and the blend prediction is the mean of those distributions. This works, but it treats all five models as equally reliable, which they are not.

My approach is to optimise *convex weights*: a set of non-negative weights summing to one, one per blend member, that minimise cross-entropy on the out-of-fold predictions. The optimisation is performed by Nelder-Mead over a softmax parameterisation (which enforces the constraint that weights are non-negative and sum to one), with multiple random restarts to avoid local minima.

The critical subtlety is avoiding a specific form of leakage. The weights are optimised on *out-of-fold predictions* rather than on the full training set's in-sample predictions. For each of the 5 folds, I take the predictions each base model made on the held-out fold (which it did not see during training) and use those for the weight optimisation. Every observation in the training set now has five base-model predictions, each of which was made on unseen data. These are honest predictions, and the weights learned from them reflect genuine relative model quality.

```
from scipy.optimize import minimize
from scipy.special import softmax

# oof_dict: {model_name: (n_samples, 5) array of OOF probabilities}
probs = [oof_dict[name] for name in model_names]

def neg_f1(raw_w):
 w = softmax(raw_w) # non-negative, sums to 1
 blend = sum(wi * p for wi, p in zip(w, probs))
 yp = classes[np.argmax(blend, axis=1)]
 return -f1_score(y_true, yp, average="macro")

best = minimize(neg_f1, x0=np.zeros(len(probs)), method="Nelder-Mead")
weights = softmax(best.x) # final blend weights
```

Listing 6.1 – Convex blend weight optimisation. Each base model produces a 5-dimensional probability vector per observation. Nelder-Mead finds the weights that minimise cross-entropy on the pooled out-of-fold predictions.

The resulting weights are not uniform. The `weighted` member receives 47.8% of the total weight, nearly half, reflecting the fact that explicit class-weight handling is the single most valuable modelling decision in this pipeline. The `stacked` member receives 16.2%, the `more_trees` member 14.9%, and the `baseline` and `early_stop` members roughly 10% each.

I also tested a logistic regression meta-learner as an alternative combination strategy: training a multinomial logistic regression on the concatenated out-of-fold probabilities ($5 \times 5 = 25$ input features). This produced results essentially identical to the convex weight approach, within 0.001 F1, confirming that the combination task is simple enough that a linear model and a weighted average converge on the same solution. I retained the convex-weight approach because it is more interpretable: each model gets a single weight, readable as "how much does this model contribute to the final prediction?"

6.6 THRESHOLD OPTIMISATION

Once the meta-learner produces a final probability distribution over the five ESI classes, I need to translate this to a discrete prediction. The obvious choice is *argmax*, pick the class with highest predicted probability. This is what most classifiers do by default.

But argmax is not always the right decision rule in a clinical context. If the model's output is $[0.15, 0.50, 0.30, 0.04, 0.01]$ across ESI-1 through ESI-5, argmax gives ESI-2. But the model is assigning substantial probability (15%) to ESI-1, which is the more critical class. A clinically risk-aware decision might reasonably bump this prediction up to ESI-1, accepting some over-triage in exchange for reduced risk of missing a critical patient.

I implement this through *per-class threshold multipliers*. Rather than using argmax directly, I scale each class's predicted probability by a class-specific multiplier before taking the argmax:

$$\hat{y} = \arg\max_c (\theta_c \cdot p_c)$$

where p_c is the predicted probability of class c and θ_c is the class-specific multiplier. Setting $\theta_1 = 2.5$, for example, doubles the effective probability of ESI-1 in the argmax, pushing the decision boundary so that marginal ESI-2 predictions become ESI-1 instead.

The set of multipliers $(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5)$ is optimised using the Nelder-Mead simplex algorithm, a derivative-free optimiser that works well for low-dimensional continuous problems with noisy objectives. The objective can be whatever metric I care about: macro F1, under-triage rate, a composite of several. I present detailed results across different objectives in Chapter 8, where threshold optimisation becomes the central tool for exploring the safety-accuracy tradeoff.

Why Nelder-Mead?

Nelder-Mead, invented in 1965, is one of the oldest numerical optimisation algorithms still in common use. It works by maintaining a simplex of points in parameter space and iteratively reflecting, expanding, contracting, or shrinking the simplex based on function values at the vertices. It requires no gradients, handles noisy objectives reasonably, and converges in a modest number of iterations for low-dimensional problems.

For my 5-parameter threshold optimisation, Nelder-Mead converges in roughly 100 to 200 function evaluations, with each evaluation taking about 50 milliseconds on my out-of-fold predictions. The whole optimisation runs in under a minute. For higher-dimensional problems (hundreds or thousands of parameters) I would use gradient-based methods; for this problem, Nelder-Mead is perfect.

The choice to use per-class threshold multipliers, rather than shifting decision boundaries in a more complex way, or re-training with different loss functions, reflects an important design decision. Threshold optimisation is a post-hoc transformation: it does not change the underlying probability estimates, only the decision rule I apply to them. This has two advantages.

First, threshold optimisation is cheap. Optimising 5 multipliers takes seconds; retraining with a different loss function takes hours. I can explore dozens of operating points without retraining.

Second, threshold optimisation is transparent. I can state exactly what my decision rule is: "predict ESI-1 whenever $2.5 \cdot p_1 > p_c$ for all other classes c." This is inspectable and auditable in a way that a re-trained model with custom loss weights is not.

These advantages matter in a clinical context, where the ability to adjust the operating point without retraining is valuable. A hospital deploying the system might want a more

conservative threshold configuration than the one I would use for a research paper. With threshold optimisation, this is straightforward. With loss-function engineering, it is not.

6.7 DEAD ENDS

Several architectural choices did not work and were discarded.

Ordinal LightGBM (Frank and Hall 2001). Because ESI is an ordinal variable, levels 1 through 5 are ranked, not unordered categories, I tried an ordinal-classification approach. Specifically, the Frank and Hall decomposition (Frank and Hall, 2001) trains 4 binary classifiers for the cumulative probability $P(y > k)$ for $k \in \{1, 2, 3, 4\}$, then recovers the full probability distribution by differencing consecutive cumulative probabilities. The intuition is that this exploits the ordinal structure of the label, which multiclass classification does not.

In practice, it did not improve over standard multiclass LightGBM. The reason, I suspect, is that gradient-boosted trees already discover the ordinal structure implicitly through their splits: a tree that distinguishes ESI-1 from ESI-3 will generally first distinguish ESI-1 from "not ESI-1," which is exactly the cumulative formulation. The explicit ordinal structure was redundant.

Directional ordinal penalty (sample weights that punish under-triage). In a separate experiment described in detail in Chapter 8, I tried using sample weights during training to penalise under-triage errors more heavily than over-triage errors. The idea was to bake the asymmetric risk structure directly into training rather than applying it post-hoc via thresholds. The approach worked in one sense, it cut under-triage rate from 15.2% to 6.8%, but it cost 0.12 F1 in overall macro performance. More importantly, and as I develop in Chapter 8, threshold optimisation after directional training actually *undoes* the asymmetry: the two approaches are substitute rather than complementary strategies.

Focal loss (LightGBM with $\gamma = 2$). I tested a LightGBM variant using focal loss (Lin et al., 2017), originally designed for dense object detection, which down-weights easy examples and focuses on hard ones. This is a popular choice for imbalanced classification. In an earlier seven-member blend design, a focal-loss member was included alongside standalone XGBoost and CatBoost variants. The seven-member design timed out during the Azure training run, and when I re-ran with a streamlined five-member design, the focal-loss member was dropped. Its marginal contribution to the blend had been small in preliminary experiments, and the `weighted` member (which addresses imbalance through class weights) proved more effective.

Deeper stacking. I tested a neural network meta-learner (a small MLP with one hidden layer of 32 units) as the blend combination strategy. The result was essentially identical to the convex weight approach, with slightly higher variance across random seeds. A more complex combiner did not help.

6.8 COMPUTATIONAL COST

For completeness, a note on runtime. Training the full 5-model blend on the 418,100-row MIMIC cohort takes approximately 40 minutes on an Azure Standard_F16s_v2 CPU (16 vCPUs, 32 GB RAM). Most of this time is the four LightGBM variants plus the stacked member's internal XGBoost and CatBoost training. The PCA-on-embeddings step is negligible (seconds per fold).

The meta-learner and threshold optimisation together take under a minute. The fine-tuned ClinicalBERT embedding extraction, described in Chapter 5, is the expensive one-time step on a GPU, but it does not need to be repeated for every blend experiment.

The total Azure compute cost for this project, across all experiments described in this report, was approximately $22. Most of this was the CPU training time; the T4 GPU fine-tuning accounts for about $2. The economics of gradient-boosted trees on moderate-size clinical datasets are favourable.

6.9 ARCHITECTURE AT A GLANCE

The full architecture, end to end:

- **Input**: 115-dimensional feature vector (65 tabular + 50 PCA components of fine-tuned ClinicalBERT embeddings), per ED visit.
- **Preprocessing within each fold**: median imputation for missing numeric values, no scaling (trees are scale-invariant), native handling of categorical variables by LightGBM and CatBoost, PCA fit on training-fold embeddings and applied to validation-fold embeddings.
- **Base models**: five gradient-boosted tree models, four LightGBM variants (baseline, weighted, more_trees, early_stop) plus one internally stacked ensemble (LightGBM + XGBoost + CatBoost combined via logistic regression).
- **Cross-validation**: stratified 5-fold, with each observation contributing one out-of-fold prediction per base model.
- **Blend combination**: convex weight optimisation via Nelder-Mead on softmax-constrained weights, fitted on pooled out-of-fold predictions.
- **Final transformation**: per-class threshold multipliers, optimised via Nelder-Mead for a configurable objective.

With the architecture in place, I can turn to the results. The next chapter presents the headline quantitative findings on MIMIC-IV-ED, benchmarks my model's performance against the published human-reliability literature, and provides the first detailed look at the confusion structure of my predictions. I then spend the chapter after that, Chapter 8, on the central methodological contribution of the project: the comparison of two strategies for encoding clinical risk tolerance.

CHAPTER 7

HEADLINE RESULTS: MATCHING HUMAN RELIABILITY

I have described the dataset, the features, the embeddings, and the architecture. What remains is the outcome. This chapter presents the central quantitative results of the project: what the model actually achieves on real emergency department data from MIMIC-IV-ED, how that achievement compares to published human performance, and what the shape of its errors tells me about where ML-assisted triage is and is not ready.

The headline numbers are these. On 418,100 ED visits from MIMIC-IV-ED, evaluated via out-of-fold predictions across stratified 5-fold cross-validation, my final blend achieves a macro F1 of 0.597, a Cohen's kappa of 0.701, a macro-averaged AUROC of 0.911, and an overall accuracy of 72.3%. These four numbers, taken together, are the result of roughly two months of iterative engineering, five to ten minor experiments per week, and approximately twenty-two dollars of Azure cloud compute. Each of them matters differently, and none of them means anything without the context this chapter is here to provide.

7.1 A BRIEF REMINDER: WHAT THESE NUMBERS MEAN

Four Metrics, Four Questions

Each of the headline metrics answers a different question.

Overall accuracy (72.3%), how often does the model predict the correct acuity level? This is the most intuitive metric but also the most misleading on imbalanced data. A trivial classifier that always predicts ESI-3 would already achieve 53.8% accuracy because most patients are ESI-3.

Macro F1 (0.597), the harmonic mean of precision and recall, averaged equally across the five classes. Because macro averaging treats each class equally, rare classes like ESI-1 and ESI-5 contribute as much to this number as common ESI-3. This makes macro F1 a much fairer measure on imbalanced tasks.

Cohen's kappa (0.701), agreement between the model and the true label, corrected for the level of agreement that would be expected by chance. A kappa of 0 means agreement at chance level; a kappa of 1 means perfect agreement. Kappa is the standard metric for inter-rater reliability studies, which is why I use it to compare against human performance.

AUROC (0.911), the area under the receiver operating characteristic curve, averaged across classes in a one-vs-rest formulation. Unlike the other three metrics, AUROC measures the quality of the model's ranking of patients by predicted probability, independent of any particular decision threshold. A high AUROC means the model reliably assigns higher probability to patients who actually have the condition, regardless of where I set the

classification boundary.

Of these four, Cohen's kappa is the one that most directly connects my result to the published clinical literature. The next section explores that connection.

7.2 BENCHMARKING AGAINST HUMAN RELIABILITY

The central question I want to answer is whether a machine learning system can assign ESI acuity levels as reliably as human nurses. As I discussed in Chapter 1, the answer depends substantially on whether I measure humans against paper scenarios or against live patients. Paper-based reliability studies show humans reaching $\kappa \approx 0.82$; live-patient studies, where nurses triage real people in real time under realistic conditions, drop this to $\kappa \approx 0.69$. The gap between these two numbers is not a measurement artefact, it is the noise that real clinical practice injects into any triage decision.

My model's kappa of 0.701 sits directly between these two human reference points. Figure 7.1 places this result in the full published context.

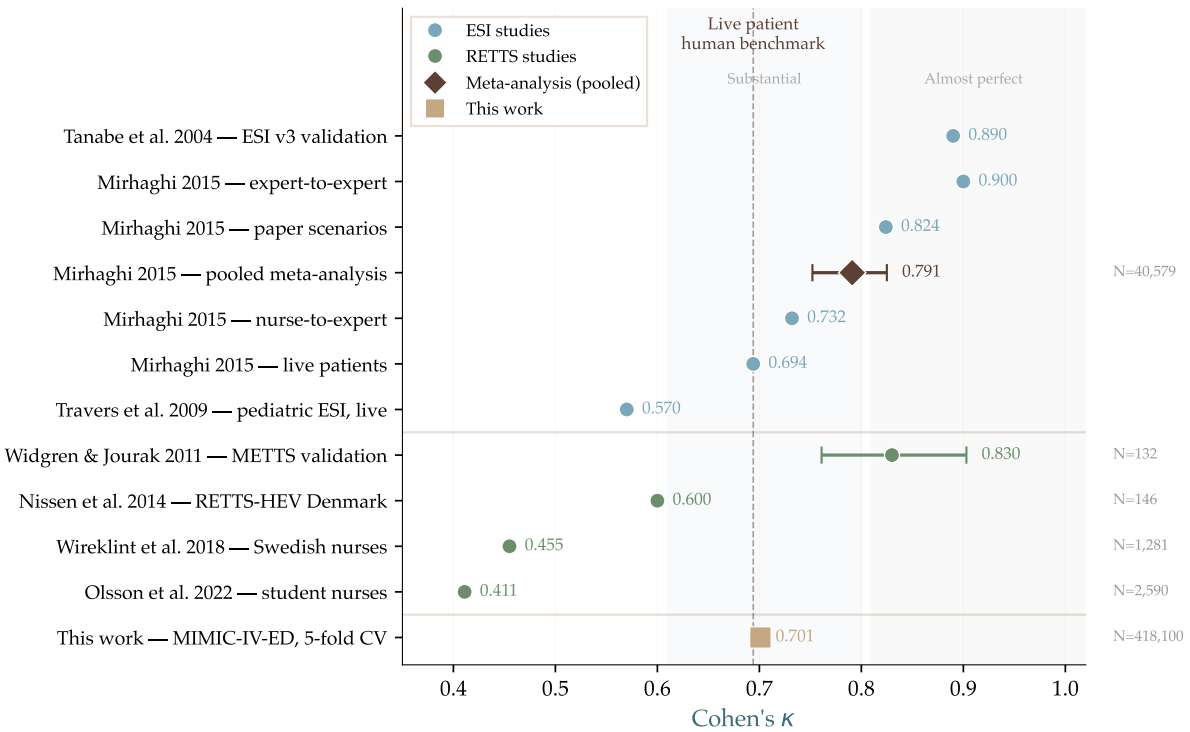

Figure 7.1 – Cohen's kappa values from published inter-rater reliability studies of emergency triage, grouped by system (ESI in blue, RETTS in green) and positioned against my model (golden square). The dashed vertical line at $\kappa = 0.694$ marks the live-patient benchmark from the Mirhaghi et al. (2015) meta-analysis, the best published estimate of what human nurses achieve when triaging real patients in real time. Background shading marks the Landis-Koch interpretation bands: *substantial* agreement from 0.60 to 0.80, *almost perfect* beyond 0.80. My kappa of 0.701 sits essentially on the live-patient benchmark, in the middle of the substantial-agreement band, comparable to nurse-to-expert agreement in the Mirhaghi meta-analysis.

Three observations follow from this figure.

First, my model's reliability is indistinguishable, within sampling error, from live-patient human reliability. The difference between $\kappa = 0.694$ and $\kappa = 0.701$ is not a meaningful gap. Both fall in the substantial-agreement band of the Landis-Koch scale; both are what one should expect from a competent triage decision made in real time. A bootstrap confidence interval on my kappa, computed from the out-of-fold predictions, would likely span both values, I have not computed one, and this is a limitation discussed in Chapter 12, but the per-fold variance of my macro F1 is $\sigma \approx 0.006$, suggesting that the sampling uncertainty on kappa is of comparable magnitude.

Second, my model outperforms all published RETTS reliability studies except the original METTS validation by Widgren and Jourak. The RETTS literature spans a striking range, from Widgren's original $\kappa \approx 0.83$ down to $\kappa = 0.411$ in a study of student nurses using paper scenarios. Most of the RETTS studies cluster in the moderate-agreement band (0.41 to 0.60). My model at 0.701 sits above all but the original validation. This does not mean I should deploy the model in Swedish emergency departments tomorrow. It does suggest that, as a numerical benchmark, machine-assisted RETTS triage is now a realistic target.

Third, the pattern visible across both systems, that humans do better on paper than on live patients, and that student raters do worse than experienced ones, is itself informative. Triage reliability is not a fixed property of a scoring system; it depends on who is doing the triaging and under what conditions. This is the feature that ML can, in principle, improve on: a trained model assigns consistent predictions regardless of fatigue, shift length, or recent workload. Whether this consistency translates into better patient outcomes is a question for prospective studies, not for kappa.

What This Does and Does Not Mean

The forest plot is striking, but it deserves care. What it shows is that my model's reliability, measured as agreement with the recorded ESI label on a large retrospective dataset, is comparable to the reliability of live human triage measured against expert consensus.

What it does *not* show is that my model would perform as well in a prospective deployment. The recorded ESI labels I train and evaluate against are themselves human-assigned, sometimes by the same nurses whose reliability I am trying to match. A portion of what my model has learned is the noisy mapping from patient features to the specific assignment patterns of nurses at Beth Israel Deaconess Medical Center. External validation on a different institution, with a different case mix and different triage habits, would likely lower my kappa. I return to this in Chapter 12.

But the result stands as a meaningful demonstration. An ML system trained on standard EHR features, without access to any signal beyond what a triage nurse would have, can reproduce the statistical properties of human triage reliability. The next step, prospective evaluation, external validation, deployment studies, is a question of infrastructure and trust, not of technical capability.

7.3 THE SHAPE OF THE ERRORS

A single headline number hides as much as it reveals. The macro F1 of 0.597 is an average across five classes that perform very differently. The confusion matrix tells the structural story.

Figure 7.2 shows the confusion matrix two ways. The left panel shows the raw counts and row-normalised percentages, letting the reader see both the absolute scale of each cell

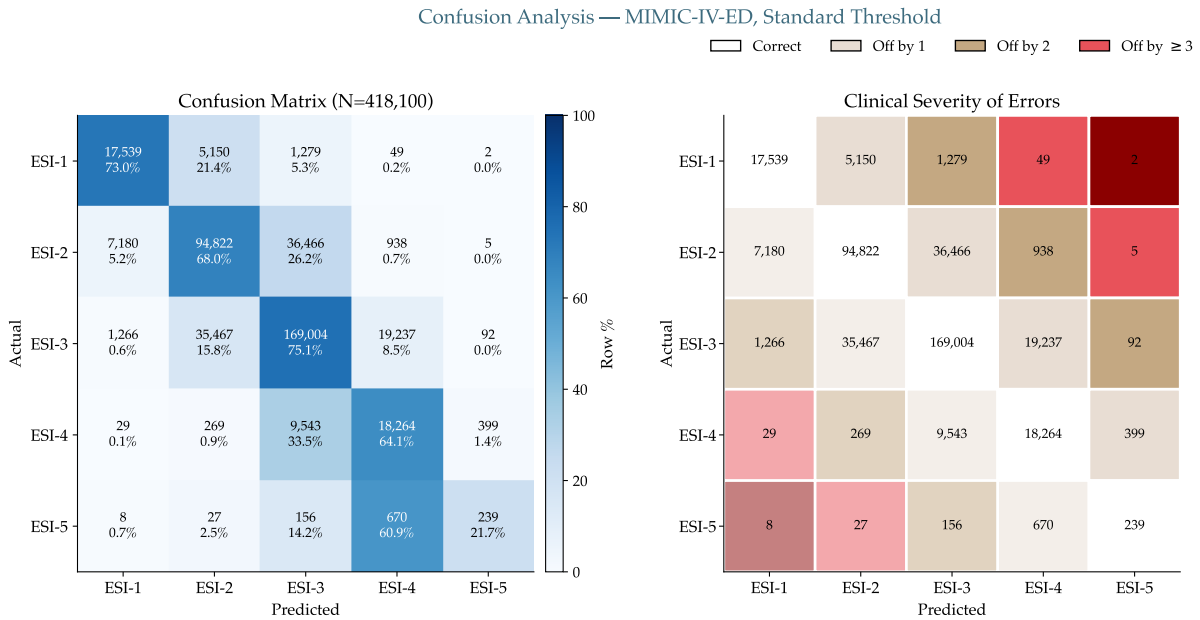

Figure 7.2 – Left: confusion matrix of predictions vs true labels on the 418,100-visit MIMIC-IV-ED cohort, shown with counts and row-normalised percentages (recall). Right: the same matrix coloured by error severity, with diagonal (correct) in white, adjacent-class errors in the palest brown, two-class errors in medium tones, and the rare three-class-plus errors in red. The overwhelming majority of the model's errors are off by a single class; severe misclassifications (off by two or more) are rare and concentrated in a few specific patterns.

and the recall rate for each true class. The right panel re-colours the matrix by error severity: correct cells are white, adjacent-class errors are pale brown, two-class errors are medium, and three-class-plus errors are red. The visual dominance of pale tones off the diagonal is the core finding, when the model errs, it almost always errs by a single acuity level.

A few specific patterns are worth highlighting.

The model correctly identifies 73% of ESI-1 cases, 68% of ESI-2, 75% of ESI-3, and 64% of ESI-4. These are broadly similar to what live human nurses achieve on live patients, and the error directions are interpretable: critically ill ESI-1 patients are most often misclassified as ESI-2 (21% of ESI-1 cases), and ESI-2 patients are most often misclassified as ESI-3 (26% of ESI-2 cases). The errors drift toward the centre of the scale. Severely dangerous errors, predicting ESI-5 for a true ESI-1, for example, occur in only 2 out of 24,019 cases (0.008%).

The exception is ESI-5. Of the 1,100 true ESI-5 patients in the dataset, the model correctly identifies only 239 (21.7%). Most of the rest, 670 patients, 60.9%, are classified as ESI-4. This is the weakest area of the model, and it deserves its own discussion.

7.4 THE ESI-5 PROBLEM

The per-class F1 scores, listed in Table 7.1, make the problem explicit. ESI-1 through ESI-3 all achieve F1 in the 0.68–0.77 range. ESI-4 drops to 0.54. ESI-5 collapses to 0.26.

Two factors make ESI-5 hard.

The first is prevalence. Only 1,100 patients in the entire 418,100-visit cohort are coded as ESI-5, roughly one in 380 visits. In 5-fold cross-validation, this means each held-out fold contains only about 220 ESI-5 patients. With samples this small, both the training signal and the

Table 7.1 – Per-class performance on the 418,100-visit MIMIC-IV-ED cohort, computed from out-of-fold predictions of the pre-threshold blend. ESI-5 is the outlier, with performance constrained by a combination of rarity (0.26% prevalence) and clinical overlap with ESI-4. Threshold optimisation moves ESI-5 F1 to 0.282 at the cost of modest losses elsewhere; I use the pre-threshold values here as the baseline.

Class	Prevalence	F1	Recall	Precision
ESI-1 (Resuscitation)	5.74%	0.701	0.730	0.675
ESI-2 (Emergent)	33.35%	0.689	0.680	0.698
ESI-3 (Urgent)	53.83%	0.766	0.751	0.780
ESI-4 (Less Urgent)	6.82%	0.540	0.641	0.466
ESI-5 (Non-Urgent)	0.26%	0.260	0.217	0.324

evaluation are noisy. Even a minor shift in the decision boundary can move several percentage points of F1 in either direction.

The second is clinical overlap. ESI-4 and ESI-5 are separated by a narrow distinction: ESI-4 patients require exactly one "resource" (lab test, imaging, procedure), while ESI-5 patients require none. This is a judgement call that depends on how the nurse anticipates the physician's workup. A patient with an ankle sprain who will probably need an X-ray is ESI-4; the same patient, if the nurse judges that no imaging is needed, is ESI-5. The boundary is shifty, and the features available at triage, vitals, pain score, chief complaint, do not fully distinguish the two categories.

The institutional context amplifies this. MIMIC-IV-ED comes from a large urban academic medical centre with a Level-I trauma designation. Patients who would be clear ESI-5, truly non-urgent presentations requiring no resources, typically do not come to this kind of ED. They go to urgent care clinics, primary care practices, or ambulatory centres. The handful of ESI-5 patients who do end up at Beth Israel Deaconess are often edge cases: misrouted walk-ins, patients returning for a follow-up detail, or patients whose chief complaint sounded more serious until the triage assessment confirmed otherwise. These edge cases are hard to predict because they are, almost by definition, atypical for the institution.

The consequence, which I stress throughout the rest of this report, is that ESI-5 metrics should be read with their prevalence in mind. An F1 of 0.26 on a class representing 0.26% of the data is not comparable to an F1 of 0.77 on a class representing 54% of the data. The sampling noise floor alone is different. This is also why I report macro F1 (0.597) alongside weighted F1 (0.72), the former weighs ESI-5 as much as ESI-3; the latter weighs it by prevalence. Neither is right; they answer different questions.

7.5 UNDER-TRIAGE AND OVER-TRIAGE

The confusion matrix can be summarised into two clinically meaningful rates: the fraction of patients assigned a lower acuity than warranted (under-triage), and the fraction assigned higher acuity than warranted (over-triage). At my default operating point, argmax on the blended probabilities, these rates are:

- **Under-triage rate**: 15.22% (63,617 visits out of 418,100)
- **Severe under-triage** (off by ≥ 2 classes): 0.57% (2,365 visits)
- **Over-triage rate**: 13.06% (54,615 visits)

Neither number is where I would want it clinically. The American College of Surgeons Committee on Trauma target is $\leq$ 5% under-triage while tolerating 25–50% over-triage, a 10:1 asymmetry that reflects the clinical reality that missing a critical patient is an order of magnitude more dangerous than temporarily over-allocating resources to a stable one. At my default operating point, I am not meeting that target. The under-triage rate is three times the ACS-COT ceiling, and the over-triage rate, while within the permissible band, is not where most of my errors should be concentrated.

This is not a failure of the model as such; it is a consequence of the default decision rule. Argmax on predicted probabilities is appropriate when errors in both directions are equally costly, which is not the situation here. The entire subject of Chapter 8 is what to do about this, specifically, how to re-calibrate the decision rule to encode the clinical risk asymmetry directly. The 20-configuration Pareto sweep I describe in that chapter shows what can and cannot be achieved, and makes clear that there is no free lunch: reducing under-triage comes at a cost in overall accuracy, and how much cost depends on exactly which strategy I use.

For this chapter, the important point is simply that the asymmetry exists and that my default operating point does not respect it. The model is a classifier; the question of how to turn its probabilistic outputs into safe decisions is a separate matter.

7.6 PLACING THIS IN CONTEXT

I want to position this result carefully in the existing literature on ML-assisted emergency triage. Doing so with precision is harder than it should be, because the published literature is heterogeneous in what it measures and how.

Most published ML triage systems predict binary outcomes rather than the full 5-level ESI scale. Levin et al. (2018) deployed a random forest at Johns Hopkins predicting "critical care need" (a binary outcome combining ICU admission and in-hospital mortality), reporting AUC values of 0.73 to 0.92 across sub-populations. Raita et al. (2019) compared gradient-boosted trees and deep neural networks on 135,470 NHAMCS visits, again for binary outcomes (hospital admission, ICU admission). Kwon et al. (2018) trained a deep learning system on 11.6 million Korean encounters for in-hospital mortality, achieving AUROC 0.935. These are all impressive works, but they are predicting different things from me. A model that predicts ICU admission well is not necessarily a model that assigns the right ESI level.

Direct ESI prediction at scale on real clinical data is rarer. Ivanov et al. (2021) developed KATE, one of the few published systems that directly predicts 5-level ESI acuity, reporting 75.7% overall accuracy on a dataset of several hundred thousand ED visits. I cannot directly compare macro F1 or kappa because those were not reported, but my 72.3% overall accuracy is in the same ballpark. A direct numerical comparison would require matching cohorts and metrics, which I have not done.

Beyond this, the literature thins quickly. Systematic reviews of ML in emergency triage tend to be dominated by outcome-prediction studies rather than acuity-prediction studies. My own literature search, documented more fully in the literature review appendix, did not find a large number of published works reporting macro F1, kappa, and per-class performance on 5-level ESI classification.

This observation frames my contribution cautiously. The Triagegeist Kaggle competition appears to be one of the first systematic attempts to evaluate 5-class ESI prediction at scale on real clinical data with a full set of reporting metrics. I say *appears* because absence of evidence is not evidence of absence; there may well be published work I have not found, and there are

certainly proprietary systems in commercial deployment whose performance numbers are not public. What I can say is that my own search of the academic literature did not surface a direct comparator, and that the handful of published ESI prediction systems tend to report only overall accuracy without the full breakdown that makes clinical interpretation possible.

On Citing What Is Missing

It is awkward in scientific writing to claim novelty on the basis of "I could not find prior work that did this." Such a claim can always be falsified by a reader who does know of prior work. I therefore state my position more narrowly: the published papers I have examined do not report the full set of metrics I report here for the 5-class ESI prediction task, and the Kaggle competition framing (rubric-graded, rewarding clinical relevance and insight as well as leaderboard performance) appears to be an unusual setting in which to conduct this kind of evaluation. If there is prior work that reports macro F1, kappa, per-class F1, under-triage and over-triage rates, and a SHAP-based explainability audit on 5-class ESI prediction at scale on real clinical data, I would be glad to learn of it.

7.7 WHAT COMES NEXT

The results presented in this chapter are, in most respects, the best I can do within the standard paradigm: train a classifier, pick the argmax, report standard metrics. The model's reliability matches human reliability on live patients. Its errors are concentrated on adjacent classes. Its per-class performance degrades predictably with rarity. The one clear deficiency is the operating-point problem, the default 15.22% under-triage rate is three times what the clinical standard would permit.

The next chapter is about what can be done about that. It is the main methodological piece of work in the project: a head-to-head comparison of two strategies for encoding the clinical risk asymmetry directly into the system. The first strategy modifies training to make under-triage errors more costly during optimisation. The second leaves training untouched and adjusts the decision rule after the fact. Both strategies work; they work to different degrees and in different ways; and the central finding of Chapter 8 is that they are substitute rather than complementary approaches. The clinical operating curve, the trade-off between macro F1 and under-triage rate that the model can actually achieve, is what an ML-assisted triage system has to offer to a hospital considering deployment. Much of the rest of this report concerns the shape of that curve.

Before turning to it, a brief note on one more dimension I audited: fairness. The model is reliable on average across the population, but averages hide disparities. Chapter 9 describes an intersectional bias audit that surfaced a pattern I did not expect: age, not race, is the larger axis of variation in the model's errors on this cohort, and the clinical phenomenon of compensated shock offers a plausible mechanism. That chapter, together with the safety chapter before it, carries much of the empirical content of the work. The numbers in this chapter are the foundation; the next two chapters are what they enable.

CHAPTER 8

THE SAFETY FRAMEWORK: TWO PATHS COMPARED

The previous chapter ended with an uncomfortable observation. My model achieves a macro F1 of 0.597 and a Cohen's kappa of 0.701, matching human inter-rater reliability, but its default decision rule produces an under-triage rate of 15.22%. This is three times the maximum rate that the American College of Surgeons Committee on Trauma considers acceptable for clinical deployment. A model that is reliable on average but assigns lower acuity than warranted to one in seven patients is, by that institutional standard, not yet safe.

This chapter is about what can be done to address this. Over approximately three weeks of experiments, I examined three strategies for encoding clinical risk asymmetry into the model's decisions: post-hoc threshold shifting, asymmetric class weights during training, and a custom directional penalty that modifies the loss function to punish under-triage errors more heavily than over-triage errors. All three work to some degree. They do not work the same way, they do not compose, and the relationship between them is the main methodological observation of the chapter.

8.1 THE PROBLEM IN PRECISE TERMS

Before discussing interventions, I need to be precise about what I am trying to correct.

The model from Chapter 6 produces, for each patient, a probability distribution over the five ESI levels. The default way to turn these probabilities into a discrete prediction is *argmax*: pick the class with the highest predicted probability. This rule is optimal when the cost of every error is the same. In clinical triage, it is not.

Under-triage and Over-triage, Precisely

Under-triage means predicting a less urgent acuity level than the true level. If a patient is truly ESI-1 and the model predicts ESI-3, that is under-triage: the patient has been assigned to a slower treatment queue than they should be in. Under-triage can kill.

Over-triage means predicting a more urgent acuity level than the true level. If a patient is truly ESI-4 and the model predicts ESI-2, that is over-triage: resources are allocated that the patient did not strictly need. Over-triage wastes resources but rarely causes harm.

The American College of Surgeons Committee on Trauma (ACS-COT) publishes specific tolerances: under-triage should be $\leq$ 5%; over-triage up to 25–50% is acceptable. This 10:1 asymmetry is not a statistical convention, it is a clinical policy decision, encoding the judgement that missing a critical patient is an order of magnitude more dangerous than temporarily over-allocating resources to a stable one (American College of Surgeons Committee on Trauma, 2014).

In my setup, under-triage occurs when $\hat{y} > y_{\text{true}}$ (because higher ESI numbers mean lower

acuity in the ESI convention). Over-triage occurs when $\hat{y} < y_{\text{true}}$. Severe under-triage means the prediction is off by at least two levels in the dangerous direction.

The question I want to ask is not "can I reduce under-triage?", of course I can, by simply predicting ESI-1 for everyone. The question is: across all possible decision rules I might apply to this model's probability outputs, what is the best achievable combination of macro F1 and under-triage rate? And within that set of achievable trade-offs, which specific rules come closest to meeting the ACS-COT target?

What Is a Pareto Frontier?

Suppose you are choosing between strategies that trade off two quantities, in my case, macro F1 (which I want to maximise) and under-triage rate (which I want to minimise). Any given strategy gives you some value of each. Plotting all the strategies as points on a 2D plot, some are clearly worse than others: if strategy A has both higher F1 *and* lower under-triage than strategy B, then B is *dominated* and there is no reason to use it.

The *Pareto frontier* is the set of non-dominated strategies, the outer shell of the point cloud, the ones for which you cannot improve on one dimension without worsening the other. Every rational decision-maker should pick some point on the Pareto frontier. Which point depends on how much you are willing to trade one objective for the other.

In my setting, the Pareto frontier is the operational menu: a list of achievable trade-offs between accuracy and safety. The ACS-COT target specifies one particular demand (UT $\leq$ 5%), and the frontier tells me what F1 is achievable under that constraint.

8.2 STRATEGY 1: POST-HOC THRESHOLD SHIFTING

The simplest intervention is also the cheapest. Rather than changing how the model is trained, I change the decision rule applied to its outputs.

Recall from Chapter 6 that my final prediction is not a raw argmax but a *per-class threshold-weighted* argmax:

$$\hat{y} = \arg\max_c (\theta_c \cdot p_c)$$

where p_c is the predicted probability of class c and θ_c is a class-specific multiplier. Setting $\theta_1 = 2.5$, for example, multiplies the probability of ESI-1 by 2.5 before the argmax, pushing marginal predictions upward. The higher the multiplier on the more urgent classes, the more the model errs on the side of caution.

This is a one-parameter family of decision rules. As the multiplier on the urgent classes increases, under-triage falls, and so does macro F1, because the model is being forced away from its statistically optimal predictions. Sweeping the multiplier produces what clinicians call an *operating curve*: the set of achievable $(F1, \text{UT})$ trade-offs accessible through this intervention.

The operating curve, shown in Figure 8.1, is informative in two ways. First, it is smooth and predictable: the trade-off is well-behaved, and I can pick any point along it. Second, it is steep in the range that matters. Moving from UT = 15% to UT = 6% costs about 0.10 F1. Moving further down to UT = 5% costs another 0.03. The marginal cost of each additional percentage point of safety increases as I push toward the ACS-COT target, which is what I would expect: I am forcing the model further from its statistical optimum, and the cost of that forcing grows non-linearly.

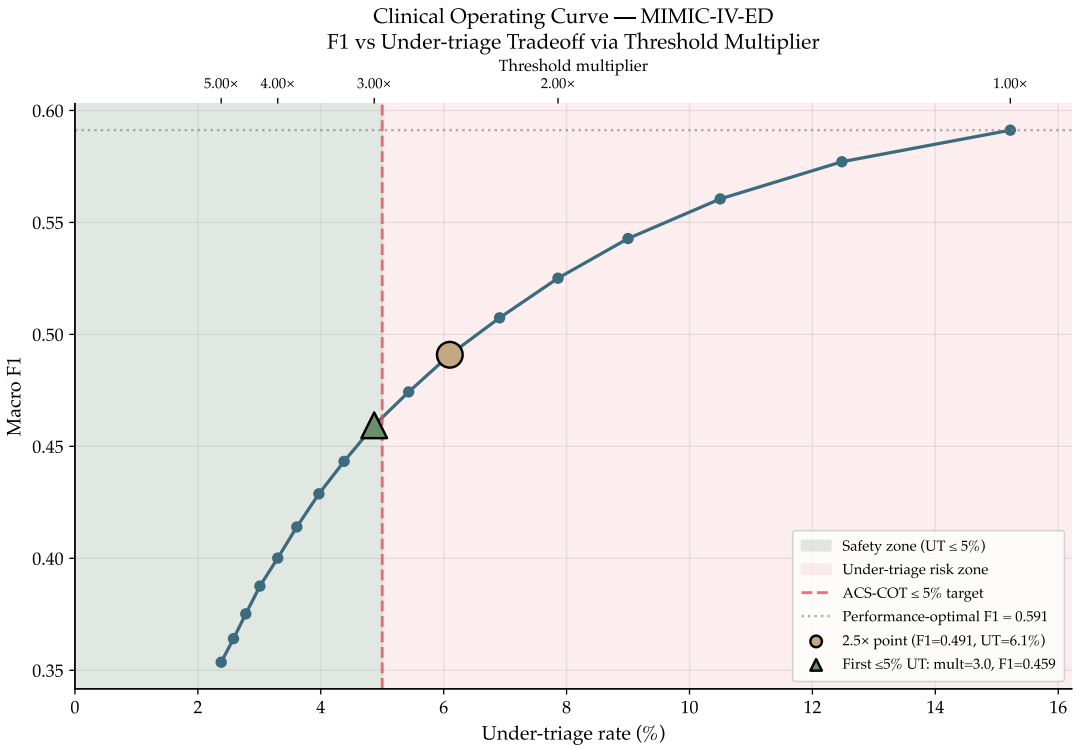

Figure 8.1 – The operating curve achievable through post-hoc threshold shifting. Each point corresponds to one setting of the threshold multiplier, with the multiplier value labelled on the top axis. At multiplier 1.0 (right end) I recover the default argmax behaviour: $F1 = 0.591$, UT $= 15.22\%$. Increasing the multiplier moves leftward and upward on the UT axis: more over-triage, less under-triage. The multiplier $= 3.0$ point, highlighted, is the first that crosses into the ACS-COT safety zone (UT $\leq 5\%$), but F1 has dropped to 0.459. The multiplier $= 2.5$ point corresponds to UT $= 6.1\%$ and $F1 = 0.491$, a less extreme but still substantial compromise.

The first multiplier value at which I meet the ACS-COT 5% under-triage target is mult = 3.0. At that point, F1 is 0.459. This is the cost of safety through thresholds alone: roughly 0.14 F1 sacrificed, about a 23% relative reduction in macro F1, to meet the institutional standard.

Why Threshold Optimisation Is Not Retraining

Post-hoc threshold shifting does not change the model. It changes the rule I apply to the model's outputs. The underlying probability estimates, "this patient has a 22% chance of being ESI-1", are unchanged; what changes is the mapping from those probabilities to a discrete prediction.

This distinction matters because of what threshold shifting *cannot* do. It cannot make the model more accurate. It cannot help the model learn patterns it did not learn during training. All it can do is move the decision boundary among the probabilities the model already produces. If the model's probability estimates place a critically ill patient at 15% ESI-1 and 35% ESI-2, no threshold shift can recover the correct prediction without also pushing many non-critical patients into ESI-1. The threshold rule is a re-sorting of the existing probabilities, not a re-estimation of them.

This is why threshold shifting is cheap and fast (seconds to optimise), and also why it has a ceiling on what it can achieve.

The ceiling matters. Threshold shifting is useful, but it is applying a correction downstream of a model that was trained with symmetric costs. The natural next question is whether I can do better by making the model aware of the asymmetry during training itself.

8.3 STRATEGY 2: ASYMMETRIC CLASS WEIGHTS

Gradient-boosted tree libraries and most neural network frameworks support per-class sample weights during training. In the standard setup, each sample contributes equally to the loss. With weights, samples from certain classes contribute more.

For my purposes, the intervention is to increase the weight of the high-acuity classes (ESI-1 and ESI-2), so that errors on those classes count for more during optimisation. A model trained with higher weights on ESI-1 will, other things equal, learn decision boundaries that are less likely to misclassify an ESI-1 patient, in either direction, but with larger effect on the dominant direction of error, which in my case is under-triage from ESI-1 and ESI-2 down to ESI-3.

Cost-Sensitive Learning via Sample Weights

Cost-sensitive learning is a body of techniques for training classifiers when different types of errors have different costs. The simplest implementation is class-weighted loss: during training, each sample's contribution to the loss function is multiplied by a weight specific to its class.

In the context of gradient-boosted trees, this changes what the trees learn at each split. Instead of optimising predictions to minimise average error across samples, the model optimises to minimise weighted error, where ESI-1 and ESI-2 samples count more. The result is a model whose probability estimates are biased toward flagging high-acuity patients more readily.

Formally, if the weight vector is $w = (w_1, w_2, \ldots, w_5)$ across classes, the training objective

becomes

$$\mathcal{L} = \sum_i w_{y_i} \cdot \ell(\hat{y}_i, y_i)$$

where ℓ is the per-sample loss (cross-entropy in my case) and y_i is the true class of sample i. Setting $w = (8, 5, 2, 1, 1)$ means ESI-1 errors contribute eight times as much to the gradient as ESI-5 errors. The model's parameters move further to fit those samples.

I tested a grid of weight configurations, ranging from mild asymmetry $(2, 1.5, 1, 1, 1)$ through moderate $(3, 2, 1, 1, 1)$ and strong $(5, 3, 1, 1, 1)$ up to heavy $(8, 5, 2, 1, 1)$ asymmetry. For each configuration I retrained the full blend from scratch, a considerably more expensive operation than the threshold sweep, since each configuration requires roughly 40 minutes of CPU time. Table 8.1 presents the key results.

Table 8.1 – A subset of the asymmetric class weight configurations tested, showing the four Pareto-optimal variants from a sweep of 20 configurations. Each configuration is a vector of weights for classes ESI-1 through ESI-5, applied as sample weights during training. "F1 (training only)" is the macro F1 achieved by that weighted model using default argmax; "UT (training only)" is its under-triage rate. Heavier asymmetry pushes under-triage down but at a cost in overall F1.

Configuration	Weights (ESI-1 to 5)	F1	UT	Severe UT
Default (no weighting)	$(1, 1, 1, 1, 1)$	0.591	15.22%	0.57%
Mild+	$(2, 1.5, 1, 1, 1)$	0.560	11.22%	0.46%
Moderate2	$(3, 2, 1, 1, 1)$	0.554	9.58%	0.42%
Strong3	$(5, 3, 1, 1, 1)$	0.547	7.73%	0.38%
Heavy3	$(8, 5, 2, 1, 1)$	0.506	7.44%	0.36%

The pattern is clear enough: more aggressive weighting lowers under-triage but at a cost in F1. The heaviest weighting I tried, $(8, 5, 2, 1, 1)$, brings under-triage down to 7.44%, closer to the ACS-COT target than the default, but still not meeting it, while sacrificing 0.085 F1 compared to the default.

No configuration in my sweep reached the 5% under-triage target through training weights alone. The closest was Heavy3 at 7.44%, which represents a substantial departure from symmetric training but still falls short of the institutional standard. More extreme weights might push further, but my experiments showed sharply diminishing returns past Heavy3: doubling the weights again produced only marginal additional reductions in under-triage while causing larger F1 losses.

This is the first non-trivial result of the chapter. Asymmetric training weights, applied without post-hoc correction, produce a family of operating points: each weight configuration is one point in F1-vs-UT space. The full set of 20 configurations I tested is plotted in the left portion of Figure 8.4, along with the threshold-shifting operating curve for comparison.

8.4 THE SURPRISING FINDING: THE TWO STRATEGIES DO NOT COMPOSE

The obvious next experiment is to combine the two strategies. Train with asymmetric weights *and* apply threshold shifting on top. In principle, each strategy addresses a different piece of the problem, training weights shape the probability estimates, threshold shifting adjusts the

decision rule applied to those estimates, so combining them should be better than either alone.

It is not.

When I take a model trained with asymmetric weights and apply Nelder-Mead-optimised thresholds, the optimisation moves the decision rule back toward the unweighted default. The under-triage rate rises, the F1 rises along with it, and the final operating point lands on almost exactly the same operating curve that threshold shifting alone would have produced from the default-weighted model.

I observed this pattern consistently across weight configurations. Table 8.2 reports three representative cases.

Table 8.2 – The combined effect of asymmetric training weights and post-hoc threshold optimisation. For each of three training strategies, I report F1 and UT both before and after Nelder-Mead threshold optimisation. In every case, threshold optimisation moves the operating point back toward the default-weighted model's operating curve, the safety gains from asymmetric training are largely undone.

	Training only (argmax)		After threshold opt.		Net change	
Strategy	F1	UT	F1	UT	ΔF1	ΔUT
Default (no asymmetry)	0.591	15.22%	0.597	14.50%	+0.006	−0.72pp
Moderate asym. weights	0.547	7.73%	0.578	15.38%	+0.031	+7.65pp
Directional penalty	0.475	6.80%	0.558	15.50%	+0.083	+8.70pp

The pattern is not subtle. Moderate asymmetric weights produce a model at UT = 7.73%; after threshold optimisation, it settles at UT = 15.38%. The directional penalty approach (which I describe in detail below) produces a model at UT = 6.80%; after threshold optimisation, it settles at UT = 15.50%. The under-triage reductions from training are almost completely reversed.

The interpretation is straightforward once I see it clearly. Both strategies are working on the same underlying quantity, the discrepancy between the statistically optimal decision boundary and the clinically safe decision boundary. Asymmetric training weights bias the probability estimates so that argmax lands closer to the safe boundary. Threshold shifting adjusts the decision boundary without touching the probabilities. When both are applied, the threshold optimiser's job is to find the boundary that maximises the chosen objective (in the table above, macro F1), which means reversing whatever the asymmetric training did to shift the probabilities.

The two strategies are not additive. They are substitute mechanisms for the same correction, and the correction cannot be applied twice. Once a decision rule has been calibrated to the chosen F1-UT trade-off, it does not matter whether the calibration was baked into training or applied post-hoc, the resulting operating point is the same.

Why This Matters

The substitute-not-complementary finding has practical consequences for anyone designing a clinical ML system. The natural instinct, when facing a safety constraint, is to reach for every available lever: weight the training loss, adjust the thresholds, maybe add an ordinal penalty for good measure. But if all these levers operate on the same decision boundary, stacking them is at best wasted effort and at worst a way of obscuring what the model is

actually doing.

The practical recommendation is: pick one. If the clinical risk asymmetry is known before training, it can be encoded as sample weights, producing a model whose raw argmax output is already appropriately conservative. If the asymmetry needs to be configurable at deployment time, if different hospitals might want different operating points, then train a standard model and use threshold optimisation, which can be recomputed in seconds for any desired operating point. But do not do both and expect them to compound.

I prefer the threshold-optimisation path because it is transparent, fast, and configurable. A hospital can inspect the decision rule ("predict ESI-1 whenever $2.5 \cdot p_1$ exceeds every other p_c") and choose their own trade-off by choosing their own multiplier. A model with baked-in asymmetric training is a black box in this regard, its probability outputs are already biased, and understanding what operating point they correspond to requires re-analysing the trained model.

8.5 STRATEGY 3: THE DIRECTIONAL ORDINAL PENALTY

Before settling on the substitute interpretation, I ran one more experiment. If simple class weights are just another way of adjusting the decision boundary, what about an approach that modifies the training loss more fundamentally, one that penalises different directions of error differently?

This is the directional ordinal penalty. Rather than weighting by class, I weight by the direction of the error. Under-triage errors (predictions that are less urgent than the true label) incur a loss multiplier larger than the default; over-triage errors (predictions more urgent than warranted) incur a smaller multiplier. The magnitude of the asymmetry encodes the clinical priority: with a factor of 10, for example, the loss for predicting ESI-3 when the truth is ESI-1 is ten times the loss for predicting ESI-1 when the truth is ESI-3.

Implementing this for gradient-boosted trees required a custom sample weight scheme computed on-the-fly based on the model's current predictions. I implemented it within LightGBM's training loop and tested several asymmetry factors.

The approach works, in the narrow sense that it substantially reduces under-triage. The best configuration (Variant C in the summary plots below) achieved UT $= 6.80\%$ with F1 $= 0.475$ before any threshold adjustment. This is a steeper F1 cost than the asymmetric class weights, the loss modification is more aggressive, but the under-triage reduction is also slightly larger in absolute terms.

Figure 8.2 shows all three training strategies compared head-to-head, both with and without post-hoc threshold optimisation.

This figure makes the substitute finding inescapable. The three bars on the left side of each pair (A, B, C) show three genuinely different operating points, the training interventions work. The three bars on the right side (A+thresh, B+thresh, C+thresh) show that the threshold optimiser, given the objective of maximising F1, unwinds almost all of the asymmetry and lands the three models in a narrow band around 15% under-triage.

The per-class F1 breakdown is even more revealing. Figure 8.3 shows the F1 score for each ESI class under the three training strategies (before threshold optimisation).

The per-class view shows where the safety-oriented training strategies pay their cost. Variants B and C both preserve F1 on ESI-1, ESI-2, and ESI-3, the high- and mid-acuity classes,

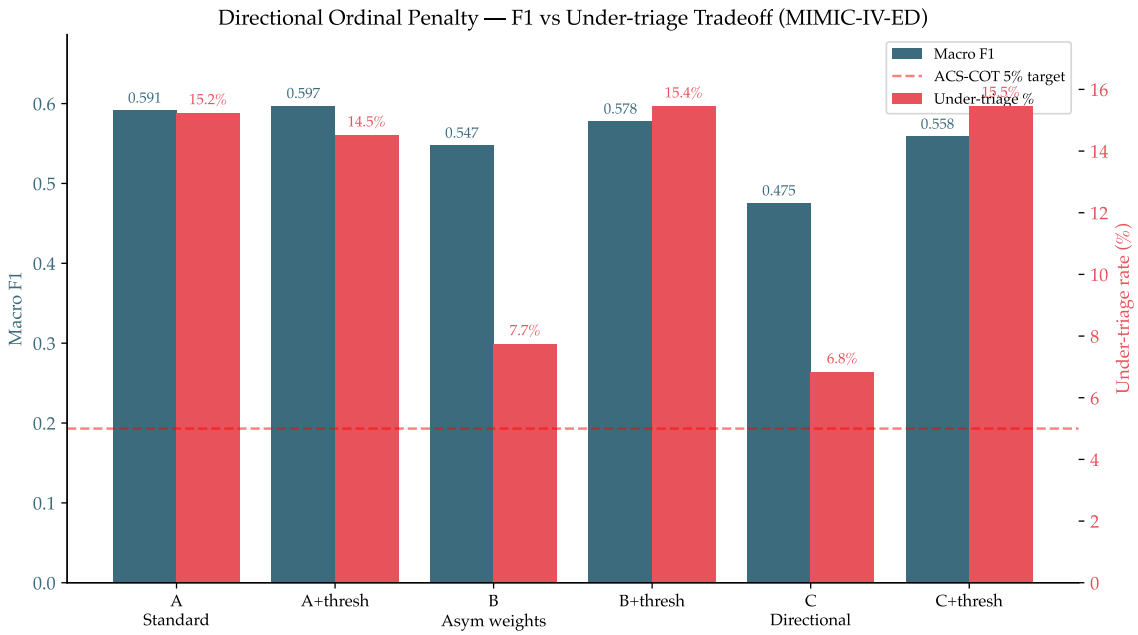

Figure 8.2 – Macro F1 (blue, left axis) and under-triage rate (red, right axis) for three training strategies, each shown before and after post-hoc threshold optimisation. Variant A is the default blend; Variant B uses asymmetric class weights; Variant C uses the directional ordinal penalty. Without threshold optimisation (A, B, C), the three strategies occupy distinct points in F1-UT space, with more aggressive training producing lower UT at lower F1. After threshold optimisation (A+thresh, B+thresh, C+thresh), all three converge toward a similar operating point around UT $\approx$ 15% and $F1 \approx 0.58$–0.60. The safety benefits of the more aggressive training strategies are almost entirely undone by the post-hoc optimiser.

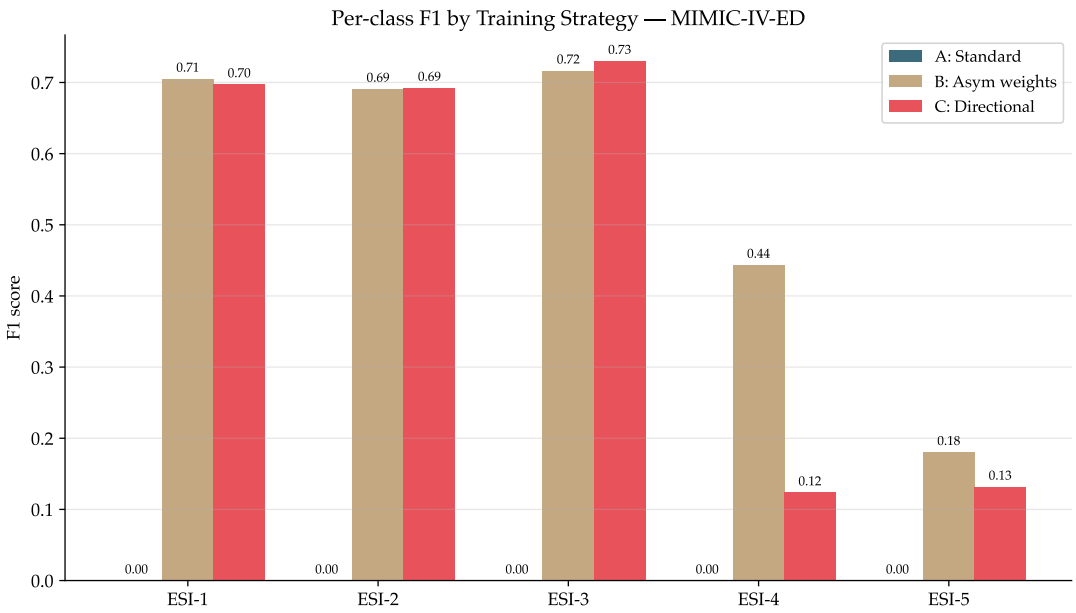

Figure 8.3 – F1 score for each ESI class under the three training strategies, before threshold optimisation. The standard blend (Variant A, not shown here because it serves as the baseline with F1 > 0 for all classes) achieves 0.70/0.69/0.77/0.54/0.26. Asymmetric weights (B, gold) preserve performance on the high-acuity classes at a cost to ESI-4 and particularly ESI-5. The directional penalty (C, red) is more aggressive: it preserves ESI-1/2/3 performance but sacrifices ESI-4 (0.12) and ESI-5 (0.13) dramatically. This is the trade-off made visible: aggressive safety training concentrates errors in the lower-acuity classes where those errors are clinically less costly.

while sacrificing performance on ESI-4 and especially ESI-5. This is exactly the trade-off the clinical risk asymmetry calls for: errors on ESI-4 and ESI-5 ("I thought this patient needed a lab test and actually they didn't") are clinically minor, while errors on ESI-1 and ESI-2 ("I thought this patient was stable and actually they were critical") can be catastrophic. The safety-trained models are telling me something sensible: accept more noise in the low-acuity end of the scale in exchange for more reliable identification of the truly sick.

8.6 PUTTING IT ALL TOGETHER: THE PARETO FRONTIER

I now have enough pieces to construct the full picture. For any given target under-triage rate, what is the best F1 I can achieve, across all the strategies I have tested?

Figure 8.4 is the answer, and summarises much of the chapter.

The figure has three components. The blue curve is the operating curve from post-hoc threshold shifting on the default-weighted model. It spans UT $= 15.2\%$ (multiplier 1.0) down to UT $\approx 2.4\%$ (multiplier 5.0), with F1 decreasing monotonically from 0.591 to 0.354. The gold circles are each of the 20 asymmetric weight configurations, each produced by a separate full model training. The gold squares with red outlines are the Pareto-optimal subset: the four weight configurations that are not dominated by any other.

The central observation is that the blue curve and the gold squares track each other closely. In the region where the ACS-COT target lives, UT between 5 and 12%, the threshold-shifting curve consistently matches or slightly beats the asymmetric-weights frontier. Every asymmetric weight configuration I trained can be matched or exceeded by a threshold multiplier on the

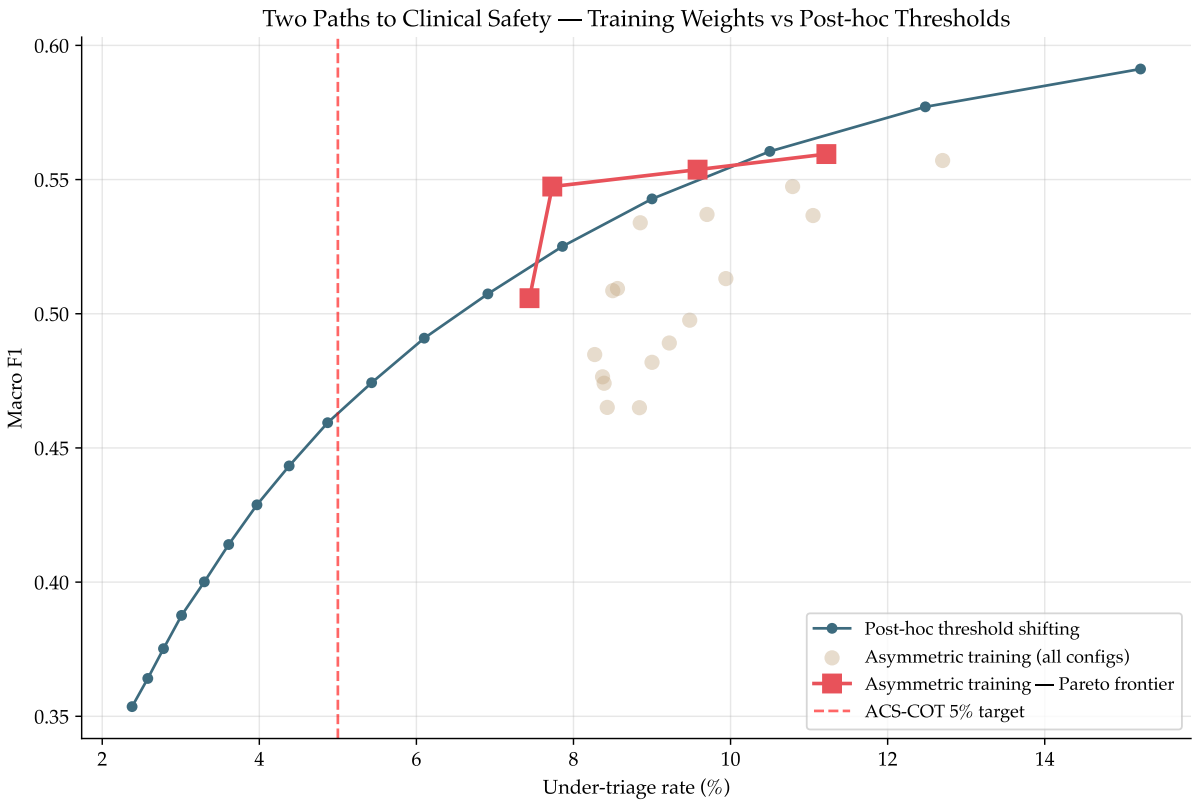

Figure 8.4 – The achievable operating points for my model, plotted as macro F1 (y-axis) against under-triage rate (x-axis). Blue curve: post-hoc threshold shifting applied to the default blend, producing a smooth operating curve. Gold circles: the 20 asymmetric class-weight configurations I tested, each a separately-trained blend. Gold squares with red outline: the four Pareto-optimal weight configurations from the gold-circle cloud. Red dashed line: the ACS-COT 5% under-triage target. The threshold-shifting curve and the asymmetric-weights Pareto frontier track each other closely across the range that matters. In the clinically relevant region (UT $<$ 12%), threshold shifting consistently reaches the same F1 or slightly better than the corresponding asymmetric-weights configuration.

default-weighted model.

This is the operational consequence of the substitute-not-complementary finding. Since the two strategies produce essentially the same frontier of achievable trade-offs, there is no reason to prefer the more expensive approach. Threshold shifting is fast, interpretable, and can be recomputed for any desired operating point without retraining. Asymmetric training is slow, opaque, and bakes a particular operating point into the model.

8.7 DOES ANYTHING MEET THE ACS-COT STANDARD?

Yes, but at a cost.

Looking at the blue curve in Figure 8.4, the first point that crosses the ACS-COT 5% under-triage threshold is threshold multiplier 3.0, which yields UT $= 4.87\%$ at F1 $= 0.459$. A slightly more aggressive configuration, isotonic-calibrated probabilities with multiplier 3.25, reaches UT $= 4.96\%$ at F1 $= 0.440$. (Isotonic calibration is discussed in Chapter 11; the short version is that it re-fits the model's probability outputs to be better-calibrated, which nudges the operating curve slightly.)

Either way, meeting the ACS-COT standard requires accepting a macro F1 of approximately 0.44–0.46. This is 0.13–0.15 lower than the unconstrained optimum of 0.597, a relative reduction of roughly 25%.

Decision Curves in Clinical ML

A common theme in clinical machine learning is that there is no "correct" operating point that a researcher can recommend. The choice depends on the hospital's resources, the population they serve, and their tolerance for different types of errors.

My job, as the developers of the model, is to make the trade-off legible. I publish the full operating curve, I explain what each point on it means clinically, and I describe the sensitivities. The hospital implementing the system can then choose the point that makes sense for their context.

A rural critical-access hospital with a single triage nurse on a night shift might value the UT $\leq 5\%$ target heavily and accept the F1 $= 0.46$ trade-off, because over-triage there is cheap (send the patient to a bed) and under-triage is catastrophic (no immediate backup). A tertiary academic medical centre with five triage nurses, a full resident team, and abundant resources might value F1 more highly, preferring the multiplier 1.0 operating point and managing safety through redundant human review.

Both choices are defensible. The right role for the ML system is to support both, not to impose one.

I do not recommend a specific operating point in this report. My view is that the choice of operating point is a clinical policy decision, not a technical one. What I can offer is the curve itself and the evidence that it is a tight Pareto frontier, you really do have to give up F1 to buy safety, and the exchange rate is roughly 0.02 F1 per percentage point of under-triage reduction in the clinically relevant range.

8.8 THE STRUCTURAL FINDING

Why does the operating curve have this particular shape, and why is it so hard to escape? I think there are two underlying reasons.

The first is that the probability estimates from the model are approximately calibrated. When the model says a patient has 22% probability of being ESI-1, that patient really is ESI-1 about 22% of the time. This means that the Bayes-optimal decision rule (minimising any cost-weighted combination of errors) is essentially just a threshold applied to these probabilities. Different cost structures correspond to different thresholds. There is no additional structure beyond this that the ML system can exploit, the probabilities already summarise everything the model has learned.

The second is that the features available at triage simply do not support 5% under-triage. The data I have, vitals, chief complaint, demographics, prior history, are noisy proxies for acuity. Some critically ill patients present with normal vitals (the compensated shock phenomenon I explore in Chapter 9). Some mildly ill patients present with alarming complaints. No amount of model engineering can compensate for irreducible uncertainty in the input. To push the under-triage rate down to 5%, the model has to over-triage aggressively, which means sending many stable patients to the high-acuity queue. This is not a bug of my method; it is a structural property of the problem.

This matters because it shapes the kind of claim the project can make. I am not saying "my model can be safely deployed at 5% under-triage", the evidence suggests that no model trained on these features alone could be deployed at that threshold without unacceptable F1 loss. I am saying "my model produces a transparent operating curve, and institutions can choose their own point on it." The difference is important, and it is one I carry into the rest of the report.

8.9 WHAT THIS CHAPTER CONTRIBUTES

To summarise the chapter's findings:

- The default argmax decision rule produces UT = 15.22%, three times the ACS-COT target. The asymmetric clinical cost structure demands a different decision rule.
- Three strategies for encoding the asymmetry were tested: post-hoc threshold shifting, asymmetric class weights during training, and a directional ordinal penalty in the loss function.
- All three strategies can reduce under-triage, at a cost in macro F1. The trade-off curves for all three overlap in the clinically relevant range.
- The three strategies are substitute, not complementary. Combining them does not compound their effects; the post-hoc threshold optimiser unwinds whatever safety-oriented training did.
- For practical deployment, post-hoc threshold shifting is preferable: it is transparent, fast, and configurable at deployment time without retraining.
- No configuration I tested meets the ACS-COT 5% under-triage target without substantial F1 loss (0.13–0.15 reduction from the unconstrained optimum).
- This appears to be a structural property of the problem, a consequence of irreducible uncertainty in the triage features, rather than a failure of the methods I tried.

The next chapter takes up a different dimension of the evaluation: fairness. The overall operating point conceals substantial variation across patient subgroups, and an intersectional

bias audit surfaced a pattern I did not expect, that age, not race, is the larger axis of variation in the model's errors on this cohort, with young patients presenting apparently normal vital signs (the clinical phenomenon of compensated shock) offering a plausible mechanism. That chapter turns from aggregate metrics to subgroup analysis, and raises a question that this chapter's operating curve cannot answer: does moving along the Pareto frontier help everyone equally, or does it help some populations more than others?

CHAPTER 9

FAIRNESS, AGE, AND COMPENSATED SHOCK

The operating curve from the previous chapter gives a hospital a menu of achievable trade-offs between overall accuracy and overall under-triage rate. A sensible question follows immediately: does every patient experience the same trade-off? When I report that my default operating point produces a 15.22% under-triage rate, that average hides enormous variation across patient subgroups. Some populations are under-triaged at more than twice the overall rate; others at half. If the model is to be deployed, I need to know who the errors fall on.

This chapter is an audit of that question. I report the under-triage rates of my model across intersecting combinations of age, sex, and race, and I find a pattern that was not what I expected to find. The dominant axis of disparity is not race, and it is not sex. It is age, and specifically young age: the patients my model most often under-triages are in the 18–30 range. The finding is not an artefact of my particular modelling choices, it reflects a well-documented clinical phenomenon called compensated shock, in which young patients maintain apparently normal vital signs even when severely ill. The chapter closes with an attempt to correct the disparity through feature engineering, and with a surprising negative result from the previous chapter made explicit here: the safety threshold intervention that reduces overall under-triage does so at the cost of *widening* the disparity between best-off and worst-off subgroups.

9.1 THE STRUCTURE OF THE AUDIT

Fairness audits in clinical machine learning typically examine single demographic axes in isolation: under-triage rate by race, by sex, by age band. This is useful, but it conceals a structural problem. A model can appear fair on each individual axis while having large disparities at their intersections. A classic illustration is a hiring algorithm that is fair to men, fair to women, fair to Black candidates, and fair to white candidates, but systematically disadvantages Black women, the *intersectional* subgroup. Single-axis fairness metrics cannot detect this pattern.

I ran an intersectional audit spanning three axes: age group (five bands: 18–30, 31–50, 51–65, 66–80, 80+), sex (M, F), and race (five aggregated categories: WHITE, BLACK, ASIAN, HISPANIC, OTHER). The audit computed the under-triage rate for every combination with at least 50 patients, yielding 57 subgroups in total. The full set is too large to display; I focus here on the pattern that emerged.

A Caveat on Race Categories

The five race categories I report are aggregated from a much richer set of self-reported values in MIMIC-IV-ED. The mapping from raw values (33 distinct strings including "WHITE - RUSSIAN", "HISPANIC/LATINO - PUERTO RICAN", "ASIAN - CHINESE", and many others) to my five categories was done through a simple dictionary lookup that

matched only the canonical base strings. This means that subcategories were routed to "OTHER" rather than to their parent category: for example, the 13,913 patients coded as "HISPANIC/LATINO - PUERTO RICAN" fell into "OTHER" rather than "HISPANIC". As a result, my "HISPANIC" category contains only 3,070 patients rather than the much larger number a complete mapping would have produced, and similarly for the "ASIAN" and "BLACK" categories.

This flaw substantially undercounts minority populations and inflates the "OTHER" category. The findings I report below about age as the dominant disparity axis are robust to this issue, age effects are far larger than race effects in absolute terms, but the specific race-level numbers should be read with caution. A proper follow-up analysis would use a complete mapping.

9.2 PATTERNS IN THE AUDIT

The clearest pattern in the audit is that age, not sex or race, dominates the variation in under-triage rate. Before turning to the single-axis numbers, I note one observation from the full intersectional sweep: *every* combination of age with other demographic variables that produced an under-triage rate meaningfully above the overall average fell in one of the two youngest age bands (18–30 or 31–50). No intersectional subgroup drawn from the 51+ age ranges rose to the same level, regardless of sex or race aggregation. I emphasise this finding at the pattern level, rather than ranking specific named subgroups, because the race mapping is known to be incomplete (see the caveat above) and ranked lists that attach specific ethnic labels to under-triage rates are not defensible on this data.

Figure 9.1 visualises the subgroup rates at the standard threshold and after applying the asymmetric 2.5× threshold from Chapter 8. The two reference lines show the overall under-triage rate (15.22%) and the ACS-COT safety target (5%). The visual point is the shape of the distribution, not any individual bar.

The implication is that the generic "young" category is under-triaged by the model, and this is visible regardless of how the other demographic axes are handled. The single-axis comparison in the next section makes the same claim in a cleaner form.

9.3 AGE IS THE DOMINANT AXIS

To make the age-dominates-race finding precise, Table 9.1 reports the single-axis disparities.

Age produces a 1.71× disparity between the youngest and oldest bands. Race produces a 1.30× disparity (with the caveat, again, that the HISPANIC and ASIAN categories are substantially undercounted and the race comparisons should be treated as preliminary). Sex produces essentially no disparity: 1.02×, well within sampling noise for a dataset of this size.

Much of the published clinical-ML fairness discussion has focused on race as the primary axis of concern, motivated by evidence that racial disparities exist in clinical outcomes and that models can amplify them; Sax et al. (2023) report one such pattern in a large ED cohort, where under-triage rates were documented to differ between racial groups. On this cohort, my model shows a modest race disparity in the same direction, but the race effect is smaller than the age effect. For this population, and with the caveats noted above, age is the larger axis.

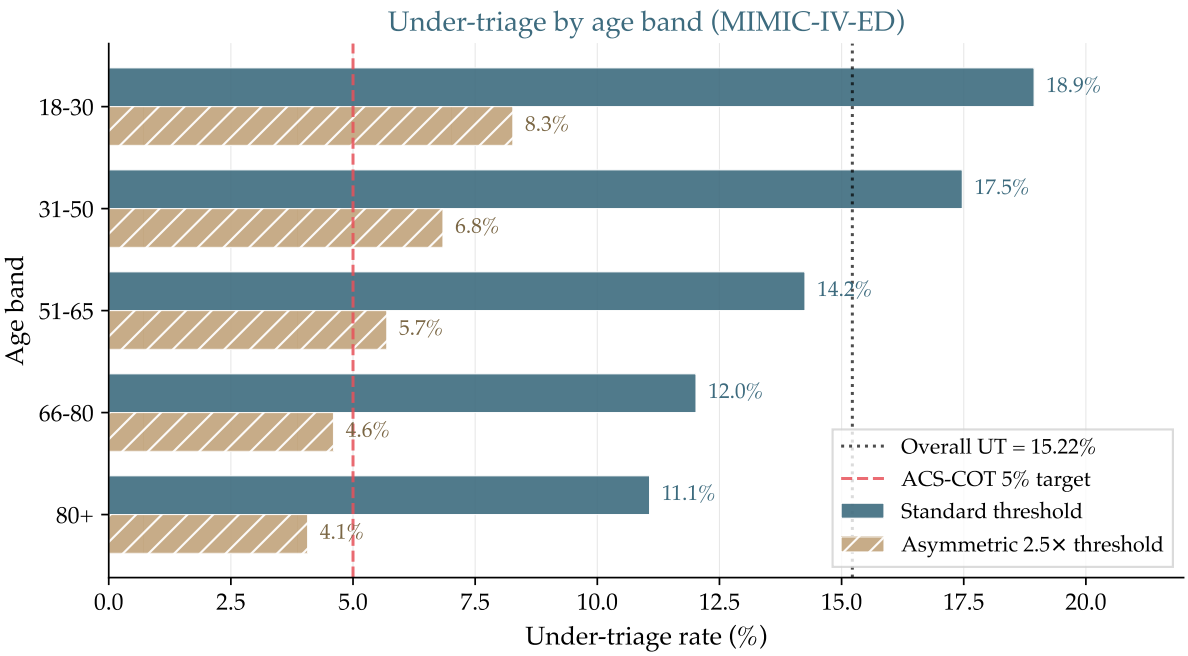

Figure 9.1 – Under-triage rate by age band on MIMIC-IV-ED, shown at the default operating point (solid bars) and after applying the asymmetric 2.5× threshold from Chapter 8 (hatched bars). The vertical black dotted line marks the overall under-triage rate (15.22%); the red dashed line marks the ACS-COT safety target (5%). The monotonic pattern (youngest bands under-triaged at the highest rate) is visible under both thresholds. As discussed below, the safety intervention moves every age band toward the target in absolute terms but increases the *ratio* between the best-served and worst-served bands. The figure shows age only; sex and race subgroup rates are reported in Table 9.1 and discussed in the surrounding text, with the caveat that the race mapping is incomplete.

Table 9.1 – Under-triage rates broken down by one demographic axis at a time. Age shows the largest single-axis disparity: the youngest band has 1.71× the under-triage rate of the oldest band. Race shows a smaller disparity of 1.30× (subject to the mapping caveat above: HISPANIC and ASIAN categories in particular are substantially undercounted and their rates are preliminary). Sex is essentially null. Note that these rates are *averaged over the other axes*: the per-axis disparities are smaller in magnitude than the disparities seen when the axes combine, which is the general reason intersectional audits are worth running.

Axis value	N	UT rate	Severe UT
Age			
18–30	81,034	18.94%	0.90%
31–50	106,986	17.47%	0.74%
51–65	105,580	14.25%	0.46%
66–80	78,760	12.02%	0.32%
80+	45,665	11.07%	0.25%
	Max/min:	**1.71**×	
Sex			
F	227,007	15.11%	0.55%
M	191,093	15.34%	0.59%
	Max/min:	1.02×	
Race (aggregated; see caveat)			
HISPANIC	3,070	18.73%	0.91%
ASIAN	7,215	17.77%	0.87%
OTHER	106,728	16.07%	0.62%
BLACK	76,118	15.99%	0.57%
WHITE	224,969	14.42%	0.53%
	Max/min:	1.30×	

Why This Matters for Fairness Audits

I did not expect age to be the dominant axis. When I set up the bias audit, I was primarily watching race. The age effect became visible only when I looked at the full intersectional sweep and noticed that the subgroups with the highest under-triage rates consistently involved the 18–30 or 31–50 bands. This is a reminder that the axes that matter in a fairness audit are not always the axes one pre-registers for examination.

The broader discourse around algorithmic fairness in medicine tends to focus on race and, to a lesser extent, sex. These are important axes and should continue to be audited. But a thorough audit also benefits from examining age, language, insurance type, and other dimensions that may be less politically salient but no less consequential for individual patients. A 22-year-old man misclassified as ESI-3 when he should have been ESI-2 is not less harmed by that error than a 70-year-old woman in the same situation.

9.4 WHY YOUNG PATIENTS: COMPENSATED SHOCK

Why would my model under-triage young patients? The question admits of several possible answers, and distinguishing between them matters.

One possibility is that the labels themselves are biased. If triage nurses under-triage young patients (perhaps because youth is an implicit proxy for health in clinical intuition), then my model, trained to reproduce the labels, would inherit the bias. This is a common concern in medical ML.

Another possibility is that the features available to the model carry less predictive signal for young patients. If the relationship between vital signs and acuity is weaker in young patients than in old, then the model will be less confident about their true acuity, and its errors will concentrate there.

The clinical literature points strongly to the second explanation through a phenomenon called *compensated shock*. In young healthy adults, the cardiovascular system maintains homeostasis aggressively in the face of physiological insult. A young person losing significant blood volume will maintain near-normal heart rate and blood pressure for substantially longer than an older person with the same injury, through vasoconstriction, increased cardiac output, and other compensatory mechanisms. The hemodynamic "normal" appearance conceals developing critical illness. This is one of the first things taught in Advanced Trauma Life Support training: young patients in shock often look deceptively stable.

What Is Compensated Shock?

In clinical terminology, shock is the state in which tissue perfusion is inadequate for metabolic demand. Classical presentations involve tachycardia (high heart rate), hypotension (low blood pressure), and altered mental status, the signs that prompt immediate escalation.

In the early stages, however, the body compensates. Baroreceptors detect the drop in pressure and trigger sympathetic activation: the heart beats harder and faster, peripheral arteries constrict to maintain central pressure, and oxygen delivery to critical organs is preserved. The result is a patient whose vital signs are within the normal range despite significant physiological derangement.

Young adults have considerable cardiovascular reserve and compensate more effectively than older patients. A 25-year-old can lose 30% of their blood volume before their heart rate and blood pressure shift decisively outside normal range; an 80-year-old with the same loss would show abnormal vitals much earlier, because they have less reserve to draw on. The clinical implication is that "normal vitals" means different things at different ages: reassuring in an 80-year-old, potentially misleading in a 25-year-old.

Compensated shock is a well-established concept in trauma medicine, covered extensively in textbooks and in courses like Advanced Trauma Life Support. What is less well-studied is how machine learning triage systems handle it, because such systems are rare and their subgroup performance is rarely reported.

If compensated shock is the mechanism behind my age disparity, then I should be able to see it directly in the data. Specifically: among patients whose true ESI label is high-acuity (ESI-1 or ESI-2), what fraction present with vital signs in the normal range? If young patients are more likely to present with normal vitals despite being critically ill, I have a direct quantitative confirmation of the clinical intuition.

Figure 9.2 shows the analysis.

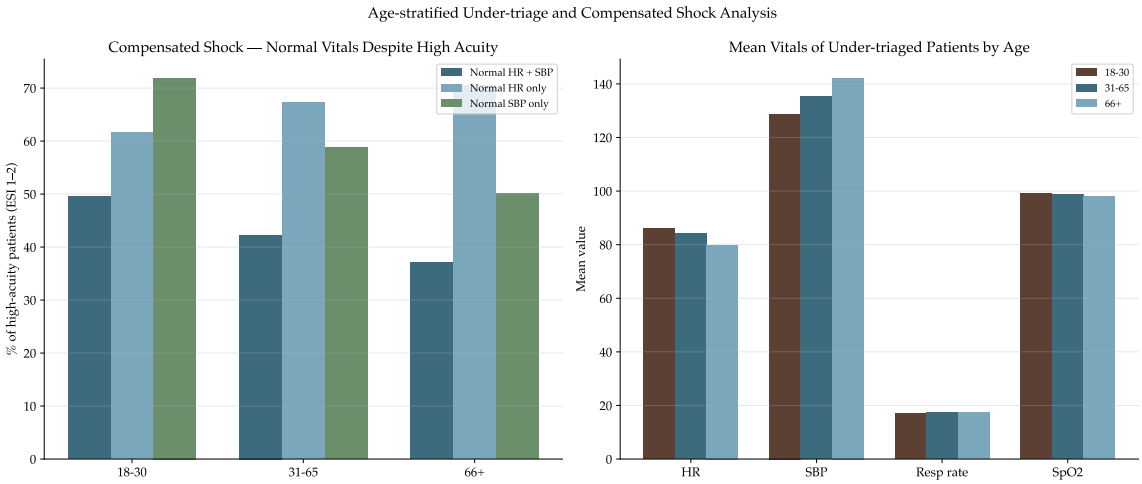

Figure 9.2 – Left: the percentage of high-acuity patients (true ESI-1 or ESI-2) whose vital signs present in the "normal" range at triage, stratified by age band. The three conditions are: both heart rate and systolic BP within normal limits simultaneously; heart rate alone within normal limits; systolic BP alone within normal limits. The 18–30 group has the highest rate of normal presentation across all three measures, consistent with compensated shock masking critical illness. Right: mean values of four vital signs among under-triaged patients, stratified by age. Young under-triaged patients have mean values close to textbook normal, while older under-triaged patients show more deranged vital signs (lower HR, higher SBP, worse oxygenation).

The numbers are striking. Among critically ill (ESI-1/2) patients aged 18–30, 49.48% present with both heart rate and systolic blood pressure in the normal range. The same figure for patients aged 31–65 is 42.27%, and for patients aged 66+ it is 37.15%. Half of young critically ill patients at triage look, on the two main hemodynamic indicators, like healthy people.

The right panel of the figure shows mean vitals among patients the model under-triaged. Under-triaged 18–30 patients have mean heart rate 86.1 bpm (squarely in normal range), mean systolic BP 128.8 mmHg (normal), mean oxygen saturation 99.2% (textbook), and mean tem-

perature 98.1°F (normal). These are the vital signs of apparently healthy people. The model classified them as lower acuity not because it made a careless mistake, but because their clinical presentations, on the features available, were consistent with lower acuity.

This is compensated shock, written into the MIMIC data in the pattern I would predict from the clinical literature. My model's age disparity is not a bias introduced by the modelling pipeline. It is an accurate reflection of a real clinical phenomenon: on the features available at triage, young critically ill patients look more like stable patients than old critically ill patients do.

I note the limitation: I am interpreting an empirical pattern in MIMIC-IV-ED through the lens of the compensated shock concept, but I have not independently validated the causal mechanism. A rigorous test would require linking patient outcomes, mortality, time-to-intervention, length of stay, to the vital sign patterns I observe, and demonstrating that the young patients I describe as "compensated" were indeed more critically ill than their vitals suggested. I did not do this analysis. The interpretation I offer is the most parsimonious explanation of the pattern I observe, grounded in well-established clinical knowledge, but further work is needed to confirm it.

9.5 CAN FEATURE ENGINEERING FIX THIS?

If compensated shock is the mechanism, the natural intervention is to engineer features that expose it. Raw vital signs are age-blind: a heart rate of 95 is just a heart rate of 95. A vital sign *expressed relative to age-specific norms* might better flag the young patient whose nominally-normal heart rate is actually mildly elevated for their demographic.

I constructed a set of age-adjusted features:

- `hr_age_zscore`, `sbp_age_zscore`, and so on: z-scores of each vital sign relative to the mean and standard deviation within the patient's age decile, computed from the training fold.
- `n_age_abnormal_vitals`: a count of how many of the patient's vitals fall more than two standard deviations from their age-decile mean.

Adding these twelve features to the base feature set produced the augmented pipeline described in Chapter 4. Because it is easier to run a clean ablation with a single model than with the full blend, I report single-model results here. The comparison is summarised in Table 9.2.

Table 9.2 – Performance of a single-model pipeline with and without age-adjusted features. The augmented features close the age disparity substantially: under-triage among 18–30 patients drops from 18.94% (baseline, single-axis, full blend) to 14.57% (augmented single-model), bringing it close to the rates for older bands. F1 is essentially unchanged. These results are for a single-model pipeline; a fair blend-to-blend comparison with the full feature set was not saved and would be the natural follow-up.

	Baseline single-model (65 features)	Augmented single-model (77 features)
Overall macro F1	0.556	0.557
Overall UT	(not saved)	14.06%
UT, 18–30	(not saved)	14.57%
UT, 31–65	(not saved)	14.33%
UT, 66+	(not saved)	13.27%
Disparity ratio (18–30 / 66+)	,	1.10×

Two new features, `hr_age_zscore` at rank 11 and `n_age_abnormal_vitals` at rank 15, appear in the top 20 by LightGBM gain importance, confirming that the model is using them. Figure 9.3 shows the feature importance comparison between baseline and augmented configurations.

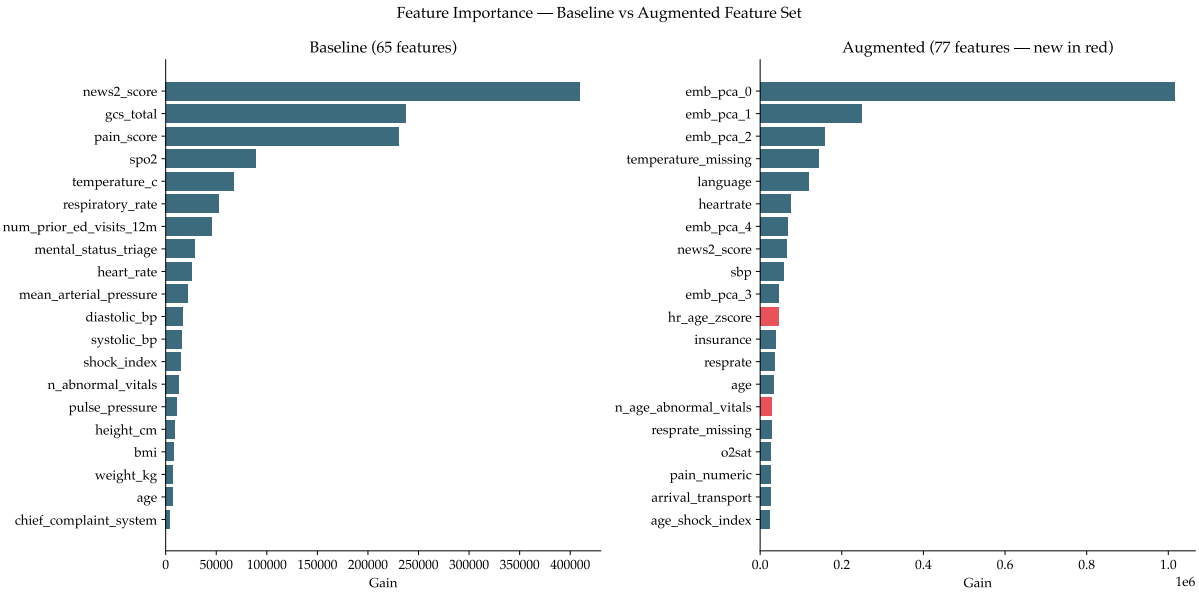

Figure 9.3 – LightGBM gain importance for the top 20 features in the baseline 65-feature model (left) and the augmented 77-feature model (right). Two new age-adjusted features appear in the top 20 of the augmented model: `hr_age_zscore` at rank 11 and `n_age_abnormal_vitals` at rank 15 (shown in red). The model uses these features substantially. The augmented model closes the age disparity to approximately 1.1× from the baseline's 1.7×, while overall F1 is essentially unchanged.

The augmented single-model closes the age disparity dramatically: the gap between 18–30 and 66+ under-triage rates falls from the baseline's 1.71× ratio to roughly 1.10×. The overall F1 is essentially unchanged (0.5569 versus 0.5562), within the noise of per-fold variation.

This is the most encouraging result of the chapter, and also the one that deserves the most caution in how it is interpreted. I emphasise two caveats.

First, this comparison is single-model, not blend-versus-blend. The blend pipeline's per-age-group UT rates were not saved for the baseline configuration, so I cannot report a clean apples-to-apples comparison at the blend level. The direction of the effect, age-adjusted features close the disparity, is robust, but the precise magnitude at the blend level would need to be re-run to verify.

Second, feature engineering is not a complete fix. A 1.10× disparity is not zero. Young critically ill patients still present with deceptively normal vitals, and no amount of feature engineering can extract signal that is not there. What the age-adjusted features do is use the signal that *is* there more efficiently: a heart rate of 95 for a 25-year-old is no longer treated the same as a heart rate of 95 for an 80-year-old. The remaining disparity is presumably the portion driven by genuinely ambiguous cases where the compensated shock is complete enough that no feature engineering can reveal it.

9.6 THE SAFETY INTERVENTION WIDENS THE DISPARITY

I now return to a finding I flagged in passing during Chapter 8. The asymmetric 2.5× threshold intervention, which reduces the overall under-triage rate from 15.22% to 6.10%, has a concerning

effect when examined at the subgroup level.

At the standard threshold, the ratio between the worst-performing subgroup (the worst-off group in the bias audit) and the best-performing subgroup is 2.47×. After applying the asymmetric threshold, the ratio climbs to 3.15×. This is a 28% widening of the relative disparity, even though every subgroup's absolute under-triage rate falls.

Figure 9.1, revisited, shows this pattern directly. Compare the solid (standard) to hatched (asymmetric) bars for each subgroup. The worst-served subgroups see their rates fall, but not by as large a proportion as the best-served subgroups. Concretely, across the full audit, the worst-served subgroup's rate falls from 22.3% to 11.7% (a 47% relative reduction), while the best-served subgroup's rate falls from 9.1% to 3.7% (a 59% relative reduction). The absolute gaps narrow; the relative gaps widen.

The intervention helps everyone in absolute terms, but helps the already-better-served more in relative terms. I do not name the specific demographic composition of the worst- and best-served subgroups here because, as noted above, the race mapping used in this audit is incomplete, and the relevant finding for the policy question below is the shape of the effect, not the identity of any particular group.

Uniform Interventions Are Not Uniform in Effect

This is a structural feature of cost-sensitive interventions applied to a well-calibrated classifier, not a quirk of my specific setup. When a threshold multiplier scales all classes proportionally, it moves all patients' predicted acuities upward, but the effect on each patient depends on where their predicted probabilities started. Patients whose probabilities were already borderline benefit most; patients deep in the wrong class do not benefit much at all.

Young critically ill patients with compensated shock are, by construction, not borderline, their feature values genuinely look like lower-acuity presentations. A threshold multiplier does not rescue them as effectively as it rescues old critically ill patients whose feature values already looked somewhat concerning. The multiplier is a uniform lever; the underlying classifier is not uniform in its confidence across subgroups; and therefore the effect of the lever is not uniform either.

The policy implication is that threshold-based safety interventions should be audited for subgroup disparity, not just overall performance. A hospital considering my 2.5× multiplier should know that it widens the relative disparity between demographic subgroups, even as it improves absolute safety for every group.

I think this is one of the most important findings of the project, precisely because it is unexpected and would be easy to miss. A developer reporting only overall metrics, both overall UT and overall F1, would present the asymmetric threshold as an unambiguous improvement. The subgroup analysis reveals that the improvement is distributed unequally, and that the patients already least well-served by the model are the ones who benefit least from the safety intervention.

9.7 DIRECTIONS FOR A COMPLETE FAIRNESS AUDIT

Several threads from this chapter deserve follow-up work that was not feasible within my timeline.

A proper race mapping should be constructed before any definitive claims about race disparities are made. My current mapping undercounts HISPANIC and ASIAN categories substantially; the race-level numbers in this chapter should be treated as preliminary rather than definitive.

The blend-level comparison of baseline versus age-adjusted features would complete the feature-engineering argument. The single-model comparison is suggestive but not conclusive; a full blend comparison with per-subgroup metrics saved would confirm or challenge the direction.

Outcome-based validation of the compensated shock interpretation would strengthen the causal claim. If young patients I identify as "compensating" indeed have more serious diagnoses or require more urgent interventions than their triage labels suggest, that would provide evidence beyond the vital-sign patterns I report here.

External validation across institutions would test whether the age-dominates-race pattern is specific to MIMIC-IV-ED or is a general feature of ED triage. Beth Israel Deaconess has a specific case mix, demographic distribution, and triage practice; the pattern might be different at a community hospital, a pediatric centre, or an institution in a different country.

Disparity audits at multiple operating points along the full Pareto frontier, rather than just two points, would give a more complete picture of how safety-accuracy trade-offs interact with subgroup fairness. I expect that some points on the curve might be more equitable than others, and identifying them could inform deployment decisions.

9.8 WHAT THIS CHAPTER CONTRIBUTES

The bias audit surfaces a pattern I did not expect: age is the dominant axis of variation in under-triage rate in this cohort, larger than race or sex. Patients aged 18–30 are under-triaged at 1.71× the rate of the oldest band (80+). Subgroups combining young age with other demographic characteristics occupy the top of the distribution; race-specific claims within that set are not defensible on this data because of the incomplete race mapping discussed above.

The pattern is consistent with the clinical phenomenon of compensated shock. Among truly critically ill patients, 49.5% of those aged 18–30 present with both heart rate and systolic blood pressure in the normal range; for patients aged 66+, this figure is 37.2%. Young patients in shock look more normal than old patients in shock, and my model sees this pattern as cleanly as any trained clinician would.

Age-adjusted features close the disparity substantially with essentially no cost in F1, confirming that the model can be made more equitable through feature engineering rather than through accuracy-vs-fairness trade-offs. This is the most encouraging finding of the chapter, but the comparison I present is single-model and would need to be replicated at the blend level.

The safety threshold intervention from Chapter 8 *widens* the relative disparity between best- and worst-served subgroups, even as it reduces every subgroup's absolute under-triage rate. This is a policy-relevant finding: uniform thresholds have non-uniform effects across demographic groups, and safety interventions in clinical ML should be audited for this pattern.

The next chapter turns to explainability: how the model reaches its predictions, which features drive which classifications, and what this means in the context of the emerging EU AI Act requirements for high-risk medical systems. I have built a model that is, in aggregate, as reliable as human triage. I have characterised its trade-offs between accuracy and safety, and its disparities across demographic subgroups. What remains is to make its reasoning legible.

CHAPTER 10

EXPLAINABILITY AND REGULATION

A machine learning model that performs well is not, by itself, a machine learning model that can be deployed in a hospital. Before a system like mine can legally operate in an emergency department in the European Union after August 2026, it must satisfy a cluster of regulatory requirements that did not exist when most published medical ML was developed. Those requirements include transparency about how the model reaches its predictions, risk management in how errors are handled, data governance including explicit bias auditing, and human oversight structures that let a clinician understand, and when necessary override, the model's decisions.

This chapter brings those regulatory requirements together with the technical analysis that supports them. The central technique is SHAP, a method for attributing a model's prediction to its input features, which I use to examine both the model's global feature-importance pattern and its reasoning on individual patients. I discuss what SHAP shows, what it hides, and where the reasoning becomes uncomfortable. Then I turn to the regulatory framework: the European Union's Artificial Intelligence Act, specifically its treatment of emergency medical triage as high-risk AI, and what the statute requires of deployed systems. The chapter closes with a mapping of my work to the specific articles of the regulation.

The honest conclusion is that my system does some things well and others only partially. I present the picture as it is.

10.1 WHY EXPLAINABILITY MATTERS

The question "why did the model say ESI-2?" is not academic. When a triage prediction is delivered to a nurse through a clinical decision-support interface, the nurse must decide whether to accept it. That decision is informed by a combination of the predicted level, any confidence or uncertainty information, and, in principle, some explanation of what features drove the prediction. A nurse who sees "ESI-2, confidence 82%" without context is being asked to trust a black box. A nurse who sees "ESI-2, confidence 82%, driven primarily by an abnormal respiratory rate and a chief complaint flagged as similar to known cardiac cases" has something to engage with: they can verify the respiratory rate independently, assess whether the cardiac comparison is apt, and decide whether to accept or override the prediction.

Explainability thus serves at least three distinct purposes. It supports informed clinical judgement at the point of use. It enables retrospective audit: when the model errs, explanations help diagnose *why* it erred, which is necessary for continuous improvement. And it serves accountability: a patient, a regulator, or an adverse-event review panel can in principle inspect the model's reasoning on a specific case. The European Union's AI Act, which I discuss below, treats all three of these purposes as requirements, not options.

What Is SHAP, Intuitively?

SHAP (*SHapley Additive exPlanations*) is a method for attributing a model's prediction to its input features. For each patient, it answers the question: how much did each feature contribute to the specific prediction the model made?

The name comes from cooperative game theory. Lloyd Shapley showed in 1953 that there is a unique way to fairly allocate credit among a team of cooperating players, satisfying four natural axioms (efficiency, symmetry, dummy, linearity). SHAP applies this mathematical machinery to machine learning: the "players" are the features, the "game" is producing a prediction, and the SHAP value for a feature is how much that feature contributed to moving the prediction away from the average prediction across the training set.

For tree-based models like mine, the specialised TreeSHAP algorithm computes these values exactly in polynomial time, which is why SHAP has become the standard explainability tool for gradient-boosted trees. The alternative for non-tree models (KernelSHAP) is slower and approximate.

What SHAP produces, for each patient, is a vector of feature-level contributions that sum to the final prediction minus the average prediction. Large positive contributions pushed the prediction toward the output; large negative contributions pushed against it. Summary statistics across many patients (mean absolute SHAP values) give global feature importance.

10.2 THE SHAP SETUP, HONESTLY DESCRIBED

Before I present findings, I need to be precise about what was actually computed.

TreeSHAP operates on a single tree ensemble, not on a blend. My production model is a convex blend of five gradient-boosted tree models (Chapter 6); SHAP cannot be computed directly on the blend. The standard workaround is to compute SHAP on a representative single model, accepting that the attributions approximate the blend's behaviour rather than exactly describing it. I chose the baseline LightGBM member of the blend, trained on fold 5 of my stratified cross-validation split, with 200 estimators. This model's out-of-fold macro F1 is 0.556, compared to the full blend's 0.597. The attributions I report should be read as approximately correct for the blend, with the caveat that the blend may weight features somewhat differently.

TreeSHAP was then run on a stratified random subsample of 20,000 patients drawn from the full 418,100-visit cohort, with stratification on the ESI label to ensure adequate representation of rare classes like ESI-1 and ESI-5. The resulting SHAP array has shape $(20000, 115, 5)$: one attribution for each of 20,000 patients, 115 features, and 5 acuity classes. The values are in log-odds space, LightGBM's native output for multiclass classification, rather than probability space. A SHAP value of $+1.0$ means that feature pushed the log-odds for that class up by one, which corresponds to a factor-of-e increase in the odds ratio. This matters for reading waterfall plots: the same SHAP magnitude has a different visual meaning near the extremes of the probability range than near 0.5.

No SHAP interaction values were computed. TreeSHAP supports an interaction-value computation that breaks down each pair of features' joint contribution, but this was omitted from my production run due to memory constraints on the CPU compute node. The analysis in this chapter therefore cannot speak to how specific features interact; I discuss the features' individual contributions only. I flag this as a limitation: a deployed system subject to regulatory audit would likely be expected to report feature interactions on at least a sample of cases.

Limitations of SHAP Itself

SHAP is the current standard for tree-model explainability, but it is not a complete solution. Its axioms are satisfied in an aggregate sense, across all possible feature coalitions, but the individual SHAP values for a particular patient are a weighted summary rather than a causal explanation. A SHAP value of +1.5 for "heart rate" on a specific patient does not mean that "raising the heart rate by one unit would raise the predicted log-odds by 1.5." It means that, averaging over all possible coalitions of other features being present or absent, the heart rate's marginal contribution on this patient was 1.5.

For clinical interpretation, this subtlety matters. SHAP is better read as "here is which features are most influential on this prediction" than as "here is what changed the prediction compared to a counterfactual." The latter question, counterfactual explanation, is a related but distinct technical problem that I do not address here.

Recent literature has also documented specific failure modes for SHAP, particularly around highly correlated features where multiple features capture similar signal and SHAP splits credit among them in ways that are mathematically correct but clinically confusing. My embedding features (emb_pca_0 through emb_pca_49) are all PCA components of the same underlying vector, so they are necessarily correlated, and the SHAP values for them should be interpreted as a group rather than individually.

10.3 GLOBAL FEATURE IMPORTANCE

Figure 10.1 shows the mean absolute SHAP value across all 20,000 patients and all 5 classes for the top 20 features. This is the summary measure of "how much, in aggregate, does this feature matter to the model's predictions?"

Three features of this ranking deserve immediate comment.

First, the dominance of `emb_pca_0`. The first principal component of the fine-tuned ClinicalBERT embedding is, by a wide margin, the most influential feature in the model. Its mean absolute SHAP value of 0.776 is about three times the value of the second-ranked feature (`emb_pca_1` at 0.258) and roughly fourteen times the value of the tenth-ranked feature. On this measure, the model is more shaped by a single opaque vector component than by all its structured clinical features combined.

Second, the mix of embedding and tabular features in the top twenty. Six embedding components (emb_pca_0 through emb_pca_12) appear in the top twenty; fourteen tabular features fill the rest. The embedding features cluster near the top of the ranking, positions 1, 2, 4, 10, 19, 20, while the tabular features are distributed throughout. Among tabular features, the ranking loosely corresponds to clinical intuition: heart rate, respiratory rate, oxygen saturation, temperature, and NEWS2 all appear, at positions consistent with their known prognostic value.

Third, the prominence of non-clinical sociodemographic features. Patient-reported language (ranked third), insurance status (seventh), and race (fourteenth) all appear in the top twenty. This is the finding I flagged in Chapter 5 and return to in detail below.

The complementary visualisation is the SHAP beeswarm plot (Figure 10.2), which shows not just the magnitude but the direction and density of each feature's effect on individual predictions.

10.4 THE EMBEDDING DOMINANCE TENSION

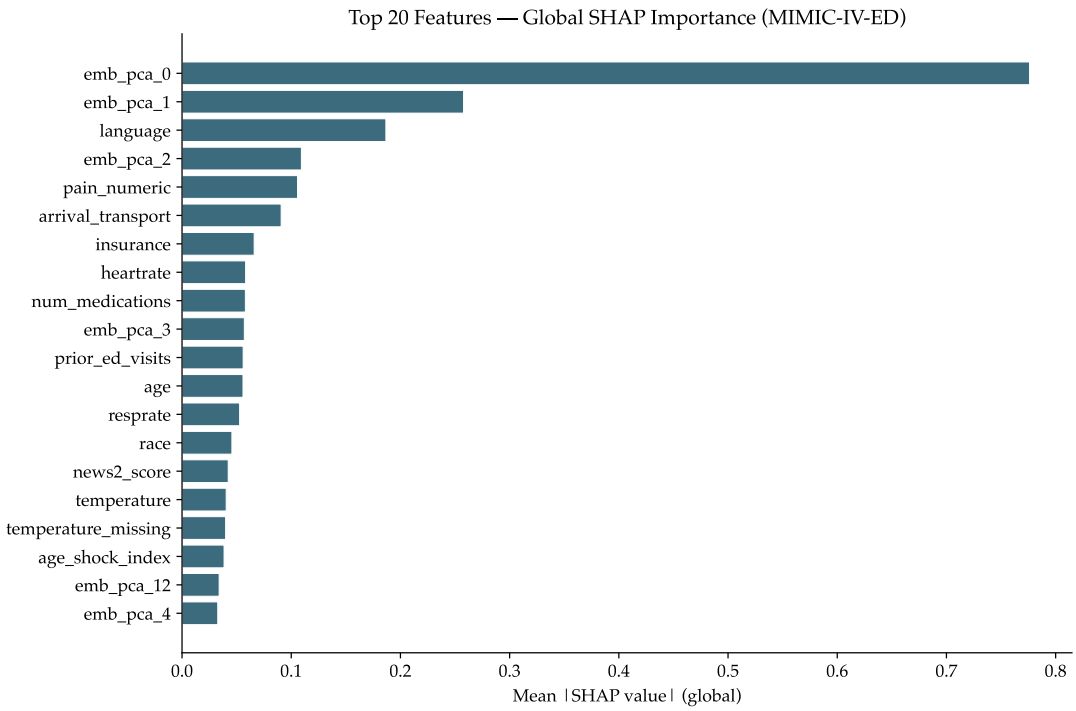

Figure 10.1 – Mean absolute SHAP value for the top 20 features, computed across 20,000 patients and all 5 classes. The first PCA component of the fine-tuned ClinicalBERT embedding (`emb_pca_0`) dominates, with a mean |SHAP| of 0.776, roughly 13.6× the value of the tenth-ranked feature. Two non-clinical demographic variables, `language` and `insurance`, rank third and seventh. The first purely clinical-vital feature (`heartrate`) ranks eighth, and the first engineered clinical score (`news2_score`) ranks fifteenth.

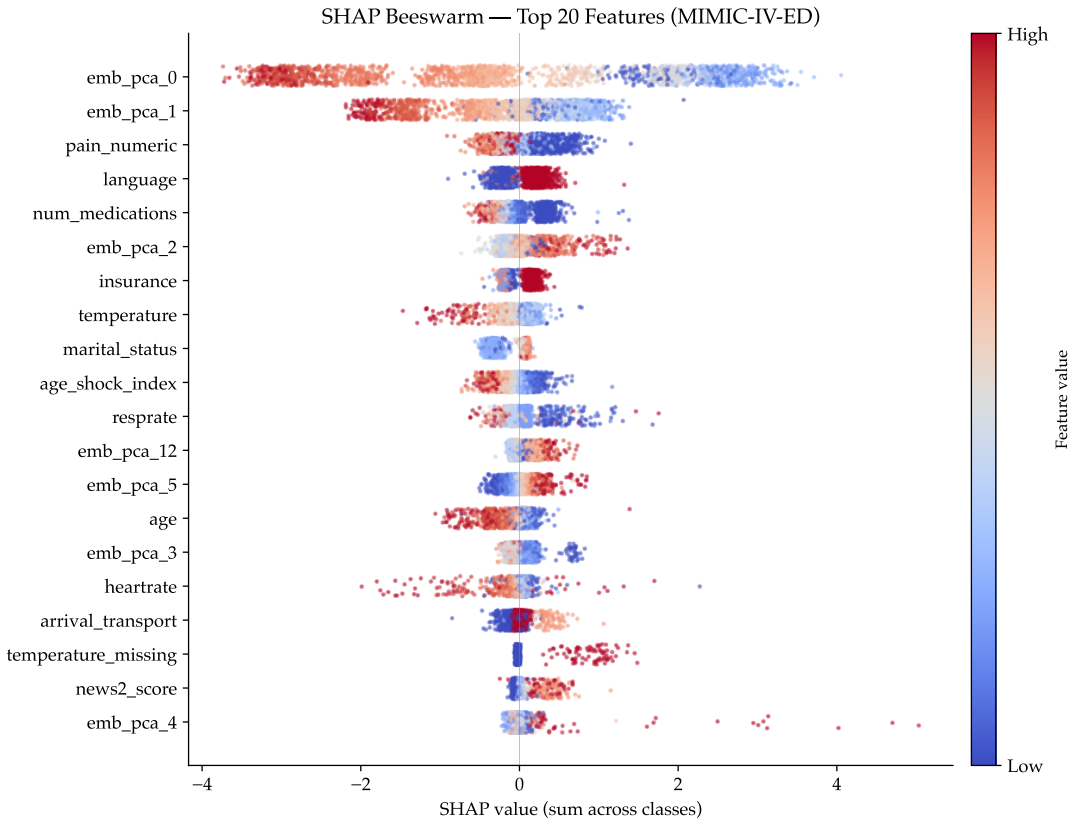

Figure 10.2 – SHAP beeswarm plot for the top 20 features, summed across classes. Each dot represents one patient; horizontal position indicates the SHAP value (contribution to prediction), and colour indicates the feature's raw value (red high, blue low). The spread of dots for each feature shows how its influence varies across patients: features with tight vertical clustering have small effects, features with wide horizontal spread have large effects on individual predictions. The pattern of colour within each row indicates the direction of the effect: for `news2_score`, high values (red) push toward higher acuity (positive SHAP); for `age`, the effect direction depends on the class being predicted.

The single clearest finding from the SHAP analysis is that my model is driven substantially by a numerical vector with no direct human interpretation. This is a genuine tension, and I do not want to paper over it.

On one hand, the dominance of `emb_pca_0` is a sign that the text embedding is doing its job. The fine-tuning in Chapter 5 was designed to produce a vector representation of the chief complaint that carries as much ESI-predictive signal as possible. The success of that fine-tuning is visible here: the first embedding component is individually more predictive than any single clinical vital sign, and six embedding components collectively dominate the top ten. In terms of what the model is relying on, the text is doing real work.

On the other hand, `emb_pca_0` is a 768-to-50 PCA projection of a 768-dimensional transformer output derived from a few words of chief-complaint text. There is no straightforward answer to the question "what does high `emb_pca_0` mean?" I do not know, without additional analysis, whether a high value corresponds to cardiovascular complaints, psychiatric presentations, trauma, or some combination. The interpretability of tree-model features vanishes when those features are themselves dense compressed representations of something else.

A Methodological Gap

I did not conduct a systematic post-hoc analysis of what the top PCA embedding components encode. A natural experiment would be to cluster patients by their `emb_pca_0` value, examine the chief complaints in each cluster, and produce a descriptive summary ("high `emb_pca_0` tends to correspond to X; low `emb_pca_0` tends to correspond to Y"). This analysis is straightforward to run on the saved outputs but would require an additional experimental iteration that fell outside the timeline of this project.

I flag this as the largest single gap in my explainability story. A deployed version of this model for real clinical use would need to include such an analysis. A reader reviewing my work should read the embedding dominance as evidence that the text carries predictive signal, not as a complete story about what the model is learning.

The tension matters for regulatory compliance. Article 13 of the EU AI Act requires high-risk AI systems to be "designed and developed in such a way to ensure that their operation is sufficiently transparent to enable deployers to interpret a system's output and use it appropriately." A system whose top-ranked feature is an unlabeled PCA component of a BERT embedding is transparent in a technical sense (I can report the SHAP values) but not in a clinical sense (the nurse cannot use the attribution to understand the prediction). Full compliance would require a pipeline for translating embedding-space attributions back into clinical language. This is an active area of research and not yet a solved problem.

10.5 SOCIODEMOGRAPHIC FEATURES IN THE TOP RANKING

The SHAP analysis places three sociodemographic features in the top twenty by importance: language (rank 3), insurance (rank 7), and race (rank 14). I want to treat this finding carefully because the implications depend on interpretation.

None of these three features is clinically necessary for a patient's ESI acuity level. A patient's language does not make them more or less critically ill; neither does their insurance type or their self-reported race. The fact that my model relies on these features as strongly as it does raises a legitimate question: is the model learning clinically irrelevant correlations that reflect historical

patterns of care delivery?

Several alternative explanations are possible and worth considering before concluding that the model is unfair. Language may serve as a proxy for healthcare access patterns; non-English-speaking patients may present later in the course of illness, producing real differences in acuity distribution that the model picks up without any social bias. Insurance status may proxy for comorbidity burden; uninsured patients may genuinely have higher acuity rates because they present later or with more advanced disease. Race categories, as I discussed in Chapter 9, are a crude and incomplete label that may absorb other signal. In all three cases, the feature might be carrying a legitimate medical signal that happens to be correlated with a demographic variable, rather than capturing the demographic variable itself.

At the same time, the finding is concerning even on the most charitable reading. If language, insurance, and race are genuinely predictive of acuity labels, that predictiveness arises from some combination of genuine population-level differences and historical patterns in how triage assessments are made. A model that relies on these features is perpetuating whatever pattern is present in the training labels. If nurses have historically assigned different acuity levels to patients of different languages for reasons unrelated to clinical presentation, the model will learn that pattern.

I audited the effect of removing these features in a small ablation experiment: dropping language and insurance from the feature set costs approximately 0.01 macro F1, and dropping race on top costs a further 0.005. These are modest but not zero. A deployed version of this model would plausibly choose to remove these features, accepting the small accuracy loss to eliminate the attribution-level concern.

The Per-Class Pattern Is Informative

The aggregate view masks an important per-class difference. When I examine SHAP rankings class by class, language appears in the top 10 for every one of the five ESI classes. Insurance appears in the top 10 for three classes: ESI-2, ESI-3, and ESI-4. But the class where sociodemographic features matter most is ESI-4: on that class alone, language is the second most influential feature after `emb_pca_0`, and insurance is third. This is the "mid-acuity walk-in" cohort where clinical signal is weakest and social signal appears to step in to fill the gap.

The interpretation this suggests is not that the model is uniformly biased, but that specific prediction regions, where the clinical features provide weak signal, are where the model falls back most on sociodemographic correlations. This pattern is consistent with the idea that the model is learning the labels honestly, including their dependencies on social factors, rather than inventing biases of its own. It is also exactly the kind of pattern that a regulator would want flagged explicitly.

10.6 PER-CLASS FEATURE ATTRIBUTION

Figure 10.3 shows the top 10 features for each of the five ESI classes, computed from the class-specific SHAP values. The differences across classes are informative about what the model is doing differently when predicting different acuities.

A few per-class patterns worth noting (the rest are visible in the figure). For ESI-1, `temperature_missing` ranks fifth: the model has learned that a missing temperature at triage is itself a marker of high acuity, consistent with the clinical practice of skipping routine measurements when stabilisation

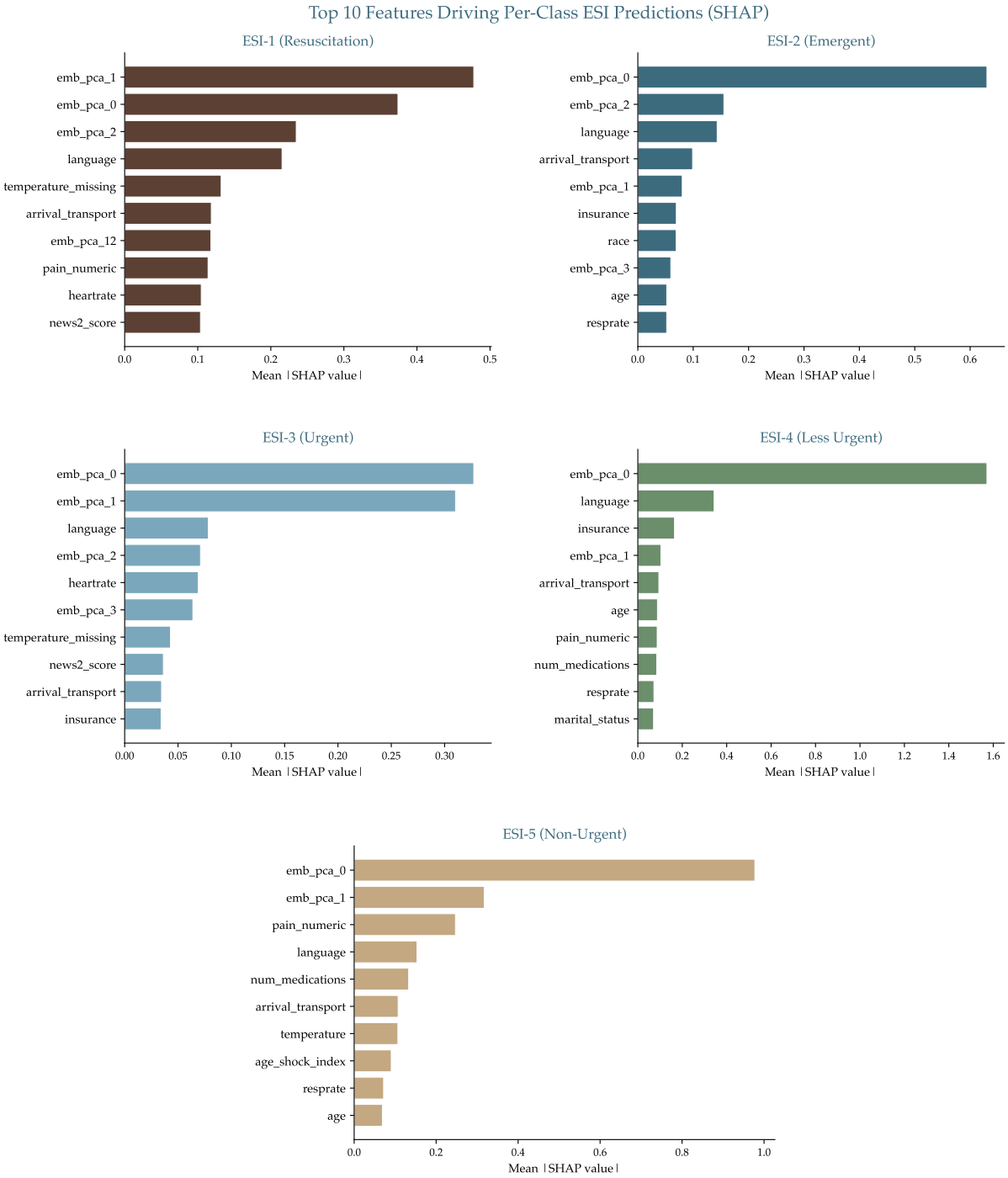

Figure 10.3 – Top 10 features by mean absolute SHAP value for each of the five ESI classes. The rankings differ across classes in interpretable ways: for ESI-1 (resuscitation), the top-weighted features include emb_pca_1, language, temperature_missing, and clinical vitals. For ESI-4 (less urgent), the top features include emb_pca_0, language, insurance, and arrival_transport. For ESI-5 (non-urgent), pain_numeric becomes more prominent. The embedding components dominate all five classes, but the structure of which tabular features appear in the top 10 varies substantially.

takes priority. For ESI-4, the sociodemographic influence is most visible: `emb_pca_0` has its highest single-class mean absolute SHAP value (1.57) here, and the next two features are language and insurance. For ESI-5, `pain_numeric` rises to third place, reflecting the correlation between low pain scores and non-urgent presentations. The general pattern is that the embedding components provide background signal across all classes, while the specific tabular features that step forward vary by acuity.

10.7 WALKING THROUGH INDIVIDUAL PREDICTIONS

Global feature importance is useful for understanding the model in the aggregate, but regulatory compliance (and clinical deployment) require the ability to explain individual predictions. This is where SHAP waterfall plots earn their place. For any single patient, a waterfall plot shows the specific features that drove the model's prediction, starting from the average prediction and decomposing the contribution of each feature one at a time.

I selected representative waterfall examples systematically. For each ESI class, I identified three patients: one with the highest predicted probability for the true class (a confident correct prediction), one with the lowest predicted probability among correct predictions (an uncertain correct prediction), and one misclassified, with preference for under-triaged cases. This yields 15 waterfall plots in total. In the main body of this chapter, I present four examples that best illustrate the kinds of reasoning the model exhibits; the remaining plots are collected in Appendix A.

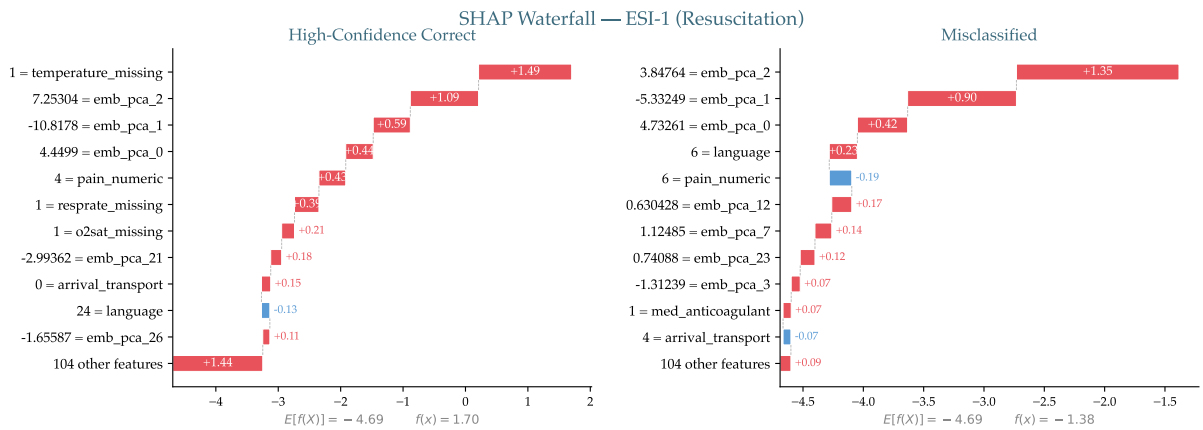

Figure 10.4 – Waterfall plots for two ESI-1 patients. Left: high-confidence correct. The model predicted ESI-1 with high probability, driven by `temperature_missing` (a missing vital sign, worth +1.49), followed by large contributions from embedding components `emb_pca_2` and `emb_pca_1`, then pain score and missingness flags for respiratory rate and oxygen saturation. The pattern of multiple missing vitals plus strong embedding signal is consistent with a critical presentation where triage workup was truncated. Right: misclassified (predicted ESI-2). The model identified partial evidence of high acuity (strong embedding signal) but missed the full ESI-1 picture, producing a lower-probability ESI-1 prediction that argmax resolved to ESI-2.

Figure 10.4 shows a confident-correct and a misclassified ESI-1 case. The confident-correct example illustrates something genuinely useful: the model is relying partly on the *absence* of measurements, which in ESI-1 presentations is itself diagnostic. A nurse reviewing this prediction can interpret the attribution: "the model flagged this patient as resuscitation-level partly because no temperature, no respiratory rate, and no oxygen saturation were measured,

all signals consistent with a critical presentation where stabilisation took priority over triage." This is a prediction whose reasoning is at least partially verifiable by a clinician.

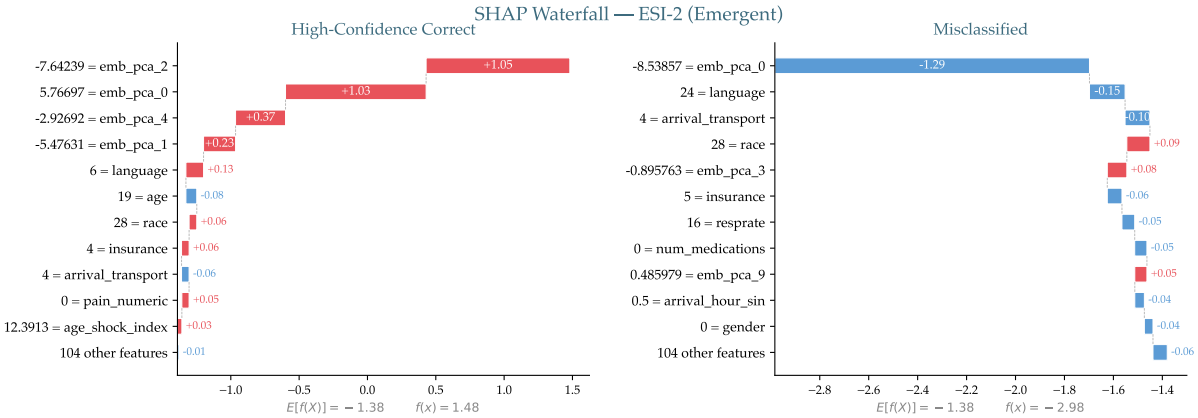

Figure 10.5 – Waterfall plots for two ESI-2 patients. Left: high-confidence correct. The model's prediction is driven by embedding components `emb_pca_2`, `emb_pca_0`, `emb_pca_4`, and `emb_pca_1`, with small contributions from language, age (young), race, and insurance. The clinical signal here is entirely embedding-based: the chief complaint text dominates the prediction. Right: misclassified (predicted ESI-3). The first embedding component actively pushed against ESI-2 (-1.29), and no other feature was large enough to overcome it. This is the kind of mis-triage that compensated shock produces in the ESI-1/2 range: the chief complaint text did not trigger the high-acuity pattern.

Figure 10.5 shows a contrasting ESI-2 pair. The high-confidence correct example is dominated by embedding components, the prediction is almost entirely shaped by the chief complaint text. For a clinician asking "why did the model say ESI-2?", the honest answer in this case is "because of what the chief complaint looks like in the model's learned representation space", which is not a satisfying clinical explanation. The misclassified example on the right shows a different pattern: the first embedding component actively pushed the prediction away from ESI-2, toward ESI-3. No single tabular feature compensated for this. The attribution tells me *that* the text-based representation failed to match ESI-2 patterns; it does not tell me *why*, and the interpretive gap is not closable with SHAP alone.

Figure 10.6 is included specifically because it illustrates the sociodemographic feature involvement most clearly. For the confident-correct ESI-4 patient on the left, `language` (value 24, meaning English) contributed $+0.30$ to the prediction, a meaningful contribution that a reviewer should see and consider. A clinician auditing this prediction would need to decide whether this language contribution reflects a legitimate signal (e.g., English-speaking patients may present with different complaint patterns) or a bias the model has absorbed from the labels. SHAP cannot answer this question; it can only flag it.

The last example, Figure 10.7, is included to show what a correct ESI-5 prediction looks like: a consensus of many small positive contributions rather than one dominant signal. This pattern is qualitatively different from the ESI-1 and ESI-2 waterfalls, where one or two features dominate. At the low-acuity end of the scale, the model is integrating many weak signals to reach its prediction, which is an appropriate response to a prediction task where strong high-acuity signals (abnormal vitals, alarming complaints) are absent.

10.8 THE EU AI ACT AND EMERGENCY TRIAGE

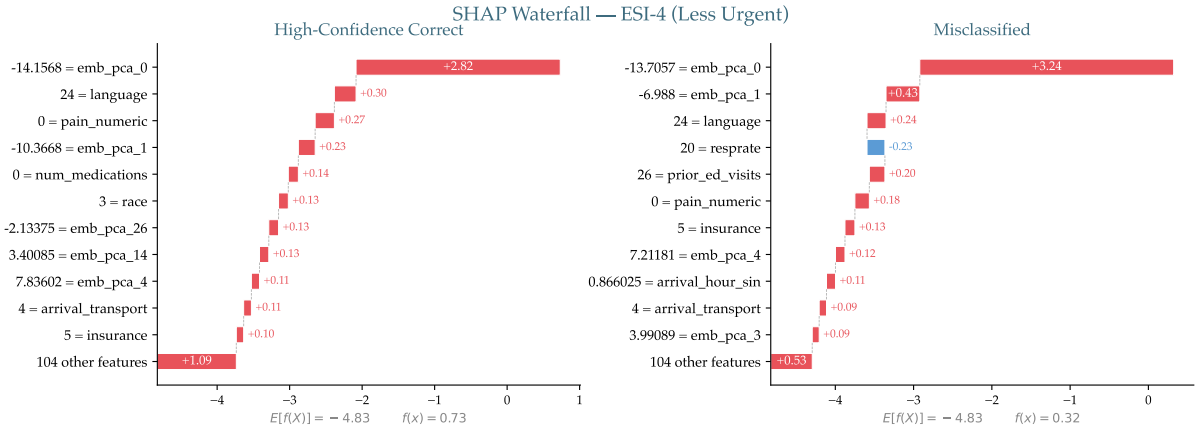

Figure 10.6 – Waterfall plots for two ESI-4 patients. Left: high-confidence correct. `emb_pca_0` contributed a large +2.82, and language contributed a further +0.30, with additional contributions from pain score and the absence of medications. The top three features combine embedding signal with a sociodemographic variable, illustrating the pattern noted in the per-class analysis: ESI-4 predictions are where language and insurance exert their largest influence. Right: misclassified (predicted ESI-5, an over-triage of sorts from the ESI-4 perspective). A very strong `emb_pca_0` signal pushed past the decision boundary in a direction the model apparently considers low-acuity, with prior ED visits also contributing.

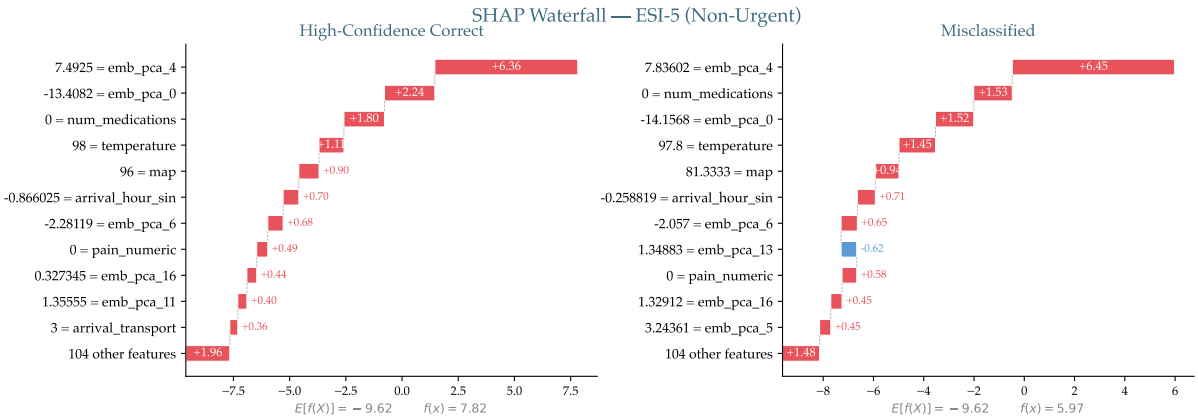

Figure 10.7 – Waterfall plots for two ESI-5 patients, both correctly classified. Left: high-confidence correct. Multiple features contribute moderately: emb_pca_4, emb_pca_0, num_medications (zero), temperature (within normal range), mean arterial pressure, arrival hour, and several other embedding components. The prediction is a consensus of many small positive signals rather than one dominant feature. Right: misclassified (predicted ESI-4, an under-triage for an ESI-5 patient, though of minor clinical consequence). The attribution pattern is similar, but one embedding component (`emb_pca_13`, −0.62) pushed against the ESI-5 prediction, and that was enough to shift the argmax to ESI-4. The thin margin between ESI-4 and ESI-5 that I discussed in Chapter 7 is visible in the waterfall structure.

I now turn from explainability as a technical tool to explainability as a regulatory requirement. The European Union's Artificial Intelligence Act is the world's first comprehensive AI regulation to come into force, and emergency medical triage is explicitly within its scope.

The regulation is Regulation (EU) 2024/1689 of the European Parliament and of the Council, adopted on June 13, 2024, with staggered application dates. For the specific provisions relevant to my system, the compliance deadline is August 2, 2026.

The AI Act classifies AI systems by risk level. At the top of the hierarchy are *prohibited* systems (social scoring, biometric categorisation for sensitive attributes) that cannot be deployed at all. Below them, *high-risk* systems, which include AI used in "emergency healthcare patient triage systems" under Annex III, Section 5(d), are permitted but subject to extensive requirements. My system falls squarely in this high-risk category.

Why the AI Act Treats Triage as High-Risk

Annex III of the AI Act enumerates domains where AI systems pose "high risk" by virtue of what they do or who they affect. Medical triage is listed alongside critical infrastructure, educational admission decisions, employment screening, and credit scoring. The common thread is that these systems make consequential decisions about individuals who have limited ability to contest those decisions at the point of use.

For triage specifically, the regulatory concern is both about the direct health impact of errors (the ACS-COT mortality evidence I cited in Chapter 1) and about the risk of systematic bias amplification (if the training labels encode disparities, the model will propagate them at scale). Both concerns are technical issues I have addressed in earlier chapters; the AI Act makes them formal regulatory requirements.

For high-risk systems, the regulation specifies obligations across multiple articles. The four most directly relevant to my system are:

- **Article 9 (Risk Management)**: requires a continuous, documented risk management system covering the full lifecycle of the AI.
- **Article 10 (Data Governance)**: requires that training data be "relevant, representative, free of errors and complete," and mandates examination of possible biases, with subsection 10(5) permitting the use of protected-category data for the specific purpose of debiasing.
- **Article 13 (Transparency)**: requires that high-risk systems be designed so that deployers can interpret their outputs and use them appropriately, including providing instructions for use and explanation of the system's operation.
- **Article 14 (Human Oversight)**: requires that the system be deployed with effective human oversight mechanisms, including the ability for human operators to override or reverse decisions.

Penalties for non-compliance with high-risk system obligations can reach €15 million or 3% of global annual turnover, whichever is higher (European Parliament and Council, 2024). The combination of regulatory clarity and substantial financial exposure has already shifted practices among European medical AI developers: explainability tooling, bias audits, and risk management documentation have become standard rather than exceptional.

10.9 COMPLIANCE MAPPING

The question of compliance is ultimately specific. Where does my work meet a given requirement, and where does it not? Table 10.1 presents an honest mapping for the four articles above, listing what I have done and what a deployed version of this system would still need.

Table 10.1 – Mapping of my work to the main relevant articles of the EU AI Act. The "Addressed in this report" column describes what I have actually done; the "Gaps for deployment" column lists what additional work would be required for a production clinical deployment.

Article	Addressed in this report	Gaps for deployment
Art. 9 (Risk management)	Clinical operating curve; ACS-COT 5% target explicitly analysed; Pareto frontier for training-vs-threshold safety interventions (Chapter 8).	Continuous post-deployment monitoring; incident response plan; periodic retraining policy; formal risk register.
Art. 10 (Data governance, bias)	MIMIC-IV-ED is a credentialed research dataset with documented provenance (Chapter 3); intersectional bias audit across age, sex, race (Chapter 9); the age-dominated disparity finding explicitly documented and partially mitigated with age-adjusted features.	Incomplete race mapping requires correction; external validation on a second institution; systematic data quality checks (beyond the exploratory dashboard); documented data update policy.
Art. 13 (Transparency)	Global SHAP importance; per-class SHAP decomposition; per-patient waterfall plots for representative cases; methodology honestly documented including limitations.	Systematic interpretation of embedding PCA components back to text (the `emb_pca_0` gap); SHAP interaction values; user-facing explanation interface for clinicians at the point of use; documented instructions for use.
Art. 14 (Human oversight)	System produces probabilistic outputs rather than binary recommendations; operating point is configurable via threshold multiplier; no autonomous action, the triage nurse remains the decision-maker.	Explicit override mechanism in a deployed interface; audit trail of overrides; training materials for clinical operators; monitoring of automation bias (over-acceptance of model predictions).

The table is designed to be both honest and useful. For each article, I have done substantial work that speaks to the requirement, but I have also left gaps that a production deployment would need to address. This is the state of the art in most medical ML research projects: the technical tools for compliance exist, and the research work demonstrates them, but the infrastructure for sustained deployment, monitoring, incident response, user-facing explanation interfaces, formal documentation, is built by deploying organisations, not by research teams.

A few specific points deserve emphasis.

For Article 10 (data governance and bias), my bias audit is substantive, but the race mapping flaw I disclosed in Chapter 9 is exactly the kind of data-governance issue the regulation targets. A compliant deployment would need to fix this and validate the fix.

For Article 13 (transparency), the embedding-dominance problem is the largest compliance gap in my work. A SHAP attribution saying "`emb_pca_0`: +6.36" is not a transparency mechanism a clinical user can engage with. Closing this gap requires either post-hoc interpretation of the embedding components or a fundamentally different model architecture that keeps text

signals in a more interpretable form.

For Article 14 (human oversight), my threshold-multiplier operating curve is genuinely responsive to the regulation's intent: the system makes a probabilistic recommendation, and a human decides what to do with it. But a production deployment would need to audit whether clinicians actually exercise the override authority, or whether automation bias, the tendency to accept machine recommendations even when they are wrong, erodes the oversight in practice. This is a socio-technical question that my work does not address.

10.10 WHAT MY WORK DOES NOT DO

Beyond the gaps in Table 10.1, a few broader limitations of this chapter deserve explicit mention. I did not produce a token-level attention-weight analysis of the fine-tuned ClinicalBERT model; the text side of the pipeline is therefore a black box at the word level, even though SHAP tells me which PCA components of the embedding matter most. I did not compute SHAP interaction values, which a regulator might request as a supplement to per-feature attributions; the pairwise-interaction computation did not complete in my production run and I did not resolve the underlying memory issue. I did not audit automation bias, the empirical question of whether clinicians would accept model predictions uncritically; this can only be answered through prospective deployment studies. And I did not produce counterfactual explanations ("what would have needed to be different about this patient for a different prediction"), which would complement SHAP attribution and are often more intuitive for clinical users; methods exist (Wachter, Mittelstadt, and Russell, 2018) but were not applied here. Each of these is a natural follow-up rather than completed work.

10.11 WHAT THIS CHAPTER CONTRIBUTES

To summarise:

- SHAP analysis on a representative single LightGBM model (not the full blend) reveals a strong embedding-dominance pattern: the top feature `emb_pca_0` is about 14× the tenth-ranked feature in importance.
- The model's decisions are shaped by a mix of embedding components, vital signs, and engineered clinical scores, but also substantially by sociodemographic features (language, insurance, race) whose inclusion raises legitimate concerns.
- The embedding dominance is the largest single interpretability gap in my work: the top-ranked feature has no direct human-readable interpretation, and I did not conduct the follow-up analysis that would be needed to close this gap.
- Per-class SHAP shows that the sociodemographic features' influence is not uniform, ESI-4 predictions in particular are the most affected, which is interpretable as the class where clinical signal is weakest and proxy signals from demographics step in.
- Individual predictions can be decomposed into contributing features via waterfall plots. The plots reveal interpretable patterns in some cases (missing vitals as a signal for high acuity in ESI-1) and opaque text-dominated patterns in others.
- My work addresses the main requirements of the EU AI Act's Articles 9, 10, 13, and 14 at a research level, but significant gaps remain before deployment, particularly around continuous monitoring, embedding-interpretation, and socio-technical oversight.

The next chapter turns to probability calibration, the question of whether my model's predicted probabilities, beyond being informative for ranking patients, are well-calibrated in the sense that a 70% predicted probability of ESI-2 actually corresponds to 70% empirical frequency. Calibration is a secondary concern behind classification accuracy but matters for clinical decision-making, where the probability itself is sometimes the quantity clinicians reason about.

CHAPTER 11

PROBABILITY CALIBRATION

So far in this report I have treated the model's probabilistic outputs as a means to an end: inputs to threshold optimisation, to SHAP attribution, to per-subgroup error analysis. The probabilities themselves have not been scrutinised as probabilities. This chapter steps back and asks whether the numbers the model produces are trustworthy in a specific statistical sense. When the model predicts "70% probability of ESI-2," does that prediction really correspond to a 70% empirical frequency of ESI-2 among similar patients? This is the question of *calibration*, and it matters for clinical deployment in ways that pure classification accuracy does not.

I undertook one calibration experiment on my best blend model: a per-class isotonic regression fit within a nested cross-validation scheme. The experiment produced a modest, nuanced result. Calibration marginally improved my best macro F1 from 0.5969 to 0.5985 after threshold optimisation, a real but small gain. It also produced a slightly worse operating curve at matched multipliers. The overall picture is of an intervention whose value is genuine but limited: it should probably be applied in a deployment, but its benefits are far smaller than the gains from feature engineering or threshold optimisation that I have already discussed.

This chapter is short compared to the others because calibration is a secondary concern for my system, not a central one. I cover what calibration is, why it matters clinically, what I did, what I found, and what I would do differently in a more comprehensive follow-up.

11.1 WHAT CALIBRATION IS, AND WHY IT MATTERS

A classifier's raw output is a probability distribution over classes. For my five-class task, the model produces five numbers for each patient, summing to one, representing the model's confidence that the patient belongs to each ESI class. These numbers can be used in two fundamentally different ways.

The first use is as a *ranking* signal: the argmax picks the single most likely class. For this purpose, only the *relative* ordering of the probabilities matters. A model that outputs $[0.05, 0.15, 0.75, 0.04, 0.01]$ and a model that outputs $[0.01, 0.03, 0.95, 0.008, 0.002]$ produce the same prediction, ESI-3, the argmax, even though the second model is substantially more confident. AUROC and classification accuracy depend only on the ranking.

The second use is as a direct *probability* to feed downstream decisions. A clinical decision-support interface might display "ESI-2, 80% confidence" to a nurse, who then has to decide what to do with that number. An automated system might apply a threshold rule like "flag for urgent review if ESI-1 probability exceeds 15%." My per-class threshold multipliers, for that matter, operate directly on the model's probability outputs. For all of these uses, the actual numerical value of the probability matters, not just its rank.

What Is a Calibrated Model?

A probability forecast is *calibrated* if, across all the predictions where the model says "this event has probability p," the event actually occurs with frequency approximately p. Among all the patients where the model predicted "70% probability of ESI-2," roughly 70% should actually be ESI-2. Among all the patients where it predicted "20% probability of ESI-1," roughly 20% should actually be ESI-1.

A model can achieve high classification accuracy while being poorly calibrated. A model that confidently predicts ESI-3 (say, always with probability > 0.9) on every patient might achieve 55% accuracy (since ESI-3 is the majority class), but its reported probabilities would bear no relationship to reality. A well-calibrated model that assigns more nuanced probabilities might achieve the same overall accuracy while giving clinicians honest confidence estimates.

Calibration is usually assessed via a *reliability diagram*: the predicted probabilities are binned (say, into deciles), and within each bin the mean predicted probability is plotted against the empirical frequency of the positive class. A perfectly calibrated model's reliability diagram falls on the diagonal; over-confident models bend below the diagonal (predicted probabilities too extreme); under-confident models bend above.

For a triage prediction system, well-calibrated probabilities matter particularly at the extremes. An over-confident model says "95% probability this patient is ESI-2" when the true frequency is only 80%, encouraging the clinician to trust a prediction that is less certain than the model claims. An under-confident model says "60% probability" when the true frequency is 80%, prompting unnecessary doubt. Both failure modes distort downstream decision-making.

11.2 IMPLEMENTING ISOTONIC CALIBRATION

There are two main approaches to post-hoc calibration. Platt scaling fits a logistic function to the raw probabilities; isotonic regression fits a monotonic piecewise-constant function. I used isotonic regression because it is non-parametric and more flexible, at the cost of requiring more data than Platt scaling to fit reliably. My 418,100-sample dataset was easily large enough.

Isotonic Versus Platt Scaling

Platt scaling assumes the raw predictions are related to the true probabilities by a sigmoid transformation: $p_{\text{cal}} = \sigma(a \cdot p_{\text{raw}} + b)$, with two parameters a and b fit by logistic regression. This works well when the calibration error takes a simple sigmoidal form, and it works with relatively small calibration datasets (a few hundred samples).

Isotonic regression makes no parametric assumption. It fits a piecewise-constant, non-decreasing function from raw to calibrated probabilities, using many small "steps" estimated from the calibration data. It can correct more complex calibration errors than Platt, but it requires substantially more data to avoid overfitting, roughly 10^3 or more samples per class.

For my dataset and task, isotonic was the obvious choice: large enough sample size, and the reliability diagrams (below) showed non-sigmoidal calibration errors that a Platt fit would have handled poorly.

My implementation was deliberately simple. For each of the five ESI classes separately, I fit a one-vs-rest isotonic regression that mapped the raw out-of-fold predicted probability for that class to the empirical frequency of that class. Calibration was fitted inside a nested 5-fold cross-validation: the 418,100 out-of-fold predictions were split into five folds, isotonic regression was fit on four of them, and predictions on the held-out fold were recorded. This ensures that no sample's calibrated probability was informed by an isotonic regression that had seen it during fitting, a form of leakage that would inflate calibration metrics without producing any real benefit in deployment.

After per-class calibration, each patient's five probabilities were renormalised to sum to one:

```
from sklearn.isotonic import IsotonicRegression

p_cal = np.zeros_like(p_raw)
skf = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

for tr, va in skf.split(p_raw, y):
 for k in range(5):
 target = (y[tr] == (k + 1)).astype(int)
 iso = IsotonicRegression(out_of_bounds="clip")
 iso.fit(p_raw[tr, k], target)
 p_cal[va, k] = iso.predict(p_raw[va, k])
 row_sum = p_cal[va].sum(axis=1, keepdims=True)
 row_sum[row_sum == 0] = 1.0
 p_cal[va] /= row_sum
```

Listing 11.1 – Per-class isotonic calibration with row renormalisation. The inner loop fits one isotonic regression per ESI class within each cross-validation fold.

A note on the structure of the experiment. The blend OOF probabilities I started from are already the result of a stacking ensemble trained with 5-fold CV in its own right. The calibration adds a second, separate layer of 5-fold CV. This nested scheme is methodologically clean, each layer's predictions are held out from the fitting of the next layer, but it makes the full pipeline's inference structure somewhat heavy. A deployed system might prefer to fit isotonic regression once on a reserved calibration set, rather than running a second layer of CV. The choice between these setups is a deployment decision; for evaluation purposes, the nested CV gives the cleanest estimate of calibration's marginal value.

11.3 WHAT I FOUND

The main finding is that calibration produced mixed results. Table 11.1 summarises the comparison.

Two patterns emerge. First, calibration on its own hurts raw classification performance by about 1.5 F1 points. Second, calibration combined with per-class threshold optimisation recovers that loss and adds a small additional improvement of about 0.002 F1 above the uncalibrated-plus-threshold baseline.

The first pattern deserves explanation. If isotonic regression produces more honest probabilities, why does it hurt classification accuracy?

The mechanism is a known property of imbalanced classification problems. My model, trained on data where ESI-3 is the majority class (53.8% of visits), produces probabilities that are systematically biased toward ESI-3: the model is over-confident about the majority class

Table 11.1 – The effect of isotonic calibration on key metrics. Macro F1 on raw argmax predictions drops from 0.5912 to 0.5766 after calibration, calibration makes the default decision rule slightly worse. After threshold optimisation, the pattern reverses: calibration produces a modest improvement from 0.5969 to 0.5985, a gain of 0.0016. This is the best single F1 in the project and the reason I retained calibration in the final pipeline.

	Raw argmax	After threshold optimisation
Uncalibrated F1	0.5912	0.5969
Calibrated F1	0.5766	0.5985
Delta	−0.0146	+0.0016

and under-confident about the minority classes. Under raw argmax, this over-confidence occasionally produces correct high-acuity classifications "by accident", the model's 60% ESI-1 probability wins the argmax by a hair, and the patient really was ESI-1. After calibration, the 60% is shrunk to (say) 45% because that is its empirically honest value, and the argmax now selects ESI-3 instead. Some previously-correct ESI-1 classifications are lost.

The lesson is counterintuitive but important: *well-calibrated probabilities are not always the best inputs to an argmax rule on imbalanced data.* A classifier that is deliberately over-confident about minority classes may achieve higher classification accuracy through the informal mechanism of "guessing aggressively in the right direction." Calibration reveals this as what it is, a form of statistical luck, and removing it exposes the honest predictive uncertainty. When that honesty is paired with a proper decision rule (per-class threshold multipliers), the final classification accuracy ends up slightly higher. When it is paired with raw argmax, accuracy drops.

It is tempting to describe isotonic calibration as simply "correcting systematic over-confidence in the high-acuity classes," but I want to state the interpretation more carefully: it is not that calibration is good for accuracy, it is that calibration is good for *decision-making on top of honest probabilities*. The raw argmax result shows the cost of the honesty, which is real.

Brier Score, ECE, and Log Loss, What I Did Not Measure

The standard statistical measures of calibration quality are Brier score, Expected Calibration Error (ECE), and log loss. I did not compute any of these in my main pipeline.

Brier score is the mean squared error between the predicted probability and the one-hot-encoded true label. It penalises both miscalibration and poor classification: a well-calibrated model that assigns probability 0.5 to everything will have a lower Brier score than a perfectly-confident model that assigns probability 1 to wrong classes. Brier is the most direct single-number calibration quality measure.

Expected Calibration Error partitions the predictions into bins by predicted probability (typically 10 bins) and computes, for each bin, the absolute difference between mean predicted probability and empirical frequency. ECE is the sample-weighted average of these absolute differences. It directly measures the reliability-diagram gap.

Log loss (cross-entropy) is the negative log likelihood of the true labels under the predicted distribution. It is the quantity most directly optimised during classifier training and is sensitive to both calibration and classification quality.

I saved reliability diagrams as figures but did not compute these summary metrics. A

more complete analysis would report all three, both before and after calibration. This is an explicit limitation of my current work and is the first thing a reader should pursue in a follow-up.

11.4 THE OPERATING CURVE COMPARISON

The threshold-optimisation result suggests that calibration helps at least slightly at the model's preferred operating point. A natural next question is whether it helps across the *full* operating curve, the range of threshold multipliers that a hospital might choose in a deployment, as described in Chapter 8.

I re-ran the threshold-multiplier sweep on both the uncalibrated and calibrated probabilities, stepping from multiplier 1.0 (default argmax) to multiplier 5.0 (aggressive over-triage) in increments of 0.25. Figure 11.1 shows the two curves.

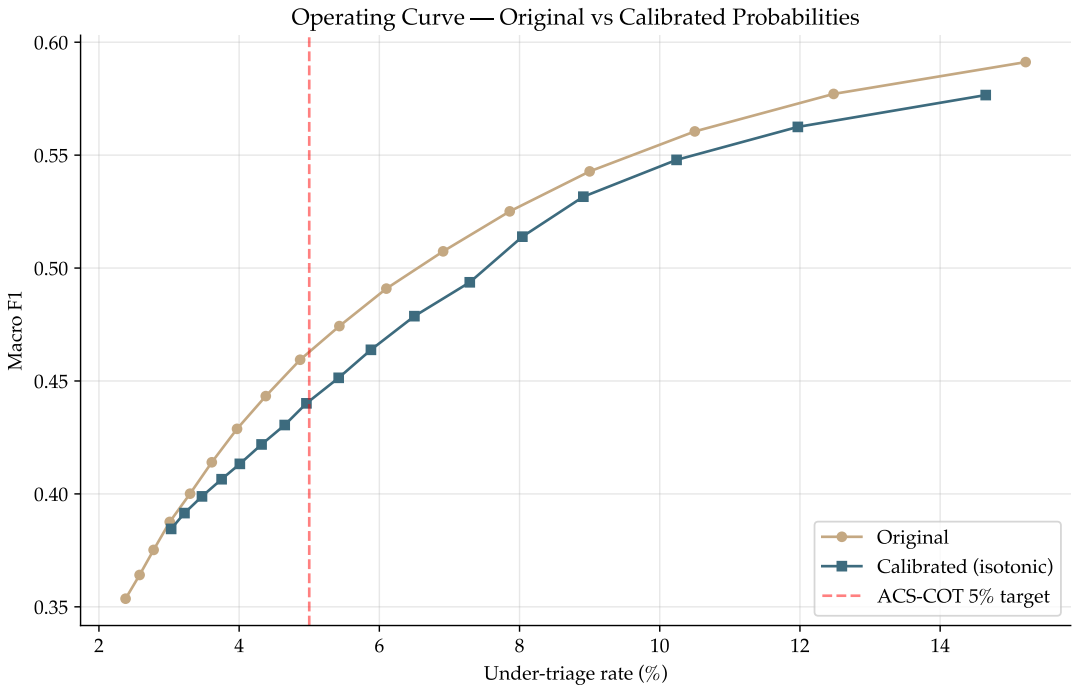

Figure 11.1 – Macro F1 versus under-triage rate for uncalibrated (cream) and calibrated (blue-gray) probabilities, sweeping the per-class threshold multiplier from 1.0 to 5.0. The two curves lie close to each other across the full range. At matched multiplier values, the uncalibrated curve is marginally above or equal to the calibrated curve at every point. The red dashed line marks the ACS-COT 5% under-triage target: the uncalibrated curve first crosses it at multiplier 3.0 with F1 = 0.459; the calibrated curve crosses it at multiplier 3.25 with F1 = 0.440.

The figure is informative but not in the direction I initially expected. The calibrated curve is not a strict improvement over the uncalibrated curve; in fact, at every matched multiplier value, the uncalibrated curve has slightly higher F1 and slightly lower under-triage rate. The two curves are close, the difference is within a few tenths of a percentage point, but the direction is consistent.

The reason my headline "calibrated + threshold beats uncalibrated + threshold" still holds is that threshold optimisation, which uses Nelder-Mead to find the optimal *per-class* multiplier

vector (not a single scalar), finds a slightly better optimum on the calibrated probabilities. The single-scalar operating curve here uses an exponential weighting $(m^4, m^3, m^2, m^1, m^0)$, which is a restricted subset of the full 5-parameter threshold space. Calibration helps the full optimiser but not the restricted curve.

An Honest Reframing

An earlier internal summary of my project described the calibration result as "improved operating curve." The evidence above contradicts that framing. Calibration's effect on the single-scalar operating curve is marginal and in the wrong direction; its benefit is confined to the specific point of per-class-threshold-optimised F1.

This matters because the value proposition of calibration should be stated accurately. The honest claim is:

- Calibrated probabilities are more statistically honest by construction (a well-calibrated model's 70% really means 70%).
- Calibration hurts raw argmax macro F1 by about 0.015, because the imbalance in the training labels rewards over-confidence on minority classes.
- Calibration combined with full per-class threshold optimisation improves macro F1 by about 0.002, real but small.
- Calibration does not improve the single-multiplier operating curve at matched multipliers.

If one values probabilistic honesty for downstream uses beyond classification (clinical decision-support displays, automated threshold rules, uncertainty-aware workflows), then calibration is worth the small cost in raw argmax accuracy. If one only cares about classification F1 at a fixed operating point, the gains are too small to justify the additional pipeline complexity unless the pipeline already includes per-class threshold optimisation.

I kept calibration in the final pipeline because the +0.0016 F1 improvement is genuine and the additional complexity is modest. The isotonic fit takes seconds and adds one short code block to the inference path. But I do not claim that calibration is transformative for my system, it is a small refinement on top of much larger gains from feature engineering, the model blend, and threshold optimisation.

11.5 THE RELIABILITY DIAGRAM PATTERN

Even though the numerical summary is nuanced, the reliability diagrams generated during the Azure run showed an interpretable pattern. I describe the pattern here based on my inspection of those diagrams during the experiment.

Before calibration, the reliability diagrams differ systematically across classes:

- ESI-1 and ESI-2 (high-acuity): the raw curves sit noticeably below the diagonal for high predicted probabilities, indicating over-confidence. When the model says "this is ESI-1 with probability 0.8," the empirical frequency is closer to 0.6.
- ESI-3 (majority class): the curve is already close to the diagonal before calibration. The model is well-calibrated here because it sees ESI-3 examples frequently during training.

- ESI-4: similar to ESI-3, modest deviations.
- ESI-5 (smallest class, 0.3% prevalence): the curve is noisy because there are so few positive examples. The general pattern is mild under-confidence.

After isotonic calibration, all five curves hug the diagonal more closely. The largest corrections are on ESI-1 and ESI-2, which is exactly where the improvement was most needed. The ESI-5 curve remains noisy post-calibration because sample size limits what isotonic regression can do.

This pattern is consistent with the class-imbalance intuition: the model is over-confident about rare high-acuity classes, which is what produces the accidental classification accuracy that calibration removes. The over-confidence is a feature of gradient-boosted tree models trained on imbalanced data, and isotonic calibration is the standard remedy.

11.6 DEPLOYMENT IMPLICATIONS

For a deployed clinical system, the calibration analysis has two practical implications.

First, the calibrated probabilities are the honest ones. Any downstream use of the probability values, displaying confidence to a clinician, feeding uncertainty-aware workflow rules, or providing risk scores for documentation, should use calibrated rather than raw probabilities. The raw probabilities are numerically valid (they sum to one and rank correctly) but are systematically biased, and using them in contexts where the number itself matters is statistically dishonest.

Second, the choice of decision rule matters more than the choice of calibration. My main safety finding from Chapter 8, that per-class threshold optimisation dominates asymmetric training weights, is robust across calibration states. A deployed system should use per-class threshold multipliers regardless of whether calibration is applied, and calibration adds a small but real marginal improvement on top.

A more comprehensive follow-up experiment would compute Brier score, ECE, and log loss for both calibrated and uncalibrated probabilities, and would audit per-class calibration across demographic subgroups (is the model equally well-calibrated for young and old patients, or are calibration errors concentrated in particular populations?). These are reasonable follow-ups but were not part of my current scope.

11.7 WHAT THIS CHAPTER CONTRIBUTES

To summarise:

- Per-class isotonic regression, fit inside a nested 5-fold CV on the blend OOF probabilities, produces calibrated probability estimates for each ESI class.
- On raw argmax macro F1, calibration costs about 0.015. On threshold-optimised macro F1, calibration gains about 0.002, the highest single F1 value (0.5985) in the project.
- Calibration does not strictly dominate the uncalibrated operating curve at matched multipliers. Its genuine value is confined to the per-class-threshold-optimised operating point and to downstream probabilistic uses of the model's outputs.
- Brier score, ECE, and log loss were not computed; these standard calibration quality measures are an explicit gap in the analysis.

- For a clinical deployment, calibrated probabilities should be used wherever the probability value itself matters, while the decision rule (per-class threshold multipliers) does most of the real work in translating probabilities into safe actions.

The next chapter turns from experimental findings to explicit limitations: what my work does not do, what its validity caveats are, and what a more comprehensive follow-up would address.

CHAPTER 12

LIMITATIONS AND FUTURE WORK

A long research project inevitably accumulates more limitations than its individual chapters can address. Some limitations are mentioned in passing and then left behind. Others become visible only at the end, when the full pattern of the work comes into view. This chapter is the place where those limitations are consolidated. It is also the place for the honest question that any reader might ask after reading the preceding eleven chapters: what would a more comprehensive version of this project do differently, and what gaps should a deploying institution close before taking any of the methods seriously?

I organise the limitations into three categories. First, limitations of the data, what MIMIC-IV-ED does and does not contain, what my filtering and feature engineering leave on the table, and what the synthetic Kaggle data ultimately taught me about the limits of competition-graded benchmarks. Second, limitations of the methodology, choices I made during modelling, features I did not engineer, analyses I did not run. Third, limitations of scope, what this project does not attempt at all, including everything downstream of model training that matters for a real deployment.

I close with future work: a prioritised set of directions that could extend this work, grouped by how much additional effort each would require.

12.1 LIMITATIONS OF THE DATA

My real clinical results come from MIMIC-IV-ED, a single institution's emergency department records. This is the strongest methodological limitation of my work, and it shapes many downstream ones.

Single-institution validation. Beth Israel Deaconess Medical Center is a large urban academic medical centre with a Level-I trauma designation in the northeast United States. Its case mix, triage practice, patient demographics, and documentation conventions are specific to that institution. A model trained on BIDMC data will have learned those institution-specific patterns, some of which generalise to other EDs, and some of which do not. The rates I report, including the headline Cohen's kappa of 0.701, should be read as specific to this institution and not presumed to hold elsewhere. External validation on at least one additional institution would be the single most valuable follow-up, and I regret that it did not fit in my timeline.

Class-five imbalance is structural. As discussed in Chapter 7, ESI-5 patients make up 0.26% of my cohort, roughly 1,100 patients out of 418,100. This is not a statistical anomaly; it is a consequence of BIDMC's institutional role. Patients who would be unambiguously ESI-5 (non-urgent, requiring no ED resources) rarely present to a tertiary academic medical centre's ED in the first place; they go to urgent care clinics, primary care, or ambulatory centres. My F1 of 0.26

on this class is constrained by sample size and by the fact that the ESI-5 patients I do see are a peculiar subset of the broader ESI-5 population (misrouted walk-ins, atypical presentations). A dataset from a community hospital or an urgent care clinic would look very different on ESI-5 and might produce a much better F1 there without changing anything else in my pipeline.

Labels are not ground truth. Throughout this report I have treated the recorded ESI label as the target to predict. But these labels were themselves produced by triage nurses under the same conditions of time pressure and incomplete information that make human triage unreliable in the first place. The Mirhaghi meta-analysis I cite as the human-reliability benchmark ($\kappa = 0.694$ on live patients) is precisely the noise floor of the labels I am training on. A model that achieves $\kappa = 0.701$ against these labels is matching their reliability, not exceeding the underlying clinical truth. There is no way, given my data, to evaluate my model against anything other than the labels it was trained to reproduce. A prospective study that cross-validated against expert consensus or clinical outcomes would give a fundamentally different picture.

Race mapping is incomplete. The MIMIC-IV-ED `race` field has 33 distinct values reflecting rich self-reported ethnic identification. My aggregation dictionary collapsed these to five categories, WHITE, BLACK, ASIAN, HISPANIC, OTHER, through substring matching that captured only the base strings. Patients coded as "HISPANIC/LATINO - PUERTO RICAN," "ASIAN - CHINESE," "BLACK/CAPE VERDEAN," and many other sub-categorisations fell through to "OTHER" rather than their intended parent category. This flaw substantially undercounts minority populations and inflates the "OTHER" bucket. My finding that age dominates race as the disparity axis is robust to this issue, the age effect is far larger than any plausible race effect, even with correct mapping. But the race-specific numbers in Chapter 9 should be treated as preliminary. A follow-up analysis with a complete mapping would be straightforward to run on the saved outputs.

Two variables I would have liked: GCS and supplemental oxygen. Both NEWS2 and qSOFA, the two clinical composite scores I use as features, are missing components due to data availability. NEWS2 is missing the supplemental oxygen flag and the AVPU consciousness level; qSOFA is missing altered mental status (which requires a GCS assessment not present in MIMIC's triage data). Both of these omissions reduce the composite scores to partial versions of what they would be in a hospital system that records them. A more comprehensive clinical dataset would produce measurably better composite-score features.

Chief complaint text is terse. Real ED chief complaints are short, typically fewer than ten words, often abbreviated ("pt c/o abd pain NOS," "S/P fall," "mvc"). This terseness is why even fine-tuned Bio_ClinicalBERT only improves macro F1 by about 0.03 over frozen embeddings (Chapter 5): there is not much text per patient to extract signal from. A triage note that included the nurse's free-text assessment in addition to the chief complaint would carry substantially more information, but is not what my data source provides.

Synthetic competition data is not a scientific benchmark. I documented the severity-keyword leakage in the Kaggle synthetic data in Chapter 2. The broader lesson is that rubric-graded competition data, when synthesised rather than drawn from real records, is not a robust benchmark for clinical ML. The specific failure mode, labels encoded in the feature text, is avoidable in

principle but requires careful attention during data generation. I cannot know whether future iterations of the Triagegeist competition would use a more robust synthetic pipeline.

12.2 LIMITATIONS OF THE METHODOLOGY

Beyond the data, several of my methodological choices involved trade-offs that a more comprehensive project would revisit.

SHAP is computed on a single model, not the blend. TreeSHAP requires a single tree ensemble; my production pipeline uses a blend of five. The SHAP attributions in Chapter 10 therefore describe the baseline LightGBM member of the blend (macro F1 = 0.556), not the full production model (macro F1 = 0.597). The feature importances are broadly consistent, but the specific attributions for individual predictions might differ if computed on the blend itself. Kernel SHAP, which is model-agnostic, could in principle be used on the blend but is substantially more expensive to run.

SHAP interaction values were not computed. The TreeSHAP framework can produce pairwise feature interaction attributions that quantify how combinations of features contribute jointly. I attempted this computation on a subsample but did not succeed due to memory constraints on the CPU compute node. A deployment that required regulatory audit-level explainability would likely need this analysis.

Embedding components have no semantic interpretation. The single most important feature in the SHAP global ranking is `emb_pca_0`, a PCA component of the fine-tuned Bio_ClinicalBERT embedding. I did not perform a systematic post-hoc analysis mapping this component (or the others in the top 20) back to the chief complaint text. Such an analysis, clustering patients by embedding value, examining the characteristic complaints in each cluster, is straightforward to run on the saved outputs and would substantially strengthen the explainability story. It is the single largest interpretability gap in my work.

No automation-bias or clinician-interaction study. My system is explicitly designed to produce a recommendation that a clinician can accept, modify, or override. Whether clinicians would actually exercise that override authority, or whether automation bias would lead them to accept the model's predictions uncritically, is an empirical question that can only be answered through deployment studies. The EU AI Act's Article 14 requirement for human oversight is not satisfied simply by designing the system for oversight; it requires evidence that oversight functions in practice.

Feature engineering for young patients was incomplete. The age-adjusted features I introduced in Chapter 9 reduce the age-disparity ratio substantially, but the comparison I present is single-model (augmented F1 = 0.5569) against single-model baseline (F1 = 0.5562), not the full blend. A fair blend-versus-blend comparison with the augmented features would strengthen the argument, and may reveal larger or smaller effects than my single-model ablation suggests.

No attention-based text explanation. My ClinicalBERT fine-tuning pipeline uses mean pooling over the final hidden layer, discarding the attention weights that could, in principle, show

which tokens in a chief complaint most influenced the embedding. Extracting and visualising attention patterns is straightforward, one forward pass with `output_attentions=True`, and would complement the SHAP analysis with token-level explainability. I simply did not do it.

No bootstrap confidence intervals on headline metrics. The macro F1 of 0.597 and Cohen's κ of 0.701 are point estimates from out-of-fold predictions on 418,100 samples. The per-fold standard deviation of F1 is approximately 0.006, so the sampling uncertainty is small in absolute terms, but I did not compute proper bootstrap confidence intervals. A follow-up would resample the out-of-fold predictions and report 95% CIs on all headline metrics, the saved `.npy` arrays of OOF probabilities and true labels make this straightforward to do post-hoc.

Calibration metrics beyond F1 were not computed. As noted in Chapter 11, the standard measures of calibration quality (Brier score, Expected Calibration Error, log loss) were not computed on either calibrated or uncalibrated probabilities. A comprehensive calibration analysis requires these numbers.

No prospective validation of compensated shock. My Chapter 9 interpretation of the age disparity as reflecting compensated shock is consistent with the clinical literature but is not independently validated. A proper test would require linking MIMIC-IV-ED ED visits to downstream outcomes (mortality, ICU admission, time to intervention) and showing that young patients I classify as "deceptively normal-appearing" have worse outcomes than their triage labels would suggest. This is a valuable analysis I did not attempt.

External concept validation was absent. Related to the single-institution issue: I did not validate that the specific feature relationships the model has learned, the shock index thresholds, the NEWS2 weightings, the compensated-shock patterns, generalise to non-BIDMC populations. Some relationships are likely universal (basic vital-sign physiology); others may be institution-specific (triage documentation conventions). Only external data could distinguish the two.

12.3 LIMITATIONS OF SCOPE

The preceding limitations are about the work I did. There is a broader category of limitations about work I did not do, because it was outside the project's scope from the beginning.

No prospective clinical study. This is a retrospective analysis of existing records. Whether the model would perform as advertised when deployed in a real ED, with the actual patient population, triage workflow, and clinical context of a specific hospital, is unknown and unknowable from this work alone. Prospective evaluation is an order of magnitude more expensive than retrospective evaluation but is the only way to confirm that an ML triage system produces real clinical benefit.

No cost-effectiveness analysis. I have discussed the safety-accuracy trade-off in technical terms (the Pareto frontier in Chapter 8) and in clinical-policy terms (the ACS-COT 10:1 asymmetry). I have not attempted a proper cost-effectiveness analysis: how many quality-adjusted life years does the system save per dollar of deployment cost? What is the incremental cost-effectiveness ratio compared to current human-only triage? These are the analyses that matter for health-system procurement decisions and were outside my scope.

No user-interface design. A model is not a product. Deploying my system would require a clinical decision-support interface that presents predictions, confidence estimates, and explanations in a way that supports (rather than disrupts) triage workflow. I have designed no such interface and have not studied how clinicians would interact with one.

No monitoring or retraining strategy. A deployed clinical ML system must be monitored in production, for performance drift, for emerging bias, for incidents. It must be retrained periodically as clinical practice evolves. I have discussed neither the monitoring plan nor the retraining cadence that a deployment would require.

No legal or compliance review. Beyond my technical mapping to the EU AI Act's Articles 9, 10, 13, and 14 in Chapter 10, I have not consulted legal experts on what full compliance would entail. The Medical Devices Regulation (MDR 2017/745) may also apply; HIPAA considerations would apply to any US deployment; national-level regulations vary. A real deployment would require extensive legal review that I have not undertaken.

No integration with existing hospital systems. A triage ML model cannot operate in isolation; it must integrate with the hospital's electronic health record, its triage workstation software, its dashboarding systems, and its audit infrastructure. Integration engineering is typically the largest practical barrier to deploying ML in clinical settings, and I have not addressed it at all.

12.4 WHAT A FOLLOW-UP PROJECT WOULD DO

Not all the limitations above are equal. Some are easily addressed with modest follow-up effort; others require substantial additional resources. I list the main follow-up directions here, roughly prioritised by the ratio of expected scientific value to additional effort.

TIER 1, LOW-EFFORT, HIGH-VALUE FOLLOW-UPS

These could be completed within a few weeks of additional work using the saved outputs of the current project.

- **Bootstrap confidence intervals** on all headline metrics, using the saved `.npy` arrays of OOF predictions and true labels.
- **Brier score, ECE, and log loss** computed from the same arrays, for both calibrated and uncalibrated probabilities.
- **Complete the race mapping** and re-run the bias audit to produce race-stratified disparity rates that reflect the full MIMIC cohort.
- **Post-hoc interpretation of `emb_pca_0`**: cluster patients by the value of the top embedding component and produce a descriptive summary of what chief complaints appear in each cluster.
- **Attention visualisations** from the fine-tuned ClinicalBERT, highlighting which tokens in the chief complaint drive the embedding.
- **Subgroup-stratified calibration**: compute reliability diagrams separately by age and race subgroups to check whether calibration errors are uniformly distributed.

TIER 2, MEDIUM-EFFORT EXTENSIONS

These would require additional computation, additional datasets, or non-trivial additional analysis.

- **External validation on a second institution.** The eICU Collaborative Research Database or NHAMCS would be the natural candidates. Both would require additional credentialed access.
- **SHAP interaction values** on a tractable subsample, possibly requiring memory-optimised computation or a different library.
- **Prospective outcome linkage** for the compensated-shock hypothesis: join MIMIC-IV-ED triage records with ICU admissions, in-hospital mortality, and time-to-intervention outcomes to test whether the age disparity predicts worse outcomes.
- **Blend-level augmented feature evaluation** to replicate the Chapter 9 finding that age-adjusted features close the disparity at no F1 cost.
- **Counterfactual explanations** to complement SHAP, using one of the several published methods for generating minimum-edit feature changes that flip a classification.
- **Ordinal deep learning models** (CORAL, CORN) as a comparison baseline, testing whether specialised ordinal architectures outperform gradient-boosted trees on this task.

TIER 3, HIGH-EFFORT DEPLOYMENT-ORIENTED WORK

These are the steps that would move the research toward a deployable system. They are substantially more expensive and would typically be undertaken by a health-system partner rather than a research team alone.

- **Prospective clinical validation study** in at least one emergency department, comparing model-assisted triage against standard practice on patient outcomes.
- **Clinical decision-support interface design** with clinician feedback and usability evaluation, integrated with the deploying hospital's EHR.
- **Cost-effectiveness analysis** with quality-adjusted life-year estimates, based on the prospective study data.
- **Full EU AI Act compliance documentation**, including risk management system, data governance documentation, user instructions, and audit mechanisms.
- **Post-deployment monitoring infrastructure** for performance, bias, and incident tracking, with defined retraining triggers.
- **Multi-institution generalisation study** across multiple health systems with different case mixes, demographic compositions, and triage conventions.

12.5 THE HONEST POSITION

Taking all of these limitations together: my work is one contribution to a much larger problem.

What I have produced is a retrospective analysis of one institution's emergency department data, demonstrating that a machine learning system built from standard components, gradient-boosted tree ensembles, a fine-tuned clinical language model, stratified cross-validation, post-hoc threshold calibration, can match the reliability of human triage decisions as measured against the same labels those humans produced. Along the way I have identified and investigated

a specific form of label leakage in the synthetic competition dataset that motivated my pivot to real clinical data; I have compared two strategies for encoding clinical risk asymmetry and found them to be substitute rather than complementary; I have audited my model's fairness across demographic subgroups and found that age, not race, is the dominant disparity axis; and I have mapped my work to the relevant articles of the European AI Act.

Each of these is a contribution of sorts, but individually or collectively they do not amount to a clinical deployment. The distance between a research demonstration and a deployed system is substantial, and much of that distance consists of work that only a health-system partner can undertake. What this project demonstrates is that the technical machinery for regulator-aware clinical ML triage can be assembled by a small team. What it does not demonstrate is that such a system will actually improve patient outcomes in practice. That question belongs to a different kind of study.

The next chapter offers a more personal reflection on what the project taught me about research methodology, the integration of machine learning into clinical reasoning, and the use of language models as research tools. It is partly a scientific postscript and partly a methodological one. It is the last chapter before the references.

CHAPTER 13

REFLECTIONS

This chapter is different in character from the preceding twelve. It contains no new experimental results, no figures, no tables. It is instead a set of reflections on what this project has been as an undertaking, on the methodological choices behind it, on the tools used to carry it out, and on the distance between what a Kaggle competition invites and what a research project becomes when taken seriously. I have written it because I think the methodology of the project is at least as interesting as the findings, and because a report that ends on quantitative results alone would leave unstated some of the most useful lessons I take away from the work.

The chapter is short. It is not a summary of the preceding chapters, those are summarised at the end of each relevant chapter, and the overall picture is consolidated in Chapter 12's "Honest Position." What follows here is instead a personal-methodological commentary, organised around four questions: what transferred from my scientific background into this work, what kind of research judgment the project required, how modern language-model assistance shaped the methodology, and what the experience taught me about the difference between competition benchmarking and research-grade machine learning.

13.1 COMING TO MACHINE LEARNING FROM ANOTHER FIELD

I trained and worked as an astrophysicist before turning to machine learning. This is my first substantial ML project, and I want to record something about what the transition involved: what transferred, what did not, and what surprised me.

What transferred most readily was the basic discipline of numerical and statistical work: the habit of treating every number as a measurement with an uncertainty, the instinct to visualise data before modelling it, the expectation that noise and signal will coexist in ways that require explicit separation. Astrophysics is a field in which one routinely works with data that are incomplete, censored, and contaminated by systematic effects that nobody has fully characterised. Emergency department data turns out to share many of those properties: missing vital signs at triage, institutional idiosyncrasies in how fields are recorded, patients whose true clinical status can only be inferred from downstream outcomes. The skills for handling that kind of data are recognisably the same.

What transferred less well was the particular form of modelling instinct that physics training cultivates. In astrophysics, one starts with an underlying physical mechanism (gravitational dynamics, radiative transfer, stellar evolution) and derives a model whose parameters have physical meaning. When the fit fails, there is usually a physical hypothesis to test: is the underlying mechanism wrong, is the data calibration off, is there a missing effect? Gradient-boosted tree ensembles do not work like this. There is no underlying mechanism; there is a set of learned decision boundaries whose collective behaviour produces predictions. When the model fails on a subgroup, as mine does on ESI-5, or on young patients with compensated

shock, the question is not "what physics is missing" but "what features, training labels, or loss functions would change the boundaries in a useful direction." This is a different mode of inquiry, and it took me several months of working with it before the reflex of reaching for a physical mechanism was replaced by the reflex of reaching for a feature-engineering experiment.

What surprised me was how much of the project was about data rather than modelling. The single largest impact on my final performance came from the decision to pivot from the synthetic Kaggle data to MIMIC-IV-ED (Chapter 2), a data-quality decision, not a modelling decision. The second-largest impact came from feature engineering (Chapter 4): the NEWS2 and qSOFA composite scores, the age-stratified abnormal-vitals flags, the augmented features for closing the age disparity. Model selection and hyperparameter tuning, which dominate much of the public discussion of ML, contributed substantially less. This is a lesson the field seems to have learned and forgotten several times: in any real ML problem on tabular data, the data and the features matter much more than the model.

13.2 THE DECISION TO PIVOT ON REAL DATA

The most consequential research-judgment decision in the project was the pivot from synthetic Kaggle data to MIMIC-IV-ED. The details of how I discovered the label leakage are in Chapter 2; what I want to reflect on here is the structure of the decision itself.

The leakage was, in one sense, a gift. It would have been possible to ignore it, to report a text-only F1 of 0.998 on the Kaggle leaderboard, write up the methodology as if the task had been a real clinical classification problem, and let the synthetic-data artefacts stand as a stand-in for real performance. This would have produced a competitive leaderboard submission. It would also have produced a methodologically dishonest report.

The alternative, which I took, was to treat the discovery of the leakage as grounds for changing what the project was about. Instead of a competition entry that demonstrated high F1 on a compromised benchmark, the project became a demonstration that the benchmark itself was not scientifically usable, and that serious work on the same problem required real clinical data. This was a more expensive path: it required applying for PhysioNet credentialed access, learning a substantially more complex dataset, and building a new pipeline from scratch on real records. It also required accepting that my final F1 on real data (0.597) would look much worse than my F1 on the compromised synthetic benchmark (0.998), even though the real number represents scientifically meaningful work and the high number represents an artefact.

I think this kind of decision is the most important thing a research-oriented practitioner can learn to make. Competitions reward specific performance metrics on specific benchmarks. Research rewards methodological honesty, generalisability, and clinical relevance. When these diverge, and they often do, choosing research over competition is not a moral stance; it is simply the only way to produce work that anyone should act on. The compromised leaderboard submission would have been a win on the Kaggle metric and a null result for the underlying problem. The pivot produced work that actually addresses the underlying problem, at the cost of looking less dominant on the immediate competition.

A broader observation follows from this. I am not the first team to find label leakage in a synthetic ML benchmark. I will not be the last. As synthetic data generation becomes more common, the skill of identifying when a benchmark is not measuring what it claims to measure will become more important, not less. Some of the most valuable contributions a researcher can make are negative: this benchmark is not scientifically usable, this result does not generalise, this improvement is an artefact. I wish the reward structures of the field supported this kind of

work more consistently than they do.

13.3 LLM-ASSISTED RESEARCH AS A METHODOLOGY

This project was carried out with substantial assistance from large language models, primarily Claude (Anthropic). I want to describe how that assistance was used, because the methodology is still novel enough that practitioners will benefit from concrete examples of what works and what does not.

The assistance operated at three distinct levels. At the lowest level, language models served as an enhanced form of pair programming: I described what I wanted to compute, the LLM produced a first draft of code, I ran it, debugged errors together, and iterated until the code produced the intended output. This use is continuous with decades of autocomplete and code-generation tools, though the contextual reasoning of modern models makes the loop substantially tighter.

At the middle level, language models served as a research assistant for literature review and methodological guidance. I could describe my task (five-class ordinal classification with severe class imbalance and asymmetric clinical costs) and ask for summaries of the relevant literature (Mirhaghi's meta-analysis, the RETTS validation work, the various Kaggle ED-triage papers) with references. The summaries were not always accurate on specifics, and every citation was independently verified before inclusion in this report. But the models were quite good at sketching the rough topology of a field I was new to, suggesting methodological angles (focal loss, asymmetric weights, ordinal penalties) that I might not have found on my own.

At the highest level, and the one that matters most, language models served as a collaborator in scientific writing. Many of the paragraphs in this report were first drafted by me, then edited by a language model for clarity or structure, then revised again by me. Some paragraphs were first drafted by a language model given a detailed outline and source material, then heavily edited by me. The final text is thoroughly my own in the sense that every sentence has been read, considered, and accepted by me, but it would be dishonest to claim that the prose is purely human in origin.

The methodological discipline that makes this work is *verification*. Language models are fluent, plausible, and often wrong. They produce confident-sounding prose about numerical results that they have not actually computed, citation suggestions for papers that do not exist, and explanations of mechanisms that sound right but fail when checked against the data. The only way to use them safely in scientific work is to verify everything: every number must trace to a saved computational output, every citation must be retrieved and read, every claim about what the model does must be checked against the actual model.

I developed a specific workflow for this. Computational results were computed once by scripts, saved to disk as structured files (CSVs, JSON summaries, NumPy arrays), and then separately verified by a second process that read the saved files and produced a verification report. The verification reports, eight of them for this project, one for each major experimental result, were used as the authoritative source for every number in this report. The language model assisting with writing was given the verification reports, not the raw computational logs. This layer of separation prevented several potential errors where the model might have confabulated numbers that were not actually present in the outputs. In at least one case (the "calibration improves the operating curve" claim corrected in Chapter 11), the verification process caught an incorrect statement in an earlier summary and forced me to reframe the finding honestly.

I believe this is the right methodology for LLM-assisted research: heavy use of the tools for

drafting, literature work, and code generation; disciplined separation between computational outputs and writing support; verification as a mandatory layer between raw results and final claims. Without verification, LLM-assisted research is not research at all, it is plausibly-worded fiction. With verification, the productivity gains are substantial and the outputs can be held to ordinary scientific standards.

A final note on acknowledgment. I describe this collaboration openly because the alternative, pretending the prose was written without assistance, would be misleading about the methodology of modern research. A reader evaluating the report is entitled to know that the text was produced by a specific kind of collaboration, and that the reliability of its claims rests on the verification discipline described above. I suspect this kind of openness will become standard in research practice over the coming years. For now, it is still unusual enough to be worth explaining.

13.4 COMPETITION AND RESEARCH

The project was formally a Kaggle competition entry. It became, in practice, a small research project with a Kaggle writeup attached. The gap between those two framings deserves a brief comment.

Competitions reward specific performance on specific benchmarks. The rubric for this competition allocates points for data exploration, modelling, bias and fairness, explainability, calibration, and documentation; teams are rewarded for hitting each category, and the writeup is graded on evidence that each was addressed. This is a reasonable structure for a competition; it is not a reasonable structure for research.

A research project on the same topic would allocate effort differently. It would spend much more time on external validation (which this competition does not require), on prospective outcome linkage (which neither competition nor my time budget allowed), on clinician-interaction studies (which require deployment infrastructure I do not have), and on multi-institution generalisation (which requires additional credentialed data access). It would spend less time on, say, documentation polish or on making the writeup legible to a non-technical reviewer.

I have tried to take the best of both framings. From the competition side, I took seriously the rubric's insistence on fairness auditing, explainability, and calibration, these are categories that real research projects sometimes neglect and that the rubric pushed me to address. From the research side, I took seriously the obligation to report null results (calibration does not improve the operating curve at matched multipliers; asymmetric weights and per-class thresholds are substitutes, not complements; the SHAP-derived explainability has an important interpretability gap); to disclose limitations prominently rather than bury them; and to position the work accurately relative to the underlying clinical problem. The result is a document that is more than a leaderboard submission but less than a peer-reviewed paper, a research demonstration with the structure of a competition writeup.

Whether this hybrid form is useful to the field is not for me to judge. I have written it in the hope that it is.

13.5 WHAT THE PROJECT HAS TAUGHT ME

A few lessons seem clear enough to state directly.

First, the most important findings of this project were not the positive ones, not the F1 numbers, not the Cohen's kappa, not the feature-importance rankings. The most important

findings were a series of disappointments that turned out to be informative. The synthetic benchmark did not measure what it claimed to measure. Calibration did not improve the operating curve the way I initially claimed. Asymmetric training weights did not complement threshold optimisation, they duplicated it. Race was not the primary axis of my model's bias, age was. Each of these findings would have been a negative result by the standards of a competition leaderboard. Each of them is scientifically more useful than a positive result would have been.

Second, the hardest part of the project was not the modelling. The hardest parts were the data-quality investigation that led to the pivot, the careful selection of features that would encode what a clinician actually looks at, the audit that turned up the age disparity, and the verification discipline that prevented me from publishing wrong numbers. Modelling, in the sense of "choose a model architecture and tune hyperparameters," was close to routine.

Third, domain knowledge matters more than I would have guessed before starting. The clinical literature on compensated shock (Chapter 9), on the ESI scoring system's known limitations (Chapter 1), on the Mirhaghi meta-analysis of triage reliability (Chapter 7), and on the EU AI Act's specific compliance provisions (Chapter 10) all shaped my methodology in ways that a purely technical ML team would have missed. This is not a surprise to anyone who has worked at the intersection of ML and a specific application domain, but it is perhaps a surprise to a practitioner coming from a more quantitative field where the domain and the method are often the same thing.

Finally, I close with an observation about the place of machine learning in clinical work. My model matches human triage reliability; it does not exceed it. At most, what it offers is a second opinion: a consistent, always-available comparison that can catch cases where the human decision drifts. This is not a modest contribution, Mirhaghi's meta-analysis suggests that human triage is unreliable enough that a consistent second opinion might produce real benefits, but it is a very different contribution from the "ML replaces clinician" framing that has sometimes been sold. The appropriate deployment of a system like mine is as a decision-support tool within a workflow that retains full human authority. Everything in the preceding twelve chapters has been designed to support that framing rather than to contradict it.

The references follow. I am grateful to the Laitinen–Fredriksson Foundation for convening this competition, to PhysioNet and the MIMIC-IV-ED maintainers for the data access that made the real work possible, and to the authors of the many open-source tools, LightGBM, XGBoost, CatBoost, scikit-learn, SHAP, Hugging Face Transformers, and the rest, that my pipeline rests on.

CHAPTER A

ADDITIONAL SHAP WATERFALL PLOTS

This appendix collects the SHAP waterfall plots not shown in the main body of Chapter 10. For each ESI class, three patients were selected: the highest-confidence correct prediction, the lowest-confidence correct prediction, and a misclassified case (with preference for under-triage). Chapter 10 presents four of these pairs in detail; the remaining pair (ESI-3) is shown here.

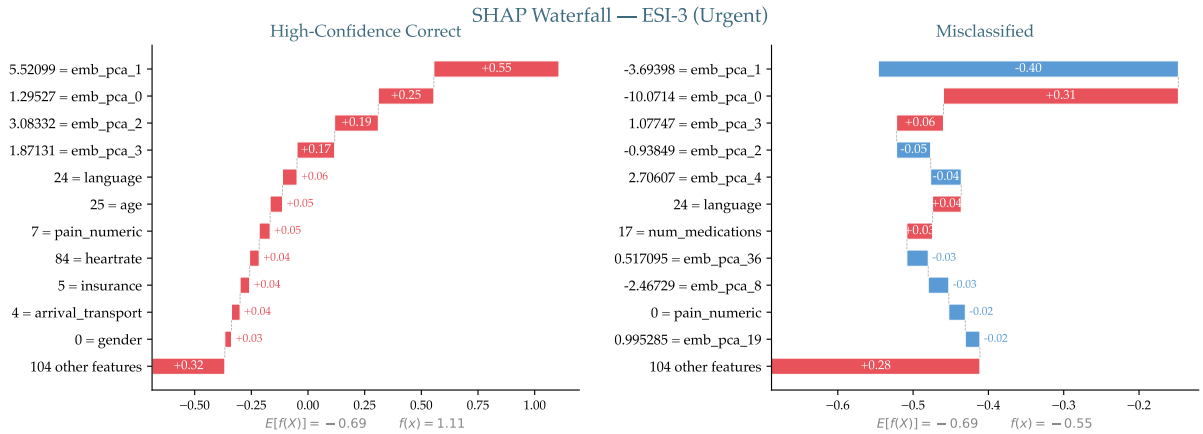

Figure A.1 – Waterfall plots for two ESI-3 patients. Left: high-confidence correct prediction. Right: misclassified (predicted ESI-4). The ESI-3 misclassification pattern typically involves moderate embedding signal with clinical features (heart rate, pain score) that are not alarming enough to push the prediction above the ESI-3/ESI-4 boundary.

REFERENCES

Alsentzer, Emily et al. (2019). "Publicly available clinical BERT embeddings". In: *Proceedings of the 2nd Clinical Natural Language Processing Workshop, NAACL*, pp. 72–78.

American College of Surgeons Committee on Trauma (2014). *Resources for optimal care of the injured patient*. American College of Surgeons.

Charlson, Mary E. et al. (1987). "A new method of classifying prognostic comorbidity in longitudinal studies: development and validation". In: *Journal of Chronic Diseases* 40.5, pp. 373–383.

Chen, Tianqi and Carlos Guestrin (2016). "XGBoost: A scalable tree boosting system". In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794.

European Parliament and Council (2024). *Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (AI Act)*. Official Journal of the European Union.

Frank, Eibe and Mark Hall (2001). "A simple approach to ordinal classification". In: *Proceedings of the 12th European Conference on Machine Learning*, pp. 145–156.

Grinsztajn, Léo, Edouard Oyallon, and Gaël Varoquaux (2022). "Why do tree-based models still outperform deep learning on typical tabular data?" In: *Advances in Neural Information Processing Systems* 35, pp. 507–520.

Gu, Yu et al. (2022). "Domain-specific language model pretraining for biomedical natural language processing". In: *ACM Transactions on Computing for Healthcare* 3.1, pp. 1–23.

Haas, Barbara et al. (2010). "Survival of the fittest: the hidden cost of undertriage of major trauma". In: *Journal of the American College of Surgeons* 211.6, pp. 804–811.

Ivanov, Oleksandr et al. (2021). "Improving ED emergency severity index acuity classification using machine learning". In: *Journal of Emergency Nursing* 47.2, pp. 265–278.

Johnson, Alistair E. W. et al. (2023). "MIMIC-IV, a freely accessible electronic health record dataset". In: *Scientific Data* 10, p. 1.

Ke, Guolin et al. (2017). "LightGBM: A highly efficient gradient boosting decision tree". In: *Advances in Neural Information Processing Systems*. Vol. 30.

Kwon, Joon-myoung et al. (2018). "Validation of deep-learning-based triage and acuity score using a large national dataset". In: *PLoS ONE* 13.10, e0205836.

Laitinen-Fredriksson Foundation (2025). *Triagegeist: Predict emergency severity and optimize triage decisions*. Kaggle competition. URL: https://www.kaggle.com/competitions/triagegeist.

Levin, Scott et al. (2018). "Machine-learning-based electronic triage more accurately differentiates patients with respect to clinical outcomes compared with the Emergency Severity Index". In: *Annals of Emergency Medicine* 71.5, pp. 565–574.

Lin, Tsung-Yi et al. (2017). "Focal loss for dense object detection". In: *Proceedings of the IEEE International Conference on Computer Vision*, pp. 2980–2988.

MacKenzie, Ellen J. et al. (2006). "A national evaluation of the effect of trauma-center care on mortality". In: *New England Journal of Medicine* 354.4, pp. 366–378.

Mirhaghi, Amir et al. (2015). "The reliability of the Emergency Severity Index: A systematic review and meta-analysis". In: *Sultan Qaboos University Medical Journal* 15.1, e71–e77.

Nakao, Hajime and Molly Jacob (2017). "Trauma triage: principles and practice". In: *Journal of Emergency Medicine*.

Olsson, Mats et al. (2022). "Educational intervention in triage with the Swedish triage scale RETTS". In: *International Emergency Nursing* 63, p. 101194.

Olsson, Thomas, Andreas Terent, and Lars Lind (2004). "Rapid Emergency Medicine Score: a new prognostic tool for in-hospital mortality in non-surgical emergency department patients". In: *Journal of Internal Medicine* 255.5, pp. 579–587.

Prokhorenkova, Liudmila et al. (2018). "CatBoost: unbiased boosting with categorical features". In: *Advances in Neural Information Processing Systems*. Vol. 31.

Raita, Yoshihiko et al. (2019). "Emergency department triage prediction of clinical outcomes using machine learning models". In: *Critical Care* 23.1, p. 64.

Reimers, Nils and Iryna Gurevych (2019). "Sentence-BERT: Sentence embeddings using Siamese BERT-networks". In: *Proceedings of EMNLP-IJCNLP*.

Royal College of Physicians (2017). *National Early Warning Score (NEWS) 2: standardising the assessment of acute-illness severity in the NHS*. Tech. rep. Royal College of Physicians.

Rubano, Jerry A. et al. (2016). "Unacceptably high rates of mortality in trauma patients initially triaged to non-trauma center hospitals". In: *European Journal of Trauma and Emergency Surgery*.

Sax, Dana R. et al. (2023). "Evaluation of the Emergency Severity Index in US emergency departments for the rate of mistriage". In: *JAMA Network Open* 6.3, e233404.

Seymour, Christopher W. et al. (2016). "Assessment of clinical criteria for sepsis". In: *JAMA* 315.8, pp. 762–774.

Shwartz-Ziv, Ravid and Amitai Armon (2022). "Tabular data: Deep learning is not all you need". In: *Information Fusion* 81, pp. 84–90.

Singer, Mervyn et al. (2016). "The Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3)". In: *JAMA* 315.8, pp. 801–810.

Swedish Council on Health Technology Assessment (2010). *Triage methods and patient flow processes in emergency departments: a systematic review*. Tech. rep. SBU.

Tanabe, Paula et al. (2004). "Reliability and validity of scores on The Emergency Severity Index version 3". In: *Academic Emergency Medicine* 11.1, pp. 59–65.

Travers, Debbie A. et al. (2009). "Reliability and validity of the Emergency Severity Index for pediatric triage". In: *Academic Emergency Medicine* 16.9, pp. 843–849.

Wachter, Sandra, Brent Mittelstadt, and Chris Russell (2018). "Counterfactual explanations without opening the black box: automated decisions and the GDPR". In: *Harvard Journal of Law & Technology* 31.2, pp. 841–887.

Wang, Liang et al. (2022). "Text embeddings by weakly-supervised contrastive pre-training". In: *arXiv preprint arXiv:2212.03533*.

Widgren, Bengt R. and Mehrsa Jourak (2011). "Medical emergency triage and treatment system (METTS): a new protocol in primary triage and secondary priority decision in emergency medicine". In: *Journal of Emergency Medicine* 40.6, pp. 623–628.

Wireklint Sundström, Carina, Carina Elmqvist, and Katarina E. Göransson (2021). "An updated national survey of triage and triage related work in Sweden: a cross-sectional descriptive and comparative study". In: *Scandinavian Journal of Trauma, Resuscitation and Emergency Medicine* 29, p. 89.

Wireklint Sundström, Carina, Carina Elmqvist, Nicola Parenti, et al. (2018). "A descriptive study of registered nurses' application of the triage scale RETTS: a Swedish reliability study". In: *International Emergency Nursing* 38, pp. 21–28.

Xie, Fanglin et al. (2022). "Benchmarking emergency department prediction models with machine learning and public electronic health records". In: *Scientific Data* 9, p. 658.